

Generic Future Inextendibility with Square-Integrable Connection Near a Fixed Kerr Spacetime

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September 23, 2026

Abstract

For each fixed rotating subextremal Kerr background with mass $M > 0$ and rotation $0 < |\mathfrak{a}| < M$, we prove that the smooth vacuum data near its complete two-ended bridge whose full maximal globally hyperbolic development admits a future $C^0 \cap W_{\text{loc}}^{1,2}$ extension form a meagre set in the weighted smooth topology. The extension metric is continuous and nondegenerate, and its weak connection is locally square-integrable. The neighborhood requires smallness of only the tenth weighted seminorm, while the topology tests every finite order.

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1 Introduction

Strong cosmic censorship asks whether initial data determine a spacetime that cannot be continued beyond its maximal globally hyperbolic development. The regularity allowed for a continuation is part of the question. Kerr’s smooth inner horizon makes this distinction especially important [2]: a continuation can preserve the metric while losing the regularity needed to interpret its connection. Here we prove a local, Baire-category form of this assertion near each fixed rotating subextremal Kerr bridge, at the threshold of locally square-integrable connection coefficients.

1.1 Data and future extensions

Fix a Kerr vacuum solution of mass $M > 0$ and signed rotation parameter $\mathfrak{a}$ with $0 < |\mathfrak{a}| < M$. The letter a used later for a double-null metric coefficient is unrelated to $\mathfrak{a}$. Let (Σ, h_*, K_*) be its smooth complete two-ended bridge data: the $t = 0$ hypersurface through the bifurcation sphere in the two-ended Boyer–Lindquist extension, with its inherited future orientation. Thus $\Sigma \simeq \mathbb{R} \times \mathbb{S}^2$; write $d_* = (h_*, K_*)$. Fix a smooth weight $\rho \geq 1$ equal to the radial coordinate outside a compact set on each end. Write ∇_* and $|\cdot|_*$ for the connection and tensor norms of h_* . For smooth symmetric covariant two-tensors u, v , put

$$p_m(u, v) = \sum_{j=0}^m \left(\sup_{\Sigma} \rho^{1+j} |\nabla_*^j u|_* + \sup_{\Sigma} \rho^{2+j} |\nabla_*^j v|_* \right), \quad m = 0, 1, \dots \quad (1.1)$$

Let $\mathcal{D}$ consist of smooth pairs $d = (h, K)$ such that h is positive definite and complete, all $p_m(d - d_*)$ are finite, and

$$R(h) + (\mathrm{tr}_h K)^2 - |K|_h^2 = 0, \quad \mathrm{div}_h K - \mathrm{d}(\mathrm{tr}_h K) = 0. \quad (1.2)$$

Give $\mathcal{D}$ the relative topology generated by the seminorms (1.1) on differences, and write

$$\mathcal{U}_\varepsilon = \{d \in \mathcal{D} : p_{10}(d - d_*) < \varepsilon\}. \quad (1.3)$$

No leading asymptotic coefficient, charge value, or parity condition is prescribed beyond the constraints and the neighborhood condition. An asymptotic expansion, or existence of every possible

charge integral, is not assumed. For each $d \in \mathcal{D}$, let (M_d, g_d) be its full smooth maximal globally hyperbolic vacuum development (MGHD), with the specified future orientation. Existence and geometric uniqueness are understood in the usual smooth Cauchy theory; see [3].

Definition 1.1 (Future extension). A future $C^0 \cap W_{\text{loc}}^{1,2}$ extension of (M_d, g_d) consists of a connected time-oriented smooth four-manifold $\widetilde{M}$, a continuous nondegenerate Lorentzian metric $\widetilde{g}$ of signature $(-, +, +, +)$ whose components belong to $W_{\text{loc}}^{1,2}$ in smooth charts, and a smooth time-orientation-preserving isometric embedding $\iota : M_d \hookrightarrow \widetilde{M}$ with proper open image. There must also be a future-directed timelike C^1 curve $c : [0, 1] \rightarrow \widetilde{M}$ such that

$$c([0, 1)) \subset \iota(M_d), \quad c(1) \in \partial\iota(M_d).$$

No field equation or global-hyperbolicity condition is imposed outside the image.

The phrase *square-integrable connection* has a weak meaning here. For a merely continuous metric, first define the lowered Koszul coefficients as distributions in a smooth chart:

$$\Gamma_{ijk}^b = \frac{1}{2}(\partial_j g_{ik} + \partial_k g_{ij} - \partial_i g_{jk}).$$

The distributional identity $\partial_k g_{ij} = \Gamma_{ikj}^b + \Gamma_{jki}^b$ shows that $g \in W_{\text{loc}}^{1,2}$ exactly when these lowered coefficients have locally L^2 representatives. On a coordinate subdomain compactly contained in a larger chart, continuity and nondegeneracy bound g and g^{-1} . Raising or lowering the first index therefore preserves L^2 , giving the usual weak Christoffel array. This interpretation uses distributional derivatives, not derivatives that happen to exist almost everywhere for a continuous function. For metrics in Definition 1.1, the weak coefficients agree almost everywhere with the smooth connection on the open image of M_d .

This is the regularity framework for distributional curvature used by Geroch–Traschen and LeFloch–Mardare [12, 16]; the forward implication from $W_{\text{loc}}^{1,2}$ to an L_{loc}^2 Levi-Civita connection is stated in [16, Proposition 4.4]. Smooth chart transitions preserve this local regularity on compactly contained subdomains. None of these facts supplies a trace of the connection on an individually prescribed curve.

Theorem 1.2. *For every fixed $M > 0$ and $0 < |\mathfrak{a}| < M$, there is $\varepsilon = \varepsilon(M, \mathfrak{a}) > 0$ such that the set of $d \in \mathcal{U}_\varepsilon$ for which the full development (M_d, g_d) admits a future extension in the sense of Definition 1.1 is meagre in $\mathcal{U}_\varepsilon$. Consequently, future $C^0 \cap W_{\text{loc}}^{1,2}$ -inextendible data contain a residual subset of $\mathcal{U}_\varepsilon$ in the weighted smooth topology (1.1).*

Here meagre means a countable union of relatively nowhere dense sets. Only the tenth seminorm is required to be small. Every finite higher seminorm can be tested in the category argument, and constants at fixed higher orders may depend on the chosen smooth datum. The neighborhood and its constants may depend on the fixed Kerr center. No nonrotating or extremal case, estimate uniform as $|\mathfrak{a}|$ tends to 0 or M , or genericity outside these neighborhoods is asserted. Nor is inextendibility asserted to be open. The topology is Baire, as recorded in [22, Lemma 2.4]; the nowhere-denseness proof below establishes the asserted countable covering directly.

Section 2.1 reduces the fixed background to mass one and positive rotation. The reduction preserves each finite seminorm order, including the tenth-order neighborhood in the theorem.

1.2 Historical context and scope

Penrose’s cosmic-censorship programme connects gravitational collapse with predictability [27]; its strong formulation concerns global hyperbolicity and the instability of Cauchy horizons [26,

Section 12.3.2]. The mass-inflation analysis of Poisson–Israel and the charged and rotating models of Ori explain how blueshift can produce an inner singularity with comparatively weak tidal effects [28, 24, 25]. Dafermos established nonlinear stability and instability results for the spherically symmetric Einstein–Maxwell–real-scalar-field system, including the coexistence of a continuous metric and an unbounded Hawking mass under appropriate decay assumptions [7, 8]. Luk–Oh subsequently proved generic C^2 future inextendibility for admissible two-ended asymptotically flat data in the spherically symmetric Einstein–Maxwell–real-scalar-field system [19, 20]. These results clarify why the topology of initial data and the regularity of extensions are both essential parts of a censorship statement.

Christodoulou identified square-integrable connection coefficients as a natural threshold when formulating strong cosmic censorship in terms of weak solutions [4, p. 9]. Luk constructed vacuum spacetimes without imposing symmetry, with continuously extendible intersecting null singular boundaries and non-square-integrable Christoffel coefficients in the construction’s coordinates [18]. For rotating black holes, the continuous extension problem has a different character from the differentiable one. The interior stability theorem of Dafermos and Luk shows that appropriately controlled interior Cauchy data have a maximal Cauchy development admitting a continuous Lorentzian extension across a nontrivial portion of a Cauchy horizon [9]. Holonomy provides a geometric way to test stronger extension regularities: Sbierski developed obstructions at locally Lipschitz regularity in [30], and later gave a nonsymmetric criterion based on signed integrated curvature blow-up stable under perturbations of the testing fields [31, Theorem 3.14]. Recent vacuum interior results of Gurriaran and of Luk and Sbierski establish weak null singularities and exclude locally Lipschitz metric extensions through the covered Cauchy-horizon portions under suitable smallness and prescribed asymptotic assumptions [13, 21]. Gurriaran starts from an interior spacelike hypersurface; Luk and Sbierski use characteristic data. These results motivate studying a continuous metric with an L^2 connection, where neither pointwise bounded connection coefficients nor their traces on prescribed curves are available. Cameron–Sbierski have recently established obstructions to $C^0 \cap W_{\text{loc}}^{1,s}$ extensions, $s > 1$, in the strongly spherically symmetric category [1]. Their local criterion also requires equivalence with a reference continuous boundary; their applications to charged black-hole interiors use additional rigidity to remove that restriction. The argument below instead controls connection mass over families of observation positions and closes a stopped transport estimate in an arbitrary ambient chart.

The three near-Kerr articles have the same embedding and future-exit conventions. On a common restriction of their data neighborhoods at the same fixed positive-rotation center, their permitted ambient metric classes satisfy

$$C^2 \subset C^1 \subset C^0 \cap W_{\text{loc}}^{1,2}.$$

Excluding the largest class gives the strongest of these three inextendibility conclusions. Each article remains a distinct result and may choose its own neighborhood. The present proof uses intermediate companion results with the hypotheses stated below; neither companion’s headline inextendibility theorem is a premise.

Source	Role in the proof
C^2 companion [23]	Exterior and interior comparison geometry, first-runout classification for geodesics of finite full lifetime, the constrained hierarchy and beam construction, the support-preserving constraint correction, and lower-region construction.
C^1 companion [22]	Corner nullity, branch-state and canonical-tube estimates, fixed-patch refinements of the C^2 ray construction, linear and exterior preparation for $A_T = e^{-\alpha T}$ at each fixed $\alpha > 0$ through the entry cylinder, and the qualitative whole-bridge joining interface.
This article	A positive family from an arbitrary weak exit, an open observation family and whole-footprint packet adaptation, the double-null nonlinear comparison at $A_T = e^{-\kappa T/4}$, and the averaged holonomy obstruction.

The C^1 packet theorem itself gives a one-point polarization; the whole-footprint lower bound is proved here from its transport construction. The C^1 companion’s exterior estimate at these fixed decay rates stops at the entry cylinder; the deep comparison at the present amplitude is also proved here. These arguments yield the local Baire-category statement of Theorem 1.2 for each fixed rotating subextremal Kerr bridge and all future timelike exits of its full development.

1.3 The new arguments

Three points are central. First, an arbitrary extension of the stated regularity supplies a positive-measure family of timelike geodesics with finite lifetime in the full development and uniform chart and velocity bounds. The construction averages the connection near a Lipschitz first-exit graph and uses preservation of Hamiltonian section volume. It uses only the intrinsic global hyperbolicity of the original spacetime. In particular, its conclusion is obtained from the actual future exit without an assumption on which part of the comparison geometry it meets.

Second, a finite wave packet is realized by exact vacuum data. Here T is a late value of the growing optical coordinate along the observing branch, and $\kappa > 0$ is the fixed decay rate of the lapse coefficient. At the observation, that coefficient satisfies $a \asymp e^{-\kappa T}$, and the packet amplitude is

$$A_T = e^{-\kappa T/4}.$$

This is larger than $\sqrt{a}$. A linear optical change of coordinates turns the packet into a comparison metric in double-null form. The key estimate makes the actual vacuum metric differ from this template by $o(A_T)$ through every required finite derivative order. To obtain it, the nonlinear equations retain the lapse factors that compensate losses in the angular elliptic estimates. Uniform estimates on each existing finite slab then permit interpolation and continuation. The resulting experiment has a curvature block—the two-by-two array of null-angular curvature components—differing from the background by more than $2A_T$ on a fixed proportion of the observation region.

Third, the extension is tested by holonomies of a measurable family of small interior loops. Position averaging converts the chart L^2 connection bound into control outside a vanishing exceptional parameter set. A stopping argument keeps each sampled loop in its selected chart. Finite interpolation in the loop size then extracts a curvature component with a bound $o(A_T)$ on most observations, contradicting the relative packet signal. The averaging and stopping mechanism can be formulated for general smooth interior geometries satisfying the stated frame, volume, and finite-jet bounds.

The category argument compares two extendible developments. First cover possible extensions by countably many tests, each prescribing finite chart bounds and a large relative measure of initial timelike geodesics in one small ball. If one test were dense in a nonempty open set of data, choose

an extendible background there and add a sufficiently late finite packet. Density and finite Cauchy stability then provide a second extendible datum whose finite observations are arbitrarily close to those of the packet experiment. Match the two developments at the same numerical terminal positions and momenta. Preservation of canonical section volume leaves a positive-measure family good for both extension witnesses. The averaged holonomy estimate makes each observed curvature block $o(A_T)$ there, while the packet makes their difference larger than A_T . This contradiction proves nowhere denseness. Every persistence assertion concerns a fixed finite observation; the argument does not require continuity of maximal geodesic lifetime.

Section 2 supplies the background geometry and branch tubes. Proposition 3.6 reduces the result to countably many extension tests. Proposition 4.1 and Theorem 5.1 construct exact finite vacuum experiments; Propositions 6.1 to 6.3 place them in the full developments and produce a curvature difference at matched observations. Proposition 7.3 supplies the averaged holonomy obstruction, and Theorem 7.4 proves the tests nowhere dense.

2 Background geometry and finite branch tubes

This section states the precise background inputs and derives the geometric properties of the finite observation tubes. The background neighborhood and branch-tube choices are made before any higher derivative order is selected.

2.1 Normalization of a fixed Kerr center

The companion interfaces use mass one and positive rotation. For $\mathfrak{a} < 0$, the fixed axial reflection $\varphi \mapsto -\varphi$, with Killing time unchanged, identifies the Kerr metric with parameter $\mathfrak{a}$ with that for $|\mathfrak{a}|$; see the Boyer–Lindquist metric in [23, Section 7.1]. It carries the bridge and its future normal to the corresponding bridge. Simultaneous pullback of the data and reference geometry preserves every covariant derivative order, as in [22, Section 2].

After the reflection, reset notation for the reflected original bridge. Let $\Psi : \Sigma_n \rightarrow \Sigma$ be the fixed identification from the dimensionless normalized bridge coordinates to this bridge. Constant spacetime scaling by M^{-2} sends the data increments to

$$\mathcal{T}(u, v) = (M^{-2}\Psi^*u, M^{-1}\Psi^*v)$$

and the center to $d_{*,n} = (M^{-2}\Psi^*h_*, M^{-1}\Psi^*K_*)$, the Kerr bridge with parameters $(1, |\mathfrak{a}|/M)$. Write $h_{*,n} = M^{-2}\Psi^*h_*$. Its connection is the pullback of ∇_* , since constant metric scaling does not change the Levi-Civita connection. A covariant tensor of rank $2 + j$ has norm scaled by M^{2+j} , so

$$\begin{aligned} |\nabla_n^j(M^{-2}\Psi^*u)|_{h_{*,n}} &= M^j(|\nabla_*^j u|_*) \circ \Psi, \\ |\nabla_n^j(M^{-1}\Psi^*v)|_{h_{*,n}} &= M^{j+1}(|\nabla_*^j v|_*) \circ \Psi. \end{aligned}$$

Let p_m^{orig} be (1.1) for (h_*, ρ) and p_m^{norm} its counterpart for $(h_{*,n}, \rho_n)$, where ρ_n is a fixed admissible normalized radial weight. The two weights are comparable under Ψ . Consequently, for each finite m there is a fixed $C_m \geq 1$ such that

$$C_m^{-1}p_m^{\text{orig}}(u, v) \leq p_m^{\text{norm}}(\mathcal{T}(u, v)) \leq C_m p_m^{\text{orig}}(u, v). \quad (2.1)$$

This is the fixed-order comparison in the scaling discussions of [23, 22]; it shows in particular that the inverse image of a normalized p_{10} ball contains an original p_{10} ball.

The fixed transformations preserve completeness, the vacuum equations, maximality, and future orientation. They also preserve Definition 1.1: smooth coordinate changes preserve the local Sobolev condition, and constant metric scaling leaves the connection and timelike exits unchanged. The induced maps of weighted constraint spaces are homeomorphisms by (2.1). Meagreness pulls back under these maps and restricts to the original open ball. We may therefore prove the theorem below for $M = 1$ and one fixed $0 < \mathfrak{a} < 1$.

2.2 The comparison development

For the normalized fixed center, take a sufficiently small compact neighborhood of physical Kerr parameters contained in the rotating subextremal range. The exact-profile construction in [23, Section 7.1] has $r_- > 0$ and positive absolute inner surface gravity there. It rescales both optical directions to one fixed rate $\kappa > 0$, for example the center's absolute inner surface gravity, and gives smooth positive relative lapse and sphere metric with smooth inverses at the inner boundary. Choose its depth $q_0 > 0$ sufficiently small for both angular gauges and for its exact-profile inequality $(|\partial_\vartheta F|^2 + \mathfrak{a}^2 \sin^2 \vartheta)_{\text{entry}} < 4\mathfrak{a}^2$, where F is the optical correction in that construction. This is possible because the fixed compact parameter range stays away from zero rotation. The deeper computing exit, exterior constants, and data threshold are chosen afterward. All these choices may depend on the fixed center; none is required to remain uniform toward zero rotation or extremality.

Choose ε sufficiently small for the exterior and interior constructions in [23, Proposition 6.4, Proposition 7.3, Proposition 7.4, Proposition 7.5, Proposition 7.6, Proposition 7.7, Proposition 8.2]. After the finite bridge and profile preparation is fixed, Proposition 6.4 derives exterior closure from (1.1)–(1.3), and Proposition 7.3 supplies the cylinder hypotheses of the interior propositions. Their finite-slab and geometric conditions are supplied by the same construction. The resulting package requires no higher-seminorm smallness and no prescribed leading coefficient or fixed-charge condition beyond the stated weighted bounds and p_{10} smallness. It gives a smooth vacuum comparison development $\mathcal{P}_d$ with the entire bridge as Cauchy surface and an open isometric embedding into M_d fixing that surface. We identify $\mathcal{P}_d$ with this image.

For clarity about dependencies, the companion uses the Kerr stationary linear estimates and zero-mode classification in [14, Theorems 5.1, 5.2, 6.6, Proposition 6.8, and Lemma 6.9]. The trapping and system estimates used in [23, Section 4, Section 5] come from [10, 15]. In that application, Hintz's Condition (2.5) controls the subprincipal margin and Proposition 3.12 supplies the pseudodifferential-inner-product reduction; the trapping hypotheses and the specialization to the asymptotically flat Kerr problem are supplied in the companion. Its finite-time nonlinear closure is a further argument there. We use the resulting background statements through the interfaces below.

The comparison development contains a lower region with a past collar, exterior computing regions on both ends, and a future double-null wedge attached at a full spacelike cylinder $\mathcal{S} \simeq \mathbb{R} \times \mathbb{S}^2$. In the gluing construction, $\mathcal{S}$ is the wedge's only relative boundary inside $\mathcal{P}_d$; see the proof of [23, Proposition 8.2]. A future timelike track that crosses $\mathcal{S}$ therefore remains in the wedge until its first runout from $\mathcal{P}_d$. In the wedge,

$$\begin{aligned} t &= (u + v)/2 \geq 0, & x &= (v - u)/2, & \mathcal{S} &= \{t = 0\}, \\ g &= -2a \, du \, dv + \gamma_{AB} (d\theta^A - b^A du)(d\theta^B - b^B du), \\ a &= q\mathcal{A}, & q &= q_0 e^{-\kappa(u+v)}, & q_0, \kappa &> 0. \end{aligned} \tag{2.2}$$

The angular fields are tensors on $\mathbb{S}^2$. A fixed finite sphere atlas is used for coordinate norms. Either branch can be arranged to have $v \rightarrow +\infty$ by interchanging the eikonals from the initial cylinder

and preparing the angular labels for the new shift-free direction. This is the label change described in [22, Proposition 4.2] and [23, Lemma 7.8]; it preserves the following estimates.

Set $y_0 = u, y_1 = v, B_0 = b, B_1 = 0$, and

$$L_i = \partial_{y_i} + B_i, \quad \mathfrak{L}_i = \partial_{y_i} + \mathcal{L}_{B_i}.$$

The second operator acts on angular tensors. The geometric fields are

$$\begin{aligned} \chi_i &= \tfrac{1}{2} \mathfrak{L}_i \gamma, & \ell_i &= L_i \log a, & \tau_i &= \text{tr}_\gamma \chi_i, \\ \xi_0 + \xi_1 &= d_\theta \log a, & \xi_0 - \xi_1 &= -a^{-1} \gamma(\partial_v b, \cdot). \end{aligned} \tag{2.3}$$

Thus $d_\theta \log a = d_\theta \log \mathcal{A}$. Angular covariant derivatives and contractions in the equations use γ unless specified otherwise. Angular Sobolev H^k norms use component derivatives in the fixed finite atlas, with fixed localizers and coordinate densities. Equivalently, one may use derivatives from one fixed smooth reference connection and a fixed reference metric, with solution-independent equivalence constants. This is the fixed-chart convention of [23, Proposition 7.4, Proposition 7.6]. In particular, the metric and inverse bounds below give uniform pointwise comparison with the reference metric.

Convention 2.1 (Fixed-order subexponential bounds). For a fixed datum and fixed finite derivative orders, a bound $Q_T \leq e^{o(T)}$ means that for every $\delta > 0$ there are C_δ, T_δ such that $Q_T \leq C_\delta e^{\delta T}$ for $T \geq T_\delta$. For a positive quantity, a two-sided bound also bounds its reciprocal. Constants can depend on the datum and the chosen orders. Uniformity for a family of changing data is asserted only when proved separately.

Proposition 2.2 (Background estimates). *For every $d \in \mathcal{U}_\varepsilon$, the comparison geometry has the following properties.*

- (i) *The metrics $\gamma^{\pm 1}$ and $\mathcal{A}^{\pm 1}$ are bounded in angular H^6 , and $\xi_i, a^{-1} \partial_v b$ are bounded in angular H^5 . There are nonnegative measurable functions $g_i : \mathbb{R} \rightarrow [0, \infty)$ with*

$$\sum_{n \in \mathbb{Z}} \sup_{[n, n+1]} g_i < \infty \tag{2.4}$$

such that

$$\|\chi_i\|_{H^4} + \|\ell_i + \kappa\|_{H^4} + \|B_i\|_{H^5} \leq C(q + g_i(y_i)). \tag{2.5}$$

The corresponding first pure longitudinal derivatives $\mathfrak{L}_i \chi_i, L_i \ell_i$, and $\partial_u b$ in direction 0, have directional summable bounds through angular H^2 , after enlarging the g_i .

- (ii) *The first ordinary longitudinal jets satisfy*

$$\begin{aligned} \|b\|_{H^5} + \sum_i \|(\chi_i, \ell_i + \kappa)\|_{H^4} &\rightarrow 0, \\ \sum_{k=0}^1 \left(\|\partial_{y_k} b\|_{H^2} + \sum_i \|\partial_{y_k} (\chi_i, \ell_i)\|_{H^2} \right) &\rightarrow 0, \\ \sum_{k=0}^1 (\|\partial_{y_k} \gamma\|_{H^4} + \|\partial_{y_k} \log a + \kappa\|_{H^4}) &\rightarrow 0 \end{aligned} \tag{2.6}$$

as $t \rightarrow \infty$ and $|u|, |v| \rightarrow \infty$. The displayed norms are bounded everywhere in the wedge.

(iii) For fixed $U \in \mathbb{R}$, $C > 0$, and each fixed ordinary coordinate derivative order, the fields

$$\gamma^{\pm 1}, \quad \log \mathcal{A}, \quad b, \quad a^{-1} \partial_v b \quad (2.7)$$

have $e^{o(T)}$ bounds on

$$u \leq U, \quad v \leq CT, \quad t \geq 0. \quad (2.8)$$

The assertion holds for either optical ordering and on restricted coordinate-path segments. Fixed derivatives of a , $\sqrt{a}$, and $\partial_v b$ retain respectively the factors a , $\sqrt{a}$, and a , with subexponential coefficient bounds.

Proof. These are [23, Proposition 7.4, Proposition 7.5, Proposition 7.6], with cylinder data supplied by Proposition 7.3 and exterior closure supplied by Proposition 6.4 there; see also [22, Proposition 4.2]. The low characteristic estimates give (2.5) and summable cell suprema. Proposition 7.5 includes the same-direction derivatives, which are needed in (2.6). After enlarging the envelopes to include these jets, replace each envelope on $[n, n+1)$ by its supremum on $[n, n+1]$. This gives a measurable step function whose sum of closed-cell suprema increases by at most a factor two. The frozen profile terms have the stated q bounds and are absorbed in the right sides. Proposition 7.6 provides (2.7) at every fixed order. The last statement follows by differentiating $a = q\mathcal{A}$ and the relative shift identity in (2.3). None of these results assumes a bound for parallel-frame curvature on an observing geodesic. $\square$

For the linear packet, one also uses the normalized vectors

$$\hat{e}_i = a^{-1/2} L_i, \quad \hat{e}_A = \partial_A, \quad Z_i = L_i, \quad Z_A = \sqrt{a} \partial_A.$$

Write V_α for the connection matrix of ∇_{Z_α} in this frame. These matrices are bounded; in the bulk $\min(t, v, -u) \rightarrow \infty$ their limits are

$$V_0 \rightarrow D_0, \quad V_1 \rightarrow -D_0, \quad V_A \rightarrow 0, \quad D_0 = \text{diag}(-\kappa/2, \kappa/2, 0, 0). \quad (2.9)$$

The components of a Riem in normalized slots and the differentiated connection arrays entering the wave operator are bounded and tend to zero there; their fixed ordinary jets cost $e^{o(T)}$. This is [22, Lemma 4.5], with its explicit slot formulas, equivalently [23, Lemma 9.3]. It is a consequence of Proposition 2.2 and the vacuum identities, with no bounded-curvature hypothesis on the track. This intermediate input is the one used in the later linear transfer.

2.3 Seeds, section measure, and first runout

A seed $w \in T\Sigma$ determines a future unit initial velocity whose normal and tangential components are

$$(\sqrt{1 + h(w, w)}, w).$$

The maximal geodesic with that velocity is always understood in the full M_d . Identify $T\Sigma$ with $T^*\Sigma$ by $w \mapsto h(w, \cdot)$, and let μ_d be the pullback of its canonical symplectic volume. In position and spatial covariant momentum coordinates this is ordinary coordinate volume. It is a smooth positive measure; on a fixed precompact seed chart it depends continuously on the initial metric.

We will repeatedly use its section version. On the future unit mass shell of the geodesic Hamiltonian, the restriction of the symplectic form has the flow direction as kernel. A single-valued transverse hitting map therefore preserves the restricted two-form and its third exterior power. Thus the measure on a time-level section is canonical position–momentum volume, even when hitting time varies with the initial state. This statement concerns smooth flow strictly inside M_d .

Proposition 2.3 (First runout and corner removal). *Every seed whose future geodesic has finite maximal proper lifetime in M_d crosses $\mathcal{S}$. Its first runout from $\mathcal{P}_d$ is either a branch, where one optical coordinate has a finite limit and the other tends to $+\infty$, or a corner, where both tend to $+\infty$. For each fixed datum, seeds whose first runout is a corner form a μ_d -null set.*

Proof. The full-lifetime classification is [23, Corollary 8.6]. Corner removal is [22, Proposition 3.10]; its section-volume argument applies to any smooth positive measure on $T\Sigma$. The geometric estimate used there is [23, Lemma 8.7]. Its hypotheses are the low interior bounds of Proposition 2.2 and a future unit timelike wedge track starting sufficiently far in both optical coordinates. It imposes no curvature bound or regular-extension assumption. These statements classify the first comparison runout; that event may precede the full geodesic endpoint in M_d . $\square$

2.4 Open branch tubes and launch feet

The next lemma records uniformity in a neighborhood of a seed, as well as the fixed-size angular footprint needed by the packet. For a unit timelike tangent $\dot{c}$, write $\pi_u = g(\dot{c}, \partial_u)$ and $P_A = g(\dot{c}, \partial_{\theta^A})$ for its covariant momenta. The remaining component π_v is determined by the future unit mass-shell equation. Thus (u, θ, π_u, P) are canonical state coordinates on a transverse v section.

A packet launch patch is a relatively compact coordinate patch in $(v - T, \theta)$ on $\mathcal{S}$. We call it admissible when the covectors dv have backward null rays crossing the entry, horizon and original initial hypersurfaces transversely, escaping into the selected exterior end, with uniformly nondegenerate geometry on the fixed-depth entry collar. The required fixed-order coefficient and flow bounds are those of [23, Proposition 9.6, Proposition 9.7]. Shrinking a patch with these properties preserves them.

Lemma 2.4 (A finite observation tube). *Fix a branch seed, with optical ordering $u \rightarrow u_* \in \mathbb{R}$, $v \rightarrow \infty$. There is a relatively compact open neighborhood $\mathcal{N}$ of this seed, as small as desired, with the following properties.*

- (i) *All seeds in $\overline{\mathcal{N}}$ have an initial compact flow tube from Σ to a fixed late v section, followed by tracks contained in a common compact u range and a sphere coordinate chart. The section hits in the earlier flow tube are transverse and single-valued. In canonical state variables (u, θ, π_u, P) , both momenta are bounded and*

$$p = -(\pi_u + b^A P_A) = a\dot{v} \tag{2.10}$$

has positive upper and lower bounds. Each track has branch runout in this ordering.

- (ii) *For a sufficiently small fixed interval J about zero and an unbounded sequence of T , there are product observation boxes $\mathcal{Q}_T$ of fixed side lengths in $(u, v - T, \theta)$ containing all hitting positions of these tracks at $v \in T + J$. The closure of a fixed enlargement stays in the common u and angular-chart margins. The angular feet on $\mathcal{S}$ of the L_0 segments from every point of that enlarged closure lie, with a fixed interior margin, in one admissible small packet launch patch. The patch size is fixed by the low-order ray and collar construction, independently of later derivative and accuracy choices.*
- (iii) *For fixed $u_2 > u_*$, $c_2 > 0$, and sufficiently small $c_0 > 0$, the neighborhood, J , and box sizes can be chosen so that all entry-to-observation track portions and enlarged observation boxes fit with strict outer margins in*

$$\begin{aligned} 0 \leq t \leq L_T &= (T + u_*)/2 + c_0, \\ u \leq u_2 - c_0 t/T, \quad v &\leq T + c_2 - c_0 t/T. \end{aligned} \tag{2.11}$$

The slice x intervals have a positive lower length, uniformly for large T . No margin from the entry surface $t = 0$ is required.

Proof. We first recall the individual branch-state argument of [22, Lemma 4.3, Lemma 7.3]. Let $\Xi = \dot{\theta} - b\dot{u}$ and $P_A = \gamma_{AB}\Xi^B$. Unit normalization gives

$$2aiv = 1 + |\Xi|_\gamma^2, \quad p = -g(\dot{c}, L_0) > 0. \quad (2.12)$$

The low connection identities imply $|\log p| \leq C(\dot{u} + |\Xi|_\gamma)$: the shape term is controlled by $|\Xi|_\gamma^2/p \leq 2\dot{u}$, and lapse and torsion coefficients are bounded. On a branch, $\int du < \infty$ and $\int a dv < \infty$. By (2.12),

$$\int |\Xi|_\gamma d\tau \leq C \left(\int du \int a dv \right)^{1/2} < \infty.$$

Hence p, p^{-1} are bounded, and the angular track has finite length because b is bounded and $\dot{\theta} = \Xi + b\dot{u}$. It has a limiting point on the sphere and eventually stays in a fixed chart.

Solving the unit mass shell for π_v gives the reduced v -time Hamiltonian

$$H_v = -\pi_v = \frac{a(1 + \gamma^{AB}P_AP_B)}{2p}, \quad \frac{du}{dv} = \frac{a(1 + \gamma^{AB}P_AP_B)}{2p^2}. \quad (2.13)$$

Its angular Hamilton equation and the low angular coefficient bounds give

$$\left| \frac{dP}{dv} \right| \leq C(p + |P|) \frac{du}{dv}.$$

Gronwall in the increasing bounded coordinate u bounds P . It follows that Ξ and $\dot{u}$ are bounded. In particular $\pi_u = -p - b^AP_A$ is bounded.

Choose a compact state neighborhood of a sufficiently late part of this track, with p bounded away from zero and with strict u and angular margins. Its radius can be fixed independently of v : the low first derivatives of b are uniformly bounded, so (2.10) has a uniform state-Lipschitz bound on a bounded momentum set. On this neighborhood, every fixed state derivative of (2.13) is bounded by $Ce^{-\kappa v}e^{o(v)}$. This follows from Proposition 2.2, since each derivative of a retains its factor and the state is in a compact set. The Hamiltonian field and its Lipschitz bound are therefore integrable in v . Choose the initial late section so the remaining integrated field bound is smaller than the state margins. Gronwall then keeps all sufficiently close states in that neighborhood for every later v . Their u coordinates converge, v is unbounded, and they have branch runout. Compact earlier smooth flow, followed by one further shrinking, transports this neighborhood to $\mathcal{N} \subset T\Sigma$ with the stated closure margins. This proves (i).

For (ii), let $\Phi_v(u, \theta)$ be the angular foot at $u = -v$ of the L_0 characteristic through (u, v, θ) . The angular variational equation gives

$$\|D_\theta \Phi_v\| + \|(D_\theta \Phi_v)^{-1}\| \leq C \exp \left(C \int_{-v}^u (q + g_0(s)) ds \right) \leq C. \quad (2.14)$$

Here $\int_{-v}^u q ds \leq q_0/\kappa$ and (2.4) controls the other integral. Differentiation in the terminal u contributes a bounded factor b . Differentiation in v contributes the moving endpoint term $b(-v, v, \cdot)$ and an integral of $\partial_v b$ multiplied by the bounded angular variational maps. Since $\partial_v b = O(a)$ and $\int_{-v}^u a ds \leq C$, these terms are uniformly bounded as well. Thus the foot map is uniformly Lipschitz in (u, θ, v) on the tubes in question.

Shrink $\mathcal{N}$ and the width of J so that the feet stay in an arbitrarily small fixed neighborhood of the central foot at $v = T$. The central feet lie on the compact sphere. An unbounded subsequence

therefore lies in one chart and one small patch with interior margins. The fixed-patch ray and collar estimates in [22, Section 4.5, Corollary 4.8], using [23, Proposition 9.6, Proposition 9.7], apply uniformly to all centers in a sufficiently small low-order patch. Their patch size is independent of later finite beam orders; higher accuracy changes those orders and constants. The integrable state-Lipschitz bound from (i) makes the hitting positions uniformly close to the central positions when $\mathcal{N}$ is small; the Hamiltonian field also controls their variation across the fixed interval J . Choose fixed product-box radii larger than these variations but small relative to the launch margin. The uniform foot-map Lipschitz bound then keeps the feet of a fixed enlargement inside that margin. These choices are made while shrinking $\mathcal{N}, J$, before any later derivative order. This supplies $\mathcal{Q}_T$ and proves (ii).

Finally choose c_0 small relative to $u_2 - u_*$ and c_2 , then shrink the state neighborhood, J , and box radii so that every point of the enlarged observation boxes has strict margins from all three upper faces in (2.11). Future causal vectors in (2.2) have $du, dv \geq 0$, and t is temporal. Hence each earlier point of its track from $\mathcal{S}$ satisfies the same strict side inequalities: both $u + c_0 t/T$ and $v + c_0 t/T$ decrease toward the past. The slice interval has length

$$T + c_2 + u_2 - 2t - 2c_0 t/T.$$

At $t = L_T$ this tends to $c_2 + u_2 - u_* - 3c_0 > 0$ after the choice of c_0 ; earlier intervals are longer. This proves (iii). This finite-domain fitting condition concerns only the track up to observation; its later portion need not remain in (2.11). $\square$

2.5 Finite smooth persistence

We also record the form of smooth Cauchy stability that will be used. Given a fixed datum, a compact subset of its smooth development, compact earlier flow tubes connecting it to the initial surface, and finitely many derivative orders, nearby smooth constraint data have corresponding neighborhoods with metrics arbitrarily close in those norms. The identifications come from local wave reductions, domain of dependence, and geometric uniqueness. In particular, strict finite observation and transversality inequalities persist. This is the compact stability statement in [23, Lemma 2.2], also used in [22, Lemma 3.3], based on [11, 3].

Each use is at a fixed finite experiment. The permitted data neighborhood can depend on its time and derivative orders. Neither persistence of a common optical gauge for nearby data nor continuity of maximal geodesic lifetime is required.

3 From an arbitrary weak exit to countably many tests

We first work in a smooth, time-oriented, globally hyperbolic four-dimensional spacetime (M, g) with smooth spacelike Cauchy surface Σ . Suppose that M is identified with its proper open image in a smooth ambient manifold $(\widetilde{M}, \widetilde{g})$, where $\widetilde{g}$ is continuous, nondegenerate, and belongs to $W_{\text{loc}}^{1,2}$. Thus its Christoffel array belongs to L_{loc}^2 in every smooth chart. Its restriction to M agrees almost everywhere with the smooth Christoffel array of g . All geodesics and Hamiltonian flows below are those of the smooth metric on the full M . The compact-diamond and first-exit-graph arguments are related to [30, Proposition 2.6 and Lemma 2.12]. We give their proofs here and then average smooth geodesics in phase space to obtain the positive-measure family required at the weaker connection regularity.

3.1 Intrinsic diamonds and a first-exit graph

Lemma 3.1 (Causal homotopies with interior endpoints). *Let $p, q \in M$ and let $H : [0, 1] \times [0, 1] \rightarrow \widetilde{M}$ be continuous. Suppose that every $H(r, \cdot)$ is a locally Lipschitz future causal path from p to q , with causal tangent almost everywhere, and that $H(0, [0, 1]) \subset M$. Then the entire image of H lies in M .*

Proof. Let $A = \{r : H(r, [0, 1]) \subset M\}$. Compactness of the parameter interval and openness of M imply that A is relatively open. For $r \in A$, the path is an intrinsic causal path in M , so its image lies in the single compact intrinsic diamond $K = J_M^+(p) \cap J_M^-(q)$. The image of K in the Hausdorff manifold $\widetilde{M}$ is compact and hence closed. If $r_n \in A$ and $r_n \rightarrow r$, continuity gives $H(r, t) \in K$ for every t . Hence A is also closed, and it is nonempty. This proves the assertion. Global hyperbolicity has been used only inside M . $\square$

Lemma 3.2 (A local first-exit graph). *Suppose that a future timelike C^1 curve exits M as in Definition 1.1. There is a precompact ambient coordinate chart with coordinates $(s, z) \in \mathbb{R} \times \mathbb{R}^3$, a constant $c > 0$, a bottom height $-d_0 < 0$, a spatial ball D , and a Lipschitz function $f : D \rightarrow \mathbb{R}$, with Lipschitz constant $L < \infty$, having the following properties:*

- (i) s is temporal, and every vector $(1, \eta)$ with $|\eta| < c$ is uniformly future timelike;
- (ii) $(-d_0, z) \in I_M^+(\Sigma)$ for all $z \in D$;
- (iii) the vertical segment $\{(s, z) : -d_0 \leq s < f(z)\}$ lies in M , while $(f(z), z) \in \partial M$;
- (iv) $f(0) = 0$ and, for some $0 < \delta < d_0/4$, $-\delta < f(z) \leq \delta$. The rectangle containing these segments has an upper face above 2δ .

The metric, its inverse, the positive spatial block (g_{ij}) , and the timelike normalization on any closed narrower cone have uniform bounds and uniform nondegeneracy on this rectangle.

Proof. First the given exiting curve has a late segment in $I_M^+(\Sigma)$. Extend it to the past by a past-inextendible timelike curve in M . Its future end has no endpoint in M , since its ambient limit lies on ∂M . The resulting inextendible timelike curve meets the Cauchy surface, and its remaining future segment lies in $I_M^+(\Sigma)$.

We can replace a late part by a straight timelike first-exit segment. To see this without any assertion about the ambient causal structure, choose a small chart about the given boundary point b and a point q slightly to its future on a fixed, uniformly timelike coordinate line. By continuity of $\tilde{g}$, for a sequence of late interior points $x_n \rightarrow b$ all straight segments $[x_n, q]$ in this chart are uniformly future timelike. If one has a first exit, use it. Otherwise every such segment, including its endpoint q , lies in M . For a fixed earlier point p on the exiting curve, all sufficiently late x_n then lie in $J_M^+(p) \cap J_M^-(q)$. Compactness of this intrinsic diamond would put their ambient limit b in M , a contradiction. The starting point of the chosen first-exit segment is in $I_M^+(\Sigma)$.

Translate its first exit to $(0, 0)$ and make its direction vertical by a linear coordinate change. Choose the other coordinate vectors in the spacelike orthogonal hyperplane of that direction at the exit. After shrinking the chart, ∂_s is timelike, ds is temporal, and a fixed cone $|\eta| < c$ has the asserted strict timelike margin. The closure of the chart is taken inside a larger smooth coordinate chart; continuity and nondegeneracy give all the claimed uniform algebraic metric bounds.

Choose $d_0 > 0$ so that the central segment $[-d_0, 0] \times \{0\}$ is the final part of the selected straight segment. Its bottom has a neighborhood in $I_M^+(\Sigma)$. Take $0 < \delta < d_0/4$. For z in a sufficiently small ball, define the first-exit height by

$$f(z) = \inf\{s \in [-d_0, 2\delta] : (s, z) \notin M\}, \quad \inf \emptyset = 2\delta.$$

Compact openness along $[-d_0, -\delta] \times \{0\}$ permits shrinking the ball so that $f(z) > -\delta$. Write its radius as R and shrink it further so that $R/\delta < c/4$. If $f(z) > \delta$, both $p_z = (-d_0, z)$ and $q_z = (\delta, z)$, and their entire vertical joining segment, are in M . The broken path

$$p_z \longrightarrow (0, 0) \longrightarrow q_z$$

has spatial slopes less than $c/2$. Interpolate its spatial graph linearly with that of the vertical segment, using s as parameter. This is a fixed-endpoint homotopy of future timelike paths in the chart. Lemma 3.1 forbids the member through $(0, 0) \notin M$. Thus $f(z) \leq \delta$; in particular the clipping at 2δ is inactive. Its definition now implies $(f(z), z) \in \partial M$, even if a vertical line reenters M at a later height.

Finally shrink R so that $2R/(d_0 - \delta) < c/2$, and fix $L > 4/c$. If $f(z') > f(z) + L|z - z'|$, select a height b' with

$$f(z) + 2|z - z'|/c < b' < f(z').$$

The path from $(-d_0, z')$ through $(f(z), z)$ to (b', z') has first-leg slope less than $2R/(d_0 - \delta) < c/2$ and second-leg slope less than $c/2$. Its endpoints and the reference vertical segment lie strictly below the first exit on the higher vertical. Linear interpolation of spatial graphs again contradicts Lemma 3.1. Interchanging z, z' proves $|f(z) - f(z')| \leq L|z - z'|$. Piecewise smooth timelike paths suffice in these homotopies; the strict slope inequalities also permit smoothing the corners. The common bottom height is what keeps the first leg uniformly inside the cone. $\square$

3.2 Averaging near the graph

Lemma 3.3 (Boxes with small normalized connection mass). *Fix $K > 1$ and a ball $D' \Subset D$ from Lemma 3.2. There are $l_n \downarrow 0$ and $z_n \in D'$ such that the boxes*

$$Q_l(z) = \{(s, y) : |y - z| < Kl, \ |s - f(z)| < Kl\}$$

lie in one fixed precompact subchart and satisfy

$$l_n^{-3} \int_{Q_{l_n}(z_n)} |\tilde{\Gamma}| \, ds \, dy \longrightarrow 0. \quad (3.1)$$

Proof. On the compact chart region, L^2 integrability implies L^1 integrability. All the boxes with $z \in D'$ and small l lie in a fixed compact region, and their union is contained in a layer

$$\mathcal{L}_l = \{(s, y) : y \in D'', \ |s - f(y)| \leq K(1 + L)l\}, \quad D' \Subset D'' \Subset D.$$

For each (s, y) , the set of possible centers z has volume at most $C_K l^3$. Tonelli's theorem therefore gives

$$\int_{D'} l^{-3} \int_{Q_l(z)} |\tilde{\Gamma}| \, ds \, dy \, dz \leq C_K \int_{\mathcal{L}_l} |\tilde{\Gamma}|.$$

The graph of f has four-dimensional Lebesgue measure zero, by Fubini. Absolute continuity of the L^1 integral makes the right side tend to zero. Choosing a center with no more than the average normalized mass proves the assertion along any sequence of sufficiently small scales. $\square$

Proposition 3.4 (A positive family supplied by every weak exit). *Under the hypotheses of Lemma 3.2, there is a measurable set $S \subset T\Sigma$ of positive smooth section measure such that every future unit geodesic with seed in S has finite maximal future proper lifetime in the full M and an ambient endpoint on ∂M . The endpoints lie in one compact subset of an ambient chart. Moreover, there are finitely many precompact ambient charts $\psi_j : U_j \rightarrow V_j \subset \mathbb{R}^4$ and constants $C < \infty$, $m > 0$ with the following properties:*

- (i) *on each entire V_j , the metric and inverse metric have component norms at most C , and $\|\tilde{\Gamma}_j\|_{L^2(V_j)} \leq C$;*
- (ii) *for every seed in S and every proper time from its intersection with Σ to its full future endpoint, excluding only the endpoint itself, at least one j satisfies*

$$\text{dist}(\psi_j(\gamma(\tau)), \mathbb{R}^4 \setminus V_j) \geq m, \quad |(\psi_j)_*\dot{\gamma}(\tau)| \leq C. \quad (3.2)$$

Here the section measure is the pullback of canonical volume on $T^*\Sigma$ under $w \mapsto h(w, \cdot)$, where h is the induced metric. In particular it has the same null sets as every smooth positive measure on $T\Sigma$.

Proof. Use the graph and chart of Lemma 3.2. Constants in this proof depend on this fixed chart, but not on the small scale l . Choose small numbers $\rho_0, \varepsilon_0 > 0$ so that

$$L\rho_0 < \frac{1}{4}, \quad 2\varepsilon_0 + C_\beta(\rho_0 + 2\varepsilon_0) < c, \quad (3.3)$$

where C_β bounds the derivative of a fixed smooth bump on $(-1, 1)$ that equals one at zero. We may also require $\rho_0 + 4\varepsilon_0 < 1$. Fix $K > 3$ in Lemma 3.3. For one of its boxes, write z_c for the center and set

$$s_c = f(z_c), \quad s_0 = s_c - l, \quad s_1 = s_c + l.$$

Every point (s_0, z) with $|z - z_c| < \rho_0 l$ is strictly below $f(z)$ and above the bottom, for small l . It is in $I_M^+(\Sigma)$, since its vertical segment from the bottom lies in M and is future timelike.

The initial phase volume. At such a point parameterize initial unit velocities by slopes $\eta \in \mathbb{R}^3$, $|\eta| < \varepsilon_0$:

$$U = N(1, \eta), \quad N = [-\tilde{g}((1, \eta), (1, \eta))]^{-1/2}, \quad \pi_i = N(\tilde{g}_{i0} + \tilde{g}_{ij}\eta^j).$$

On $|\eta| \leq 2\varepsilon_0$, N has positive uniform lower and upper bounds. Differentiation in the slope variables, which uses no derivative of the ambient metric, gives

$$\frac{\partial \pi_i}{\partial \eta^j} = N\tilde{g}_{ij} + N^3(\tilde{g}_{i0} + \tilde{g}_{ik}\eta^k)(\tilde{g}_{j0} + \tilde{g}_{jk}\eta^k). \quad (3.4)$$

The spatial block is uniformly positive definite. Thus this Jacobian is uniformly positive definite and has determinant bounded above and away from zero. Its positivity also makes $\eta \mapsto \pi$ injective on the convex slope ball: integrate the quadratic form of its derivative along the segment between two slopes. The resulting open set A_l of canonical initial states (z, π) on $s = s_0$ therefore satisfies

$$c_1 l^3 \leq |A_l| \leq C_1 l^3. \quad (3.5)$$

Every initial and pre-stop momentum lies in a fixed bounded set, since both $\tilde{g}$ and U are bounded on the closed wider slope cone.

The stopped smooth flow and its exact Jacobian. From each state $x \in A_l$ follow the maximal geodesic in the full M , using s as parameter as long as it is in this chart and $|\eta| < 2\varepsilon_0$. Stop also at $s = s_1$ or at a box face. Before these stops,

$$|z(s) - z_c| \leq (\rho_0 + 4\varepsilon_0)l < l,$$

so no spatial box face can be reached; the time interval is also strictly inside the enlarged box. For $s < s_1$ let $D_s \subset A_l$ consist of the initial states whose arcs exist through s with all the pre-stop inequalities strict. Smooth dependence in the interior of M implies that D_s and the joint pre-stop domain in $(s_0, s_1) \times A_l$ are open.

The hitting map $\Phi_s : D_s \rightarrow T^*\{s = \text{constant}\}$, in coordinates (z, π) , is smooth and injective. Indeed the geodesic equations are smooth on M , $\dot{s} > 0$, and backward uniqueness recovers the same initial state from a hit before any stop. Its canonical Jacobian is exactly one. For completeness, put $\mathcal{H} = \frac{1}{2}g^{\alpha\beta}\pi_\alpha\pi_\beta$. On the future unit shell $\mathcal{H} = -\frac{1}{2}$, $\partial\mathcal{H}/\partial\pi_0 = \dot{s} > 0$; hence $\pi_0 = -h(s, z, \pi)$ is a smooth root. Implicit differentiation of the shell equation gives the reduced Hamilton equations

$$\frac{dz^i}{ds} = \frac{\partial h}{\partial \pi_i}, \quad \frac{d\pi_i}{ds} = -\frac{\partial h}{\partial z^i}.$$

Their divergence in (z, π) is zero, so

$$\Phi_s^*(d^3z d^3\pi) = d^3z d^3\pi. \quad (3.6)$$

Equivalently, the restriction of the canonical symplectic form to the unit shell has the geodesic-flow direction as kernel; the additional derivatives of a variable hitting time disappear from its pullback. This proves the same volume statement for every transverse section used below.

Let

$$I(x) = \int_{\text{before stop}} |\Gamma_M(\gamma_x(s))| ds.$$

The joint open domain and the smooth interior flow make this a measurable nonnegative function, possibly infinite. Using Equation (3.6), Tonelli's theorem, and the fixed bound on the momentum fibers gives

$$\begin{aligned} \int_{A_l} I(x) dx &= \int_{s_0}^{s_1} \int_{\Phi_s(D_s)} |\Gamma_M(s, z)| d^3z d^3\pi ds \\ &\leq C \int_{Q_l(z_c)} |\tilde{\Gamma}| ds dz = o(l^3). \end{aligned} \quad (3.7)$$

The equality of the weak and smooth Christoffel arrays is only an almost-everywhere equality on M . This suffices: the same phase change of variables and Fubini's theorem show that its exceptional chart-null set has zero time measure on almost every initial arc. Thus Equation (3.7) does not prescribe a trace of a Sobolev function on a curve.

Small integrals prevent the slope stop. Put $V = (1, \eta)$. The smooth geodesic equation in s time is

$$\frac{d\eta^i}{ds} = -\Gamma_{\alpha\beta}^i V^\alpha V^\beta + \eta^i \Gamma_{\alpha\beta}^0 V^\alpha V^\beta. \quad (3.8)$$

Consequently $|d\eta/ds| \leq C_2 |\Gamma_M|$ before the stop. By Equations (3.5) and (3.7), for small enough l a set $G_l \subset A_l$ of measure at least $|A_l|/2$ satisfies $I(x) < \varepsilon_0/(2C_2)$. On these arcs the slope remains less than $3\varepsilon_0/2$. It cannot reach the wider threshold $2\varepsilon_0$. The position estimate already excludes a chart or box-face stop.

A surviving arc would give a forbidden bump. Suppose such an arc exists inside M through s_1 . Choose $\beta(s)$ supported in (s_0, s_1) , with $\beta(s_c) = 1$ and $|\beta'| \leq C_\beta/l$, and put

$$\Delta z = z_c - z(s_c), \quad z_r(s) = z(s) + r\beta(s)\Delta z, \quad 0 \leq r \leq 1.$$

Here $|\Delta z| \leq (\rho_0 + 2\varepsilon_0)l$. By Equation (3.3), every graph $(s, z_r(s))$ has spatial slope less than c . The graphs stay in the ambient rectangle and have the same two endpoints in M . Their first member is

the assumed interior geodesic arc, whereas the last passes through $(s_c, z_c) \in \partial M$. This contradicts Lemma 3.1.

It follows that every arc in G_l has a maximal pre-stop height $s_* \leq s_1$, with no slope or chart stop. Unit normalization gives

$$0 < c_3 \leq \frac{ds}{d\tau} = N \leq C_3. \quad (3.9)$$

The elapsed proper time to s_* is therefore finite. The bounded spatial slope gives an ambient position limit in a fixed compact subrectangle of the chart. If this limit were in M , smoothness of the metric there, the bounded unit velocity, and the geodesic equation would give a limiting velocity and smooth continuation. If $s_* < s_1$, that contradicts maximality of the pre-stop interval. If $s_* = s_1$, it gives the just-excluded surviving arc. Thus the limit lies in ∂M . Any continuation in the full M would, under the continuous embedding, take this boundary value at the joining proper time, which is impossible. The finite lifetime just proved is therefore the full future lifetime. Nothing here solves an ambient geodesic equation at the boundary.

Returning to Σ and controlling the earlier arcs. We spell out why all the starting states have a past intersection with Σ . A maximal timelike geodesic in a smooth spacetime has no interior causal endpoint. Indeed, in a relatively compact smooth temporal chart about a hypothetical endpoint, all causal slopes $\eta = dz/ds$ are bounded. The time component of the geodesic equation gives

$$\frac{d}{ds} \log \frac{ds}{d\tau} = -\Gamma_{\alpha\beta}^0(1, \eta)^\alpha(1, \eta)^\beta.$$

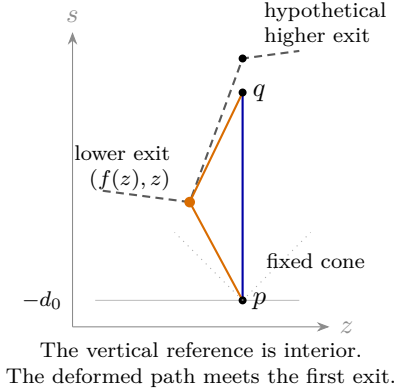
Its right side is bounded and s has a finite endpoint. Thus $ds/d\tau$ has positive upper and lower bounds. The affine duration to the endpoint is finite, the velocity is bounded, and the smooth geodesic equation gives a velocity limit and continuation. The same argument with reversed orientation handles a past endpoint. Hence maximal timelike geodesics are causally inextendible; the Cauchy property forces them to meet Σ exactly once. The starts here are in $I_M^+(\Sigma)$, so the intersection is in their past.

For each initial state in A_l , this past hit occurs in finite proper time and is transverse. Smooth dependence and the implicit function theorem give an open neighborhood on which the hit and its time are smooth. Shrink this to a precompact neighborhood on which the transverse-section hit map is a diffeomorphism onto its image and the complete earlier flow tube has compact closure in the interior unit shell. Local injectivity is sufficient: the original local time slice need not have a globally single-valued intersection with every geodesic. Second countability supplies countably many such smaller neighborhoods covering A_l . One of them meets G_l in positive measure. Restrict to that intersection.

The backward hit preserves canonical section volume by the unit-shell argument following Equation (3.6). Identifying the future unit shell over Σ with $T\Sigma$ by the velocity $\sqrt{1 + h(w, w)}n + w$, its tangential covector is exactly $h(w, \cdot)$. The image S is measurable and has positive section measure. The compact earlier flow tube is covered by finitely many ambient charts with closures in M . Taking smaller coordinate neighborhoods first gives a common positive coordinate margin. Compactness in the unit shell then bounds the velocity components in the selected charts, and bounds the earlier proper durations. Each chart can be taken precompact in a larger chart, so its metric, inverse, and Christoffel L^2 norm are finite. Add the single exit chart, where Equation (3.9) and the narrow cone supply the velocity bound and the compact subrectangle supplies the margin. Taking a maximum of the bounds and a minimum of the positive margins proves Equation (3.2) from proper time zero onward. $\square$

The two fixed-cone deformations are illustrated in Figure 1.

A common bottom fixes the cone



A small bump keeps endpoints fixed

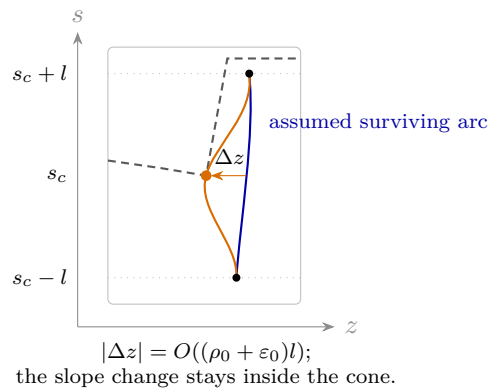


Figure 1: Two uses of the fixed timelike cone, with one spatial coordinate shown. Left: a common bottom and a point below the higher first exit would admit a path through the lower exit if the graph were too steep. Right: a sufficiently narrow geodesic surviving across a small graph box can be deformed, with its endpoints fixed, through the central exit. Both deformations contradict the intrinsic-diamond argument; the deformed paths use only the fixed timelike cone in the ambient chart.

3.3 The existential tests and their countable cover

We have obtained uniform chart witnesses along a positive-measure family of entire future geodesics. The remaining step is to localize that family in one small branch tube, where a finite packet can affect all observations. This localization must retain enough phase volume for the two data sets compared later.

Return to the near-Kerr developments. Use the single p_{10} neighborhood fixed in Proposition 2.2, and write μ_d for the section measure on $T\Sigma$ just defined. Fix a countable basis $\mathcal{B}$ of coordinate balls in $T\Sigma$, with compact closures in members of a countable atlas. The centers and radii can be taken rational in these coordinates, so the basis is fine enough for Lebesgue differentiation with any smooth positive density.

Definition 3.5 (Necessary existential extension tests). For $O \in \mathcal{B}$ and $B \in \mathbb{N}$, $B \geq 1$, let $E(O, B) \subset \mathcal{U}_\varepsilon$ consist of the data satisfying both of the following requirements.

- (i) There is an extension of M_d of the regularity and type specified in Definition 1.1, and a measurable $S \subset O$ with

$$\mu_d(S) > .99 \mu_d(O), \quad (3.10)$$

such that every seed in S has finite full future lifetime. There are at most B precompact ambient charts $\psi_j : U_j \rightarrow V_j$ witnessing Equation (3.2) along each entire such geodesic, from Σ to its full endpoint. On the whole of every V_j , each of the metric norm, inverse metric norm, and Christoffel L^2 norm is at most B . At each point a selected chart has velocity norm at most B and Euclidean distance to $\mathbb{R}^4 \setminus V_j$ at least $1/B$.

- (ii) There is an observation neighborhood $\mathcal{N}$ satisfying all three parts of Lemma 2.4 for this datum, with $\overline{O} \subset \mathcal{N}$ and the optical ordering chosen so that $v \rightarrow \infty$. One choice of its central branch track, compact state ranges, positive p bounds, transverse earlier flow tube, interval J , observation boxes, admissible launch patch, unbounded sequence of T , and constants u_2, c_2, c_0 serves every seed in $\mathcal{N}$. In particular, the entry-to-observation portions fit (2.11) with its

outer margins and positive slice-length bound. The admissible patch has the low-order ray and collar properties specified before Lemma 2.4.

All tube constants, charts, patches, and subsequences in this definition are existential witnesses for the given datum. They are not additional countable indices and are not required to work in a neighborhood of that datum.

The measurable good family in this definition may depend on the extension. For later averaging, whenever its point and velocity satisfy several chart conditions, select the least chart index. The finite collection of coordinate margin and velocity conditions is Borel on the interior unit shell, so this gives a measurable selector, also when it depends on the momentum. No closedness assertion about $E(O, B)$ is made.

Proposition 3.6 (Countable cover of all future extensions). *Every datum in $\mathcal{U}_\varepsilon$ admitting a future $C^0 \cap W_{\text{loc}}^{1,2}$ extension belongs to some $E(O, B)$. Consequently the extendible set is precisely*

$$\bigcup_{O \in \mathcal{B}} \bigcup_{B=1}^{\infty} E(O, B).$$

This cover includes extensions whose actual exit occurs after first runout from the comparison development.

Proof. Fix such a datum and extension. Proposition 3.4 provides a measurable family S_* of positive μ_d measure, finite full lifetimes, and one finite set of ambient chart witnesses. By Proposition 2.3, every seed in S_* crosses the comparison cylinder and its first comparison runout is a branch or a corner. The corner subset is null. The inputs here are precisely the finite-full-lifetime classification of [23, Corollary 8.6] and the section-volume corner removal of [22, Proposition 3.10]; the latter uses only the low-order corner estimates of [23, Lemma 8.7]. These statements have neither a regular-extension hypothesis nor a curvature bound along the selected seed.

Choose a density point w of S_* outside that null subset, with $w \in S_*$. It is a branch seed. Apply Lemma 2.4 and, if necessary, interchange the optical labels. This gives an open seed neighborhood $\mathcal{N}$ of w with all the tube, footprint, and finite-domain properties in Definition 3.5(ii). These are consequences of the background estimates and hold independently of the extension chart witnesses.

By density and the fine countable basis, choose $O \ni w$ with $\overline{O} \subset \mathcal{N}$ and $\mu_d(S_* \cap O) > .99\mu_d(O)$. To justify using countably many balls, small balls centered at a density point can be enlarged by an arbitrarily small relative amount to balls with rational center and radius. Smooth positivity of the density keeps their measures comparable, so their relative complement measure still tends to zero. Set $S = S_* \cap O$. Choose an integer B at least the number of the ambient charts, all their bounds, and the reciprocal of their common margin. This proves $d \in E(O, B)$.

Every observation invoked here lies strictly before first runout from the comparison region. The finite full lifetime and ambient chart witnesses were obtained independently for the entire geodesic in M_d . Thus first runout is used only to locate earlier observations; it need not be the ambient exit. Finally the reverse inclusion in the displayed union is part of Definition 3.5(i). $\square$

4 Preparing exact data and a double-null template

Fix a smooth datum $d \in \mathcal{U}_\varepsilon$, a branch tube together with all its witnesses from Lemma 2.4, and a finite integer $m \geq 10$. After interchanging the optical labels if necessary, its central track has $u \rightarrow u_* \in \mathbb{R}$ and $v \rightarrow \infty$. All constants in this section may depend on this fixed datum and tube.

Write $\mathcal{N}$ for its seed neighborhood. Use the witnessing constants $u_2 > u_*$, $c_2 > 0$, and $c_0 > 0$, interval J , observation boxes $\mathcal{Q}_T$, launch patch, and unbounded sequence of T throughout. The finite experiment uses

$$\mathcal{D}_T = \left\{ 0 \leq t \leq L_T := \frac{T + u_*}{2} + c_0, \quad u + \frac{c_0 t}{T} \leq u_2, \quad v + \frac{c_0 t}{T} \leq T + c_2 \right\}. \quad (4.1)$$

We construct on small enlargements and restrict inside their margins. The tube, the observation interval $v \in T + J$, and its angular footprint fit these margins along an unbounded sequence of T . The slice x -intervals have a positive lower length bound. Put

$$\alpha = \kappa/4, \quad A_T = e^{-\alpha T}, \quad \lambda = e^{\zeta T}. \quad (4.2)$$

The small constant $\zeta > 0$ is chosen only after all finite base orders and rate tolerances. In particular, A_T is substantially larger than the square root of the lapse in the observation region.

Use the observation boxes $\mathcal{Q}_T$ and their fixed enlargements from Lemma 2.4. They contain all hitting positions of the seed-neighborhood closure $\bar{\mathcal{N}}$ at $v \in T + J$. Their feet lie in one compact subpatch of the admissible launch patch, with a common interior margin along the selected sequence. Choose the launch bump below to be at least one on a neighborhood of this entire foot set.

Proposition 4.1 (Packet preparation). *For every finite $m \geq 10$, finite jet budget N , positive loss tolerance η_{out} , and $0 < \nu < 1$, one can choose $\zeta > 0$ and finite construction orders so that the following objects exist for arbitrarily large T along the indicated sequence.*

1. *There are exact smooth complete vacuum data $d_T \in \mathcal{D}$ on the whole bridge with $p_m(d_T - d) \rightarrow 0$. They agree with d on the other end, the connecting region, and an expanding protected ball on the selected end. The permitted exterior correction tail is retained.*
2. *An exact exterior evolution of d_T reaches the required part of $\mathcal{S}$, with overlapping collars and strict outflow margins. In the fixed background wave-map description its entry spacetime jets differ from $g + h^p$ by $O(A_T^2 e^{\eta_{\text{out}} T})$. Here*

$$h^p = A_T \text{ tr. rev. } g \text{ Re} \left(e^{i\lambda v} \sum_{j=0}^{J_*} \lambda^{-j} p_j \right). \quad (4.3)$$

The leading coefficient is real, trace-free, and annihilates L_0 . On the entire observation box $\mathcal{Q}_T$,

$$|(p_0)_{AB}|_\gamma \geq e^{-o(T)}. \quad (4.4)$$

At every fixed required base order the ordinary mixed coordinate jets of the normalized amplitude components are $e^{o(T)}$, and the linearized Ricci and gauge errors have arbitrarily high inverse-frequency accuracy.

3. *There is a metric $\bar{g}$ exactly of double-null form, with fields $\bar{\gamma}, \log \bar{A}, \bar{b}$ and their exact geometric derived fields. Through order N , its differences from the background fields, including $q^{-1} \partial_v b$, are $O(A_T e^{\eta_{\text{out}} T})$. Its defects in the null structure and angular equations (5.4)–(5.7) are $O(A_T^2 e^{\eta_{\text{out}} T})$. If the actual exterior induced data are written in double-null coordinates initialized by the background values on $\mathcal{S}$, their full entry jets differ from this template by the same quadratic bound.*

All entry discrepancies and all template changes and defects vanish for $v < (1 - \nu)T$. Any additional finite initial or residual jet buffers may be included in N . Positive frequency powers at these base orders are independent of the later orders used to improve linear accuracy. At the finitely many high entry orders ultimately requested by Theorem 5.1, the actual fields have $e^{o(T)}$ bounds.

For the proof, choose an internal loss $0 < \eta < \min\{\eta_{\text{out}}, \alpha/4\}$. The bootstrap estimates below are obtained with this smaller loss, after splitting it among the finitely many steps, and then imply the stated bounds with η_{out} . This distinction permits any prescribed positive output tolerance.

4.1 The homogeneous linear hierarchy and its footprint

Complex Gaussian beams and their superpositions were developed by Ralston, Tanushev, and Liu–Runborg–Tanushev [29, 32, 17]. The companion construction used here also enforces the Einstein wave-gauge constraints and supplies the fixed-order bounds on intervals growing with T . We retain the hierarchy and matching estimates needed for the larger amplitude.

We use the amplitude-independent constrained construction of [22, Proposition 4.6, Corollary 4.8], whose hypotheses are the background coefficient estimates and a branch tube; it expressly requires no bound on curvature in an observer’s parallel frame. Its proof uses the hierarchy and beam interfaces of [23, Equation (9.5), Proposition 9.6, Proposition 9.7]. We specify the part needed for a signal on a whole footprint.

Let P be the background linearized reduced vacuum operator on trace-reversed two-tensors, $C_g = \text{div}_g$, and P_1 the induced one-form wave operator. The Ricci-flat Bianchi identity is $C_g P = P_1 C_g$. For

$$k_v = g^{-1}dv = -a^{-1}L_0, \quad K_v p = p(k_v, \cdot), \quad \mathcal{T} = 2\nabla_{k_v} + \square_g v,$$

the coefficients solve

$$i\mathcal{T}p_j + Pp_{j-1} = 0, \quad iK_v p_j + C_g p_{j-1} = 0, \quad p_{-1} = 0. \quad (4.5)$$

Work on $-v \leq u \leq u_f$, where $u_f > u_2$. Multiplication of the homogeneous transport by $-a/2$ gives $\nabla_{L_0} + \frac{1}{2}\text{tr}_\gamma \chi_0$. In the normalized slots $(a^{-1/2}L_0, a^{-1/2}L_1, \partial_\theta)$ its constant diagonal weights are $\kappa(n_0 - n_1)/2$: here n_i counts the arguments of type $a^{-1/2}L_i$ in a covariant two-tensor component, so $n_0 + n_1 \leq 2$. The normalized connection formulas in [23, Lemma 9.3] make the hierarchy triangular from higher weights to lower weights. Nonnegative weights are prescribed at $u = -v$, with the components containing an L_0 argument fixed by the gauge equation; negative weights 1A, 11 are prescribed at u_f and transported backwards. The sign of each weight is favorable in its prescribed direction. Higher differentiated coefficients multiply already controlled lower jets, so each finite transport induction costs only $e^{o(T)}$. The angular characteristic and inverse characteristic jets have this same bound, including derivatives of the moving initial edge $u = -v$.

For the free leading angular data, orthonormalize a fixed smooth local coframe using the entry screen metric, obtaining e^1, e^2 . Uniform low nondegeneracy and the fixed higher coefficient bounds give $e^{o(T)}$ jets for this coframe at every fixed order. Use the real trace-free unit polarization $2^{-1/2}(e^1 \otimes e^1 - e^2 \otimes e^2)$, multiplied by the bump that is at least one on a neighborhood of the complete foot set. Take zero free angular data at higher levels. The initial polarization exists on the one small chart patch; no global choice of a nonvanishing trace-free tensor on $\mathbb{S}^2$ is required. The leading transport is real. It preserves trace and annihilation of k_v , and acts on the positive screen quotient by isometric parallel transport times a real nonzero scalar. The scalar and its reciprocal are $e^{o(T)}$, since its logarithm is an integral of the expansion coefficient. Thus the positive bump gives (4.4) at every point whose foot lies in the chosen foot set, uniformly throughout $\mathcal{Q}_T$. There is no phase adjustment at a single observation point.

Gauge compatibility follows directly by conjugating $C_g P = P_1 C_g$ with $e^{i\lambda v}$: once preceding gauge defects vanish, the next defect satisfies homogeneous one-form transport with zero initial value. Terminal prescriptions for negative weights create no additional condition, since K_v contracts precisely an L_0 argument. Smooth zero extension away from the launch band is therefore allowed.

For any fixed base order, truncating (4.5) far enough gives scaled reduced-wave and gauge residuals $A_T \lambda^{-M} e^{o(T)}$ for any prescribed M .

The exterior transfer of [22, Proposition 4.6, Corollary 4.8] supplies a real lab-coordinate tensor h^B , including the factor A_T , with

$$\|h^B\|_{C^s} \leq A_T \lambda^{s+3/2} e^{o(T)}. \quad (4.6)$$

Its wave, gauge, and entry-matching residuals can simultaneously be $A_T \lambda^{-M} e^{o(T)}$. The geometric hypotheses of that lemma hold here: the prepared optical covectors have escaping backward rays; the entry, horizon, and initial-slice crossings are transverse; the fixed-depth collar has uniform low nondegeneracy; and all fixed higher coefficient and flow jets are subexponential. These are coefficient properties of the fixed background from Proposition 2.2. The fixed small launch patch is chosen using only these low bounds. It need not shrink as approximation orders increase. For each prescribed relative annulus tolerance, a larger fixed far threshold and then large T place the initial support in that thin annulus, uniformly for all centers in the patch; this is the backward-ray estimate in [23, Proposition 9.6].

We recall why the positive exponent in (4.6) is independent of later accuracy. The constrained beams are integrated over a fixed three-dimensional center patch with normalization $(\lambda/2\pi)^{3/2}$. Each of s differentiations costs at most λ , while every finite phase and amplitude coefficient is $e^{o(T)}$ after its displayed inverse powers have been extracted. Taking absolute center integrals gives (4.6). For each beam center c , let $P_c, Q_c \in \mathbb{C}^{3 \times 3}$ be the position and momentum differential blocks of the Hamiltonian ray map acting on the initial complex phase graph. If M_0 is its symmetric initial phase Hessian with positive imaginary part, then $P_c(0) = \text{Id}$ and $Q_c(0) = M_0$. The imaginary Hessian remains positive through real caustics by the symplectic identity $P_c^* Q_c - Q_c^* P_c = 2i \text{Im } M_0$; it gives a subexponential inverse of the complex position projection. At a fixed transverse Taylor degree d , homogeneous amplitude-jet transport in ray-comoving coordinates is the central amplitude transport. Returning to fixed coordinates also applies the d -fold symmetric tensor power of $J_c(t, s)^{-t}$, where $J_c(t, s) = P_c(t) P_c(s)^{-1}$. The bounds for P_c and P_c^{-1} keep this factor subexponential for every fixed d . Thus same-degree jet couplings remain controlled when the finite accuracy order is raised. An uncanceled Taylor degree gains $\lambda^{-1/2} e^{o(T)}$ by Gaussian integration. Increasing the finite Taylor and transport orders therefore improves only inverse accuracy. The entry Gaussian expansion is triangular in amplitude level, and its new free contraction-kernel value matches the corresponding p_j . These are precisely the fixed-base assertions proved in [23, Proposition 9.7]. Linearity makes them valid at the amplitude (4.2).

4.2 Exact constraints and exterior propagation at the new amplitude

Localized deformation of the vacuum constraints was developed by Corvino–Schoen and Chruściel–Delay [6, 5]. Here the correction fixes an inner ball and retains a noncompact decaying tail. Its support and weighted bounds must remain explicit, since they determine both closeness of the complete data and the first time the perturbation can reach the interior.

In the selected exterior, t_e denotes exterior time, z lab position, and $C(t_e)$ the drift center; its co-moving radius is $r_{\text{co}} = |z - C(t_e)|$. Write $t_T \asymp T$ for the exterior time corresponding to the launch cylinder and $z_* = C(t_T)$ for its lab drift center. Choose a small relative gap $\beta > 0$ with $s_* := \sup |C'| < 1 - 2\beta$, and set

$$\rho_T = (1 - 2\beta)t_T, \quad |z - z_*| < \rho_T. \quad (4.7)$$

The thin launch annulus lies outside this protected ball with a margin. In the variables $y = (z - z_*)/\rho_T$, keep Cartesian metric components unchanged and multiply second-fundamental-form components

by ρ_T . Both constraints are then multiplied by ρ_T^2 . Extending the unperturbed rescaled datum smoothly inside the unit ball, without imposing a constraint there, gives a perturbation of flat data of size $O_s(T^{-1})$ at every fixed symbol order.

Here is the exact constraint-solver interface being used. For $s \geq 3$, let

$$\|f\|_{H_\ell^j} = \|f\|_{H^j(B_2)} + \sup_{R \geq 1} R^\ell \|f(R \cdot)\|_{H^j(\{1 < |y| < 2\})}, \quad X^s = H_1^{s+2} \oplus H_2^{s+1}, \quad Y^s = H_4^s.$$

Here H_ℓ^j is the class of locally H^j fields with finite displayed norm. In particular, the weight-one and weight-two factors include fields with saturated $|y|^{-1}$ and $|y|^{-2}$ decay at infinity; these classes are larger than the closures of compactly supported fields. [23, Proposition D.1, Lemma D.2] give a bounded operator $\mathcal{K} : Y^s \rightarrow X^s$ preserving exterior support, with $P_{\text{flat}}\mathcal{K}z = z$ for exterior-supported z , where P_{flat} is the flat constraint linearization. They also give a contraction for the nonlinear constraints when one finite background-plus-defect base norm is small. For the smooth compact beam deformation and the smooth background with all-order symbol bounds used here, the output is smooth, zero inside the protected ball, and has all-order symbol tails of weights one and two. The iteration sources may themselves have noncompact exterior-supported tails. There is no moment condition on the original defect and no fixed-charge requirement. Smoothness at higher orders requires bounded higher symbol norms, not their smallness.

Take s large enough for m and all requested comparison buffers. Let d_B be the rescaled beam seed, and let f_B be its exterior constraint defect extended by zero inside the ball. Thus $f_B = \Phi(d_B)$ outside the ball; the artificial interior need not satisfy the constraints. Linearize the constraint map at the actual rescaled vacuum background on the exterior. Gauss–Codazzi expresses this first variation as a linear combination of the reduced-wave residual and one derivative of the linear gauge defect, with smooth background coefficients. Include gauge-defect jets through $s+1$ among the fixed base orders before choosing ζ . Taylor expansion around that background then gives

$$\|f_B\|_{Y^s} \leq \text{poly}(T)e^{o(T)}(A_T\lambda^{-M} + A_T^2\lambda^{P_*}). \quad (4.8)$$

Here and below P_* depends on the fixed base orders alone. The flat operator is reserved for the correction iteration below; the linear cancellation in (4.8) uses the actual background. The rescaled seed base norm is $O_s(T^{-1}) + \text{poly}(T)A_T\lambda^{P_*}e^{o(T)}$. Choosing $\zeta P_* < \alpha/2$ makes it small. The inverse-metric and contraction balls of the cited proposition consequently apply for large T . Equivalently, solve

$$z = -f_B - N_{d_B}(\mathcal{K}z), \quad N_{d_B}(w) = \Phi(d_B + w) - \Phi(d_B) - P_{\text{flat}}w.$$

The Lipschitz constant is at most $1/2$, and the correction has the size of (4.8). The defect is smoothly zero near the inner boundary, and the support-preserving inverse and nonlinear iteration keep the correction zero on the entire inner side. Since $P_{\text{flat}}\mathcal{K}z = z$, the corrected constraint is $\Phi(d_B + \mathcal{K}z) = \Phi(d_B) - f_B$, hence zero outside the ball. On the inner side $N_{d_B}(\mathcal{K}z) = 0$ because the correction and all its jets vanish. Undoing the rescaling and restoring the original vacuum datum inside therefore gives smooth exact data on the complete bridge.

The metric stays positive and uniformly comparable to the original complete metric. Weighted differentiation after translation and rescaling costs only powers of T . For example, the seed contributes at most $\text{poly}(T)A_T\lambda^{m+5/2}e^{o(T)}$ to p_m , and the correction contributes the same polynomial type of loss times (4.8). Thus $p_m(d_T - d) \rightarrow 0$. Every higher weighted seminorm of each individual d_T is finite by the all-order symbol assertion. The exterior tail and any charge change remain part of d_T .

Initialize the exact exterior wave-map evolution with target g . Choose initial lapse and shift as for $g + h^B$; the corrected second fundamental form and exact gauge determine the remaining normal metric jets by a uniformly noncharacteristic algebraic system. Its linearization matches the constrained beam, so its discrepancy has size (4.8). Further normal jets follow from the reduced equations. Agreement on the protected region is exact.

We now justify the new amplitude in the exterior propagation. For a lab-component metric change H , the reduced equation has the form

$$\mathcal{R}_e(H)_n = (\mathcal{L}_e H)_n + ((g + H)^{-1} - g^{-1})^{\rho\sigma} \partial_\rho \partial_\sigma H_n + \mathcal{Q}_n(H, \partial H), \quad (4.9)$$

where $\mathcal{Q}$ starts quadratically and has total derivative order at most two, with fixed differentiated coefficients $e^{o(T)}$. On the uniformly nondegenerate exterior domain, $\mathcal{R}_e(h^B)$ has the size in (4.8); there is no small interior lapse in this assertion. Write $H = h^B + F$ and bootstrap sufficiently high mixed lab-coordinate error energies by $e^{-\alpha T}$. Retain the actual metric $g + h^B + F$ in the highest-order wave operator. Every other nonlinear error term has an additional packet or error factor. Put its highest error derivative in L^2 and its lower factors in L^∞ using a fixed Sobolev buffer. These factors are exponentially small, so the nonlinear energy rate is exponentially small.

The individual exterior estimate proved in [22, Section 5.4, Corollary 5.3] uses finite shrinking computing slices

$$r_{\text{co}} < R_{\text{out}}(t_e) = C_{\text{out}}(1 + T) + V_{\text{out}}(t_T + C_2 - t_e), \quad V_{\text{out}} > \sup_{\text{causal}} |dr_{\text{co}}/dt_e|. \quad (4.10)$$

Their radii are $O(T)$ on the finite comparison interval, and the side is strictly outgoing. The physical correction tail outside this computing region remains part of the complete initial datum. On these slices the estimate for square-root energies at exact orders j reads

$$E'_j \leq r_e E_j + C_j e^{o(T)} \sum_{k < j} E_k + \mathcal{F}_j, \quad \mathcal{F}_j \leq \text{poly}(T) e^{o(T)} (A_T \lambda^{-M} + A_T^2 \lambda^{P_*}). \quad (4.11)$$

For precision, its arbitrarily small integrated rate is available at this amplitude for the following reason. The actual scalar stress energy, with future slice normal and a small positive mass, has nonnegative outflow flux. Its same-order linear rate uses only first and second lab metric derivatives, which are small at large co-radius and uniformly bounded elsewhere. Higher background derivatives multiply lower exact-order energies. The first-arrival estimate, valid also for the noncompact correction since it is zero in the protected ball, is

$$(1 - 2\beta)t_T \leq (1 + \delta_R)t_e + O(R) + s_*|t_T - t_e|, \quad \delta_R \longrightarrow 0. \quad (4.12)$$

Indeed before its first hit of a fixed large co-radius R , a causal path from the changed support has lab speed at most $1 + \delta_R$; its endpoint is within $O(R)$ of $C(t_e)$. The fixed gap $1 - s_* > 0$ in (4.12) implies that the change cannot see the bounded-rate region before $t_e \geq t_T - \beta' T$, where $\beta' > 0$ can be made arbitrarily small by decreasing β and increasing R . The localized beam obeys the same conclusion. Thus

$$\int_0^{t_T + C_2} r_e dt_e \leq \eta' T$$

for any prescribed $\eta' > 0$. The bounded compact rate is independent of the chosen far threshold. Increasing that threshold later for the thin annulus therefore causes no conflict. The choices also protect the entire earlier entry interval $v < (1 - \nu)T$.

Remove the common rate from (4.11). Iteration of the finite lower-triangular hierarchy introduces only a polynomial of fixed degree in $Te^{o(T)}$, hence another $e^{o(T)}$ factor. Allocating the finitely many rate losses in advance, then choosing ζ small and M large, gives

$$\|g_T^{\text{ext}} - g - h^p\|_{C^N(\mathcal{S})} \leq A_T^2 e^{\eta T}. \quad (4.13)$$

The bound first holds with h^B on the exterior; the fixed-depth coordinate transition, spacetime trace with finite buffers, and linear entry matching replace it by h^p . It strictly improves the bootstrap since $2\alpha - \eta > \alpha$ after choosing losses small. Local reduced-wave restart inside enlarged outflow margins gives the entire finite exterior. Constraint and gauge propagation give exact vacuum. These continuation statements are the ones used in [23, Proposition 7.7, Proposition 8.2, Proposition 9.9]. This argument is the individual exterior comparison; it uses no deep wave-coordinate nonlinear estimate.

4.3 Linear optical coordinates and the relative shift

The exact evolution from the entry cylinder will use two optical functions and angles having the *same values* as the background ones on $\mathcal{S}$. Choose the neighboring eikonal roots and transport the angles so that the shift in direction 1 is zero. The entry surface consequently retains the labels $u + v = 0$. Only the linearized coordinate change is needed to construct the comparison template.

Lemma 4.2 (Linear optical gauge). *Let $Z = (z^0, z^1, z^\theta)$ be the variations of the coordinate functions, with $Z = 0$ on $\mathcal{S}$, for the input h^p . In unnormalized components $h_{ij}^p = h^p(L_i, L_j)$ and $h_{iA}^p = h^p(L_i, \partial_A)$, they obey*

$$\begin{aligned} \partial_v z^0 &= -\frac{h_{11}^p}{2a}, & L_0 z^1 &= -\frac{h_{00}^p}{2a}, \\ \partial_v z^\theta &= b\partial_v z^0 + a\gamma^{-1}d_\theta z^0 + \gamma^{-1}h_{1.}^p. \end{aligned} \quad (4.14)$$

For every fixed number of angular derivatives and r total longitudinal derivatives,

$$|\partial_{u,v}^r \partial_\theta^s Z| \leq A_T \lambda^{r-1} e^{o(T)}. \quad (4.15)$$

The fixed-new-label metric variation is $k = h^p - \mathcal{L}_Z g$. Its field variations k_γ, k_H, k_b , where $H = \log \mathcal{A}$, and the relative derivative $q^{-1}\partial_v k_b$ satisfy

$$\|(k_\gamma, k_H, k_b, q^{-1}\partial_v k_b)\|_{C^s} \leq A_T \lambda^{P_s} e^{o(T)} \quad (4.16)$$

at every fixed required order. The exponents P_s depend only on that order, not on subsequent inverse-accuracy choices.

Proof. The inverse metric is

$$g^{-1} = -a^{-1}(L_0 \otimes L_1 + L_1 \otimes L_0) + \gamma^{AB} \partial_A \otimes \partial_B.$$

Linearization of $g^{-1}(du, du) = 0$ gives $-a^{-2}h_{11}^p - 2a^{-1}\partial_v z^0 = 0$; the equation for v is analogous. For the angle condition $g^{-1}(du, d\theta^A) = 0$, the metric variation contributes $a^{-1}\gamma^{AB}h_{1B}^p - a^{-2}b^A h_{11}^p$. The coordinate variations contribute $\gamma^{AB}\partial_B z^0 - a^{-1}b^A \partial_v z^0 - a^{-1}\partial_v z^A$. Substitution of the first eikonal equation gives the last equation in (4.14), including its sign and factor.

To see the gain in (4.15), a typical term in the first equation has the form $A_T e^{i\lambda v} f(u, v, \theta)$ with fixed coefficient jets $e^{o(T)}$. At fixed u, θ ,

$$\begin{aligned} \int_{-u}^v e^{i\lambda s} f(u, s, \theta) ds &= \frac{e^{i\lambda v} f(u, v, \theta) - e^{-i\lambda u} f(u, -u, \theta)}{i\lambda} \\ &\quad - \frac{1}{i\lambda} \int_{-u}^v e^{i\lambda s} \partial_s f(u, s, \theta) ds. \end{aligned} \quad (4.17)$$

The interval has length $O(T)$; its polynomial length is absorbed in $e^{o(T)}$. Differentiation in u differentiates the lower endpoint and its coefficient as well as the integral. Each longitudinal derivative costs at most one additional λ , including the phase $e^{-i\lambda u}$ at that endpoint. Angular derivatives do not differentiate the phase. This proves the assertion for z^0 with arbitrary fixed differentiated orders. The relative coefficient h_{11}^p/a is a normalized tensor entry, so no inverse lapse was lost.

For z^1 , the leading 00 coefficient is identically zero: p_0 annihilates L_0 , and trace reversal leaves the same-null 00 component unchanged because $g_{00} = 0$. Thus h_{00}^p/a already has a factor λ^{-1} . Integration along L_0 keeps v fixed, with angular characteristic jets and inverse jets $e^{o(T)}$. Differentiating its moving initial point $u = -v$ has the same endpoint rule as above. Its interval length is again $O(T)$. This proves the bound for z^1 without requiring oscillation in the direction of that integration.

For the angular equation, h_{1A}^p is $\sqrt{a}$ times a normalized oscillatory coefficient, so (4.17) applies. The term $a\gamma^{-1}d_\theta z^0$ already has the inverse-frequency gain. Integrating its length costs only $e^{o(T)}$, also at differentiated orders. Finally,

$$\int_{-u}^v b \partial_v z^0 dv = bz^0|_{-u}^v - \int_{-u}^v (\partial_v b) z^0 dv,$$

where the lower value of z^0 is zero. All derivatives of $\partial_v b$ retain their factor a , and the remaining coefficients have subexponential fixed jets. This proves (4.15) for the angles as well.

For explicit field formulas set $W = z^\theta - z^0 b$, an angular vector field. A direct Lie derivative gives

$$\begin{aligned} k_\gamma &= h_{AB}^p - 2z^0 \chi_0 - 2z^1 \chi_1 - \mathcal{L}_W \gamma, \\ k_H &= -a^{-1} h_{01}^p - Z \log a - L_0 z^0 - \partial_v z^1, \\ k_b &= -\gamma^{-1} h_{0\cdot}^p + \partial_u W + [b, W] - z^1 \partial_v b - a\gamma^{-1} d_\theta z^1. \end{aligned} \quad (4.18)$$

Indeed $(\mathcal{L}_Z g)_{AB} = 2z^0 \chi_0 + 2z^1 \chi_1 + \mathcal{L}_W \gamma$, whereas $(\mathcal{L}_Z g)(L_0, L_1) = -Za - a(L_0 z^0 + \partial_v z^1)$. The 0A component is $\gamma_{AB}(\partial_u W + [b, W] - z^1 \partial_v b)^B - a\partial_A z^1$. The linear double-null component conditions $k_{11} = k_{00} = k_{1A} = 0$ follow from (4.14). Hence $k_H = -k(L_0, L_1)/a$ and $k_b = -\gamma^{-1} k_0$, which give (4.18). All displayed divisions by a are relative factors: h_{01}^p/a is normalized and $\partial a/a = \partial \log a$ has the given coefficient bounds. Thus these three field variations have the claimed fixed positive frequency powers. This argument alone does not estimate $q^{-1} \partial_v k_b$; that relative derivative is obtained next.

Let X_i denote the exact linear variation of ξ_i under k . The torsion sum identity gives $X_0 + X_1 = d_\theta k_H$. The linearized vacuum identities are available with arbitrary linear accuracy: diffeomorphism covariance on the vacuum background gives

$$D \text{Ric}_g(\mathcal{L}_Z g) = \mathcal{L}_Z \text{Ric}(g) = 0.$$

The algebraic and differential steps turning Ricci components into the null equations can lose fixed powers of q^{-1} , but only on the *linear residual*. On (4.1) these powers are $e^{C_s T}$ for a fixed C_s . After

ζ is fixed, increasing M in $A_T \lambda^{-M}$ makes this residual smaller than any prescribed exponential at all finitely many required orders.

Use specifically the shift-free direction-1 torsion equation of [23, Equation (7.12)]. Its linearization is

$$(\partial_v + \tau_1)X_0 = \delta\beta_1 + \delta\tau_1 \xi_1 + \tau_1 d_\theta k_H + \mathfrak{r}_{\text{lin}}. \quad (4.19)$$

Here

$$\delta\chi_1 = \tfrac{1}{2}\partial_v k_\gamma, \quad \delta\tau_1 = \text{tr}_\gamma \delta\chi_1 - \gamma^{AC} k_{\gamma,CD} \gamma^{DB} \chi_{1,AB},$$

and

$$\delta\beta_1 = \text{div}_\gamma \delta\chi_1 - d_\theta \delta\tau_1 + (\delta \text{div}_\gamma) \chi_1.$$

The operator variation in the last term uses only k_γ and one angular derivative of k_γ . Thus the entire source in (4.19) has already been estimated using (4.18); it uses neither X_i nor an unknown relative shift derivative. Also

$$\int_{-u}^v |\tau_1(u, s, \theta)| ds \leq C \int_{-u}^v (q + g_1(s)) ds \leq C.$$

At the moving entry edge $q = q_0$, the initial X_0 is a finite-jet expression with no small denominator. It has size $A_T \lambda^{P_s} e^{o(T)}$. The integrating-factor estimate therefore gives this bound for X_0 , and then for X_1 . Commutation with u and angular derivatives leaves the same top transport coefficient; differentiated coefficients multiply lower already estimated X_0 jets. Entry terms from differentiating $v = -u$ are controlled by the differentiated entry data, with normal jets also given by the equation. Derivatives in v are recovered from (4.19). This proves the finite-order assertion without a shifted transport containing an unknown k_b .

Finally linearize the defining shift relation. It gives exactly

$$\begin{aligned} q^{-1} \partial_v k_b &= \mathcal{A} \gamma^{-1} (X_1 - X_0) \\ &\quad + (k_H \text{Id} - \gamma^{-1} k_\gamma) q^{-1} \partial_v b. \end{aligned} \quad (4.20)$$

The background relative derivative and all its fixed jets are subexponential. This proves (4.16). In particular the proof never divides an ordinary error estimate for $\partial_v k_b$ by a small q . $\square$

4.4 Exact template definitions and quadratic defects

Define the template metric by the following exact assignments:

$$\begin{aligned} \bar{\gamma} &= \gamma + k_\gamma, & \bar{\mathcal{A}} &= \mathcal{A} e^{k_H}, & \bar{a} &= q \bar{\mathcal{A}}, & \bar{b} &= b + k_b, \\ \bar{g} &= -2\bar{a} du dv + \bar{\gamma}_{AB} (d\theta^A - \bar{b}^A du) (d\theta^B - \bar{b}^B du). \end{aligned} \quad (4.21)$$

For sufficiently large T , $\bar{\gamma}$ is positive definite and $\bar{\mathcal{A}}$ is positive with the background low nondegeneracy bounds. All barred derived fields are their actual geometric definitions, rather than independent truncated series:

$$\begin{aligned} \bar{\chi}_i &= \tfrac{1}{2}(\partial_{y_i} + \mathcal{L}_{\bar{B}_i}) \bar{\gamma}, & \bar{\ell}_i &= (\partial_{y_i} + \bar{B}_i) \log \bar{a}, \\ \bar{\tau}_i &= \text{tr}_{\bar{\gamma}} \bar{\chi}_i, & \bar{\beta}_i &= \text{div}_{\bar{\gamma}} \bar{\chi}_i - d_\theta \bar{\tau}_i, \\ \bar{\xi}_0 &= \tfrac{1}{2} (d_\theta \log \bar{\mathcal{A}} - \bar{\mathcal{A}}^{-1} \bar{\gamma} q^{-1} \partial_v \bar{b}), & \bar{\xi}_1 &= \tfrac{1}{2} (d_\theta \log \bar{\mathcal{A}} + \bar{\mathcal{A}}^{-1} \bar{\gamma} q^{-1} \partial_v \bar{b}), \\ \bar{c}_i &= \text{curl}_{\bar{\gamma}} \bar{\xi}_j, & \bar{Q}_i &= (K_{\bar{\gamma}}, \bar{c}_i), & \bar{m}_i &= K_{\bar{\gamma}} - \text{div}_{\bar{\gamma}} \bar{\xi}_j, \quad j \neq i. \end{aligned} \quad (4.22)$$

Here $\bar{B}_0 = \bar{b}$ and $\bar{B}_1 = 0$. In particular the torsion definitions use the relative shift derivative already controlled by Lemma 4.2.

Consider the null identities (5.4) and the angular system (5.7) as finite differential expressions in

$$\gamma, \gamma^{-1}, \mathcal{A}, \mathcal{A}^{-1}, b, H = \log \mathcal{A}, V = q^{-1} \partial_v b.$$

Their only radial lapse factors are nonnegative powers of q . For example, the cross-shape sources are $q\mathcal{A}\nabla\xi$, $q\mathcal{A}K_\gamma\gamma$, $q\mathcal{A}\xi^2$, and products of the two shapes; the lapse cross equation has $q\mathcal{A}K_\gamma$, $q\mathcal{A}\xi^2$, and shape products. The angular pair has principal coefficient $q\mathcal{A}/2$; its other terms are $q\mathcal{A}(K_\gamma\xi + \xi\nabla\xi + \xi^3)$ and $\chi_i\nabla\chi_j + \chi_j\nabla\chi_i$. The remaining angular equations use $\chi_i(K_\gamma, \nabla\xi)$ and $\nabla\chi_i\xi$. The shift identity is $\partial_v b = q\mathcal{A}\gamma^{-1}(\xi_1 - \xi_0)$. The trace equation can be written without a lapse derivative as $(L_i + \kappa)(\tau_i/\mathcal{A}) = -\mathcal{A}^{-1}|\chi_i|^2$. Angular curvature and div-curl operations involve only the sphere metric and its derivatives. Finally, ordinary differentiation of q preserves its factor. This list accounts for all the factor types in these equations and their finite differentiated versions.

It follows that substituting (4.21)–(4.22) into these expressions is a smooth finite-jet operation on a uniform low nondegeneracy set. Its background term vanishes and its linear term is the arbitrarily accurate linearized vacuum identity proved above. Its remaining terms contain at least two linear increments, so at each required order the defect is

$$O\left(A_T\lambda^{-M}e^{C_NT+o(T)} + A_T^2\lambda^{P_N}e^{o(T)}\right). \quad (4.23)$$

This separation is essential: fixed inverse- q losses have been allowed on the first, linear term only. There is no such loss on the quadratic term, by the displayed relative-field formulas. Choose ζP_N below the allocated loss tolerance, and then choose M large. Formula (4.23) becomes the quadratic bound in Proposition 4.1. The same Taylor argument bounds all template-background field differences by $A_T e^{\eta T}$.

On the entry cylinder $q = q_0$ and all eikonal roots and angular transports are uniformly noncharacteristic at low order. Their initial spacetime jets are smooth finite-jet functions of the exterior metric jets and of the prescribed coordinate values. Their linearization is (4.14), and changing metric components to the new labels has linearization $h^p - \mathcal{L}_Z g$. Thus (4.13), with a finite buffer and initially smaller loss, gives actual-minus-template entry jets of size $A_T^2 e^{\eta T}$. Nonlinear coordinate terms are quadratic finite-jet terms at this fixed depth. They do not require a comparison of the two gauges deep in the wedge. The new coordinates retain the same entry labels. The first-arrival property, causal monotonicity in double-null coordinates, and uniqueness of the optical transports show that exact changes and template changes vanish in $v < (1 - \nu)T$ wherever the local exact double-null evolution exists.

The finite choice order can now be stated without a circular increase of frequency losses. First choose the target observation and bootstrap orders. The interpolation and continuation argument in Theorem 5.1 requests finitely many higher entry and residual orders; include them and m in the base budget N . Next choose all energy loss tolerances and the support tolerance ν , including the exterior first-arrival choices. Choose $\zeta > 0$ small for every positive frequency power at these base orders. Only then enlarge the hierarchy, beam, phase-Taylor, and matching accuracy orders to absorb every linear inverse- q loss. At the fixed high entry orders, the packet changes are then exponentially small, and the background jets are $e^{o(T)}$; so the actual high entry jets are $e^{o(T)}$ as asserted. Finally take T large. All choices are finite, and each d_T is smooth at all remaining orders by the cited constraint and vacuum regularity results. This proves Proposition 4.1.

5 The finite nonlinear comparison

We now evolve the actual entry data supplied by Proposition 4.1. All comparisons in this Section use the same numerical optical and angular labels. A bar denotes the exact double-null template, a prime denotes the vacuum solution to be constructed, and an undecorated field belongs to the fixed background. In particular, the barred metric is generally not vacuum. Its equations have the controlled defects provided by Proposition 4.1. For each decoration the metric is

$$-2q\mathcal{A}du dv + \gamma_{AB}(d\theta^A - b^A du)(d\theta^B - b^B du).$$

The function q and the constants q_0, κ are common to all three descriptions; it is the positive relative factor $\mathcal{A}$ that changes.

Fix a finite desired output order s and a finite initial-neighborhood order m . Enlarging s if necessary, we assume $s \geq 30$. The domain from Proposition 4.1 is

$$\begin{aligned} \Omega_T = \{ (t, x, \theta) : 0 \leq t \leq L_T = (T + u_*)/2 + c_0, \\ u = t - x \leq u_2 - c_0 t/T, \quad v = t + x \leq T + c_2 - c_0 t/T, \quad \theta \in \mathbb{S}^2 \}. \end{aligned} \quad (5.1)$$

The fixed constants and their small enlargements are chosen as in Lemma 2.4, so the observation tube has an interior margin and every spatial interval $I_T(t)$ has length at least a fixed positive number. Its length is at most $C(T+1)$. We use these enlargements when taking traces and restarting at an endpoint. Put

$$q = q_0 e^{-\kappa(u+v)}, \quad \sigma = \sqrt{q}, \quad \alpha = \kappa/4, \quad A_T = e^{-\alpha T}.$$

Neither the constants below nor the orders chosen below require a further reduction of the fixed neighborhood $\mathcal{U}_\varepsilon$.

Theorem 5.1 (Finite double-null experiment). *For every fixed background and branch tube in Lemma 2.4, every finite m, s , and every sufficiently small fixed $c > 0$, the parameters in Proposition 4.1 can be chosen so that its exact entry data have a smooth vacuum development on Equation (5.1) in the double-null gauge initialized there. The relative lapse and the sphere metric remain positive and uniformly nondegenerate. For*

$$\mathcal{U} = (\gamma, \log \mathcal{A}, b, \chi_0, \chi_1, \ell_0, \ell_1, \xi_0, \xi_1)$$

one has, in a fixed sphere atlas and ordinary coordinate derivatives,

$$\max_{|I| \leq s} \|\partial^I(\mathcal{U}' - \bar{\mathcal{U}})\|_{L^\infty(\Omega_T)} \leq e^{-(1+c)\alpha T} \quad (5.2)$$

for all sufficiently large T in the selected experiment sequence. The same conclusion holds for $q^{-1}\partial_v b' - q^{-1}\partial_v \bar{b}$ at any preassigned finite order, after increasing the finite order budget. The individual ordinary jets of the primed relative fields through every order used in the experiment are $e^{o(T)}$, uniformly for this moving family. The exact solution and the template equal the background where $v < (1 - \nu)T$. The exact solution agrees on an entry collar, by local geometric uniqueness, with the actual exterior solution of Proposition 4.1.

Three estimates connect the prepared entry data to this conclusion. Under a low-order bootstrap, the individual exact fields first satisfy high-order bounds with arbitrarily small exponential loss. We then prove a quadratic L^2 bound for the actual-minus-template difference, keeping the lapse factors that compensate the angular elliptic estimates. Interpolation between these two bounds yields the required finite-jet error and improves the bootstrap. The estimates are proved on every existing partial slab, so a local restart completes the existence argument. The quadratic bound is Proposition 5.3; the next subsection begins with the equations needed for both estimates.

5.1 Equations, bootstrap, and finite-order data

For any of the three metrics define, with $i \neq j$,

$$\begin{aligned} a &= q\mathcal{A}, \quad B_0 = b, \quad B_1 = 0, \quad L_i = \partial_{y_i} + B_i, \quad \mathfrak{L}_i = \partial_{y_i} + \mathcal{L}_{B_i}, \quad (y_0, y_1) = (u, v), \\ \chi_i &= \tfrac{1}{2}\mathfrak{L}_i\gamma, \quad \tau_i = \text{tr}_\gamma \chi_i, \quad \ell_i = L_i \log a, \quad \xi_0 + \xi_1 = \text{d}_\theta \log \mathcal{A}, \quad \xi_0 - \xi_1 = -a^{-1}\gamma\partial_v b, \\ \beta_i &= \text{div}_\gamma \chi_i - \text{d}_\theta \tau_i, \quad c_i = \text{curl}_\gamma \xi_j, \quad Q_i = (K_\gamma, c_i), \quad m_i = K_\gamma - \text{div}_\gamma \xi_j. \end{aligned} \quad (5.3)$$

Here K_γ is Gaussian curvature, and $\text{Sym } T = (T + T^\dagger)/2$. Products of shape tensors contract their adjacent indices using γ^{-1} . The vacuum double-null equations of [23, Equation (7.12), Equation (7.14)] read

$$\begin{aligned} L_i \tau_i &= \ell_i \tau_i - |\chi_i|_\gamma^2, \\ \mathfrak{L}_j \chi_i &= a \text{Sym } \nabla \xi_j - \tfrac{a}{2} K_\gamma \gamma + a \xi_j^{\otimes 2} - \tfrac{1}{2}(\tau_i \chi_j + \tau_j \chi_i) + \chi_i \chi_j + \chi_j \chi_i, \\ \mathfrak{L}_i \xi_j &= \beta_i + \tau_i \xi_i, \\ L_j \ell_i &= a K_\gamma - \chi_i \cdot \chi_j + \tau_i \tau_j + a(2\xi_i \cdot \xi_j - |\xi_i|^2). \end{aligned} \quad (5.4)$$

The exact definition identities include

$$\partial_v \gamma = 2\chi_1, \quad \partial_v \log \mathcal{A} = \ell_1 + \kappa, \quad \partial_v b = q\mathcal{A}\gamma^{-1}(\xi_1 - \xi_0). \quad (5.5)$$

Combining Raychaudhuri with $L_i \log \mathcal{A} = \ell_i + \kappa$ gives the particularly useful identity

$$(L_i + \kappa)w_i = -\mathcal{A}^{-1}|\chi_i|^2, \quad w_i = \tau_i/\mathcal{A}. \quad (5.6)$$

Thus the constant damping survives division by the relative lapse.

For clarity, the angular system is also recorded. The symbol $*$ means a finite sum of smooth contractions with $\gamma^{\pm 1}$, with precisely the differentiated factors displayed; it conceals no additional derivative. With $Jdf = (-\partial_2 f, \partial_1 f)$ in oriented orthonormal components, one has

$$\begin{aligned} \mathfrak{L}_j \beta_i &= \tfrac{a}{2}(\text{d}K_\gamma + J\text{d}c_i) + a(K_\gamma * \xi + \xi * \nabla \xi + \xi^3) + \chi_i * \nabla \chi_j + \chi_j * \nabla \chi_i, \\ L_i Q_i &= (\text{div}, \text{curl})\beta_i + \mathcal{R}_i, & L_i m_i &= \mathcal{T}_i, \\ \mathcal{R}_i, \mathcal{T}_i &: \quad \chi_i * (K_\gamma, \nabla \xi) + \nabla \chi_i * \xi. \end{aligned} \quad (5.7)$$

The factor list can be checked without differentiating a spacetime curvature component: apply divergence minus trace differential to the shape cross equation, and use

$$\begin{aligned} L_i K_\gamma &= \text{div } \beta_i - \tau_i K_\gamma, \\ L_i \text{div } \zeta &= \text{div}(\mathfrak{L}_i \zeta) - 2\chi_i^{AB} \nabla_A \zeta_B - (2 \text{div } \chi_i - \text{d}\tau_i) \cdot \zeta, \\ L_i \text{curl } \xi_j &= \text{curl } \beta_i + \text{curl}(\tau_i \xi_i) - \tau_i \text{curl } \xi_j. \end{aligned}$$

In particular, the divergence of β_i cancels from $L_i m_i$. More explicitly,

$$L_i m_i = -\tau_i K_\gamma - \text{div}(\tau_i \xi_i) + 2\chi_i^{AB} \nabla_A (\xi_j)_B + (2 \text{div } \chi_i - \text{d}\tau_i) \cdot \xi_j,$$

which verifies all the displayed factor types of its source. These are the identities used to derive the angular system in [23, Equation (7.14)].

We work first on an arbitrary existing partial slab. Bootstrap the ordinary derivatives of $\mathcal{U}' - \mathcal{U}$ through order ten by

$$\|\mathcal{U}' - \mathcal{U}\|_{C^{10}} \leq e_T, \quad e_T = e^{-\alpha T/2}, \quad (5.8)$$

and keep the relative lapse and sphere metric in a fixed open nondegeneracy range about the background bounds. One may enlarge the fixed bootstrap order when needed for a local existence convention. Proposition 2.2 and Equation (5.8) imply uniform angular H^6 bounds for $\gamma'^{\pm 1}, \mathcal{A}'^{\pm 1}$ and H^5 bounds for the torsions. All required low angular norms of $\chi'_i, \ell'_i + \kappa, B'_i$ and the low pure longitudinal coefficient norms are majorized by

$$h_{i,T} = C_0(q + g_i(y_i) + e_T). \quad (5.9)$$

The functions $g_i \geq 0$ are fixed background envelopes with summable unit-cell suprema, enlarged to include the pure-jet controls of [23, Proposition 7.5]. The barred fields obey the same bounds after the frequency growth exponent ζ is chosen sufficiently small at the finitely many base orders. In the rest of the Section h_i denotes $h_{i,T}$.

Given any finite order N and positive loss η , the preparation provides the initial discrepancy and all required defects of Equations (5.4), (5.6) and (5.7) bounded by

$$F_T = A_T^2 e^{\eta T} \quad (5.10)$$

through that order. The geometric definitions and Equation (5.5) hold exactly for the template. Its finite jets differ from the background by $A_T \lambda^{P_N} e^{o(T)}$, including the relative shift derivative. The positive frequency exponent P_N is fixed by the base orders, independently of subsequently requested inverse-frequency accuracy. The initial high norms needed below are consequently bounded by $e^{o(T)}$ uniformly in T once $\zeta P_N < \alpha/2$. Only actual entry data, rather than the template, initialize the vacuum Cauchy problem.

5.2 Uniform high estimates for the moving experiments

Lemma 5.2 (Uniform finite-slab high bounds). *Assume Equation (5.8) on an existing initial partial slab $\Omega_T \cap \{0 \leq t \leq t_*\}$ of Equation (5.1), with the same outer margins. For every fixed finite order n there is a finite initial order $r(n)$ with the following property. Uniform $e^{o(T)}$ initial bounds through order $r(n)$ imply uniform $e^{o(T)}$ bounds through order n for all ordinary jets of*

$$\gamma'^{\pm 1}, \quad \log \mathcal{A}', \quad b', \quad \chi'_i, \quad \ell'_i, \quad \xi'_i, \quad q^{-1} \partial_v b'.$$

The constants depend only on the fixed orders, a prescribed exponential-rate tolerance, the bounded low coefficient set, the uniform envelope bounds described below, and finitely many initial norms. They do not depend on an uncontrolled high norm of the unknown solution or on the subslab endpoint.

Proof. We give the finite-subslab argument, including the dependence needed here. We reproduce the finite-slab argument of [23, Proposition 7.6, proof], with uniformly controlled envelopes. Merely applying the fixed-datum conclusion of that Proposition would not establish the present assertion.

Choose smooth cutoffs $0 \leq \psi_{i,T} \leq 1$ equal to one on the coordinate projection of Ω_T , supported in $[-C(T+1), C(T+1)]$. Extend only the scalar envelope, not the spacetime fields, and set

$$G_{i,T}(y) = g_i(y) + e_T \psi_{i,T}(y), \quad h_i = C_0(q + G_{i,T}). \quad (5.11)$$

The added L^1 mass and sum of unit-cell suprema are $O((T+1)e_T)$, uniformly bounded and tending to zero. Their tails are uniformly tight: if the added support reaches $|y| > R$, then $T+1 \geq R/C$, and hence

$$\sup_T \int_{|y| > R} e_T \psi_{i,T}(y) dy \leq C \sup_{T+1 \geq R/C} (T+1) e^{-\alpha T/2} \longrightarrow 0.$$

The same argument controls the tail supremum. Together with the fixed summable g_i , this supplies all tail thresholds uniformly. The Lipschitz potential

$$\Phi_T(u, v) = C_0 \sum_{i=0}^1 \int_{-\infty}^{y_i} G_{i,T}(r) dr + \frac{C_0 q_0}{\kappa} (1 - e^{-\kappa(u+v)}) \quad (5.12)$$

satisfies $0 \leq \Phi_T \leq C$ and $L'_i \Phi_T = \partial_i \Phi_T = h_i$ almost everywhere. In particular both directional integrals of h_i are uniformly bounded on every past transport segment in the domain.

The weighted angular curvature pair will supply stable fluxes, while trace damping controls the shape trace. We will then recover unweighted metric and torsion fields by transport, retaining the small lapse in the curvature energy throughout.

Suppress primes in this proof only. Fix a smooth reference metric on $\mathbb{S}^2$ and let D be its Levi-Civita connection, independent of u, v, T and of the solution. In this proof all angular Sobolev sizes use the tensors $D^r f$; equivalently, they use component derivatives in the fixed finite atlas of the background. The low coefficient bounds are used in this same reference convention, together with the uniform positive upper and lower metric bounds in that atlas. This is the convention for the bounded low coefficient set below. The symbol ∇ in the equations continues to mean the connection of γ .

For definiteness, put

$$|f|_k^2 = \sum_{r=0}^k \int_{\mathbb{S}^2} |D^r f|_\gamma^2 d\mu_\gamma.$$

The contraction includes every angular tensor slot; finite scalar arrays have their Euclidean contraction. These norms are uniformly equivalent to the fixed-reference Sobolev norms at each fixed order. In particular $D\gamma$ need not vanish, so $|\gamma|_k$ records the coordinate coefficient jets. At a fixed high angular order $M \geq 5$, use the actual-field sizes

$$\begin{aligned} Y_i &= |\beta_i|_M, & Z_i &= \sigma|(Q_i, m_i)|_M, & R_i &= |\chi_i|_{M+1}, & z_i &= |w_i|_{M+1}, & d_i &= |B_i|_{M+1}, \\ d_1 &= 0, & D &= d_0, & P_\gamma &= 1 + |\gamma|_{M+1}, & P_i &= |\xi_j|_M, & P &= P_\gamma + P_0 + P_1. \end{aligned} \quad (5.13)$$

Inverse metrics and relative lapse factors through order $M+1$ are bounded by $C_M P$: differentiate inversion tamely and use $d \log \mathcal{A} = \xi_0 + \xi_1$. The undifferentiated mean of $\log \mathcal{A}$ is already bounded by the low bounds for $\mathcal{A}^{\pm 1}$. The angular elliptic estimates are

$$R_i \leq C_M (Y_i + z_i + h_i P), \quad |\xi_j|_{M+1} \leq C_M (\sigma^{-1} Z_i + P). \quad (5.14)$$

To make the constant's dependence explicit, freeze the principal coefficient in sufficiently small fixed charts, identify trace-free tensors by smooth metric-dependent bases, and use the first-order elliptic estimate there. The differentiated coefficient error is bounded by $C_M(|f|_M + |\gamma|_{M+1} \|f\|_{C^1})$. Interpolation absorbs the penultimate derivative of f with a constant depending on M and the low Lipschitz/ellipticity bounds. Every high metric coefficient thus remains a variable in P ; it does not enter C_M . In the shape estimate the low factor multiplying that variable is h_i , including conversion of $\tau_i = \mathcal{A} w_i$. Equivalently, the trace-free tensor $S_i = \chi_i - \frac{1}{2} \tau_i \gamma$ satisfies $\operatorname{div} S_i = \beta_i + \frac{1}{2} d\tau_i$, and the one-form data are $\operatorname{div} \xi_j = K_\gamma - m_i$ and $\operatorname{curl} \xi_j = c_i$.

Commute Equation (5.7) with D^r , $0 \leq r \leq M$. Transport the resulting angular tensor $D^r f$ by $\mathfrak{L}_i(D^r f) = \partial_{y_i} D^r f + \mathcal{L}_{B_i} D^r f$ on all of its slots. In particular, for a scalar f the commutator is $D^r L_i f - \mathfrak{L}_i(D^r f)$, because its r derivative slots are now covariant tensor slots. In the commuted principal operators, ∇ acts on every slot. If I denotes the r derivative slots, the new gradient slot and the original one-form slot are paired by

$$(\nabla F)_{IA} = \nabla_A F_I, \quad (\operatorname{div} \mathcal{B})_I = \gamma^{AB} \nabla_A \mathcal{B}_{IB}, \quad (\operatorname{curl} \mathcal{B})_I = \epsilon_\gamma^{AB} \nabla_A \mathcal{B}_{IB},$$

where ϵ_γ is the oriented area bivector. Use the same contraction of the I slots in both members of the pair. Their densities are

$$\begin{aligned} e_{Y_i} &= \frac{1}{2} \sum_{r=0}^M |D^r \beta_i|_\gamma^2 && \text{in direction } j, \\ e_{Z_i} &= \frac{a}{4} \sum_{r=0}^M (|D^r Q_i|_\gamma^2 + |D^r m_i|_\gamma^2) && \text{in direction } i. \end{aligned} \tag{5.15}$$

The scalar to one-form weight ratio is exactly $a/2$, and the m_i current has no angular principal derivative. At each order, integration by parts on the full sphere cancels the two principal terms except for

$$-\frac{1}{2} \int_{\mathbb{S}^2} \langle D^r \beta_i, (\mathrm{d}a) D^r K_\gamma + J(\mathrm{d}a) D^r c_i \rangle_\gamma \mathrm{d}\mu_\gamma.$$

This uses $\nabla J = 0$ and the same derivative-slot contraction on both sides. Since $\mathrm{d}a = a(\xi_0 + \xi_1)$, the sum of these errors is bounded by $C_M \sigma Y_i Z_i$. The sphere integrals of the densities in Equation (5.15) are uniformly equivalent to Y_i^2, Z_i^2 , because $a = q\mathcal{A}$ and $\mathcal{A}^{\pm 1}$ are bounded.

Here are representative commutator bounds that specify the high coefficient dependence. The difference between the two angular connections is

$$(\nabla - D)^C_{AB} = \frac{1}{2} \gamma^{CD} (D_A \gamma_{BD} + D_B \gamma_{AD} - D_D \gamma_{AB}), \quad |\nabla - D|_M \leq C_M \mathbf{P}_\gamma.$$

Tame products, the low bounds for $\mathrm{d}Q_i$, and $|\beta_i|_{C^1} \leq Ch_i$ give, for $r \leq M$,

$$\begin{aligned} |D^r \{a(\mathrm{d}K_\gamma + J\mathrm{d}c_i)\} - a\{\nabla D^r K_\gamma + J\nabla D^r c_i\}|_0 &\leq C_M(\sigma Z_i + q\mathbf{P}), \\ |D^r \{(\mathrm{div}, \mathrm{curl})\beta_i\} - (\mathrm{div}, \mathrm{curl})D^r \beta_i|_0 &\leq C_M(Y_i + h_i \mathbf{P}), \\ |[D^r, \mathfrak{L}_j]\beta_i|_0 &\leq C_M(h_j Y_i + h_i \mathbf{d}_j). \end{aligned} \tag{5.16}$$

In the first line, every high derivative of a or J multiplies a low derivative of Q_i and costs $q\mathbf{P}$; the other terms cost $q|Q_i|_M \leq \sigma Z_i$. In the second line, a high derivative of γ^{-1} or $\nabla - D$ multiplies low $\mathrm{d}\beta_i$ or β_i , giving $h_i \mathbf{P}$; its scalar-current weight supplies one further σ . The fixed reference curvature contributes only lower derivatives with these same factors.

For the last line, the identity $\mathcal{L}_B f = B * Df + DB * f$ for a covariant tensor shows that the commutator has low DB_j against the top β_i jet, or a derivative of B_j through order $M+1$ against a low β_i jet. Reference-curvature terms have the additional low factor B_j . Likewise the scalar Q_i, m_i commutators cost $C_M\{\sigma \mathbf{d}_i + h_i Z_i\}$ after weighting. The order- $M+1$ trace commutator costs $C_M h_i(\mathbf{z}_i + \mathbf{d}_i)$, whereas the order- M torsion commutator costs $C_M(h_i \mathbf{P}_i + \mathbf{d}_i)$. The time-independent choice of D also gives $[\partial_v, D^r]b = 0$ in the shift equation.

The remaining differentiated products have the same tame placement. A high angular connection in $\chi_i * \nabla \chi_j$ costs $h_i h_j \mathbf{P}$; in $a\xi * \nabla \xi$ it costs $q\mathbf{P}$. The estimate for a differentiated torsion in an a term is

$$q(|\xi_0|_{M+1} + |\xi_1|_{M+1}) \leq C_M\{\sigma(Z_0 + Z_1) + q\mathbf{P}\}$$

by Equation (5.14). These observations and Leibniz placement give, apart from favorable damping, paired squared-current costs at most C_M times

$$\begin{aligned} Y_i[\sigma(Z_0 + Z_1 + \mathbf{P}) + h_j(Y_i + \mathbf{R}_i) + h_i(\mathbf{R}_j + \mathbf{d}_j) + h_i h_j \mathbf{P}], \\ Z_i[\sigma(Y_i + \mathbf{R}_i + \mathbf{d}_i + \mathbf{P}) + h_i(Z_0 + Z_1)]. \end{aligned} \tag{5.17}$$

Their transport directions are j and i , respectively. The remaining currents, in directions i , 1 , 1 , and i , have costs

$$\begin{aligned} & C_M \mathbf{z}_i h_i (\mathbf{z}_i + \mathbf{R}_i + \mathbf{d}_i + h_i \mathbf{P}), \\ & C_M \mathbf{D} [\sigma (\mathbf{Z}_0 + \mathbf{Z}_1 + \sigma \mathbf{P}) + h_1 \mathbf{D}], \\ & C_M \mathbf{P}_\gamma \mathbf{R}_1 + C_M h_1 \mathbf{P}_\gamma^2, \quad C_M \mathbf{P}_i (\mathbf{Y}_i + \mathbf{R}_i + \mathbf{d}_i + h_i \mathbf{P}). \end{aligned} \tag{5.18}$$

For the first line of Equation (5.18), the high relative lapse factor multiplies the two low shapes and therefore has coefficient h_i^2 . The shift source is $q\mathcal{A}\gamma^{-1}(\xi_1 - \xi_0)$, which gives the second line. The last two lines are the metric and torsion transports. Their unweighted sources explain why $\mathbf{R}_1$ and $\mathbf{Y}_i$ remain there. Area and norm differentiation of D^r tensors uses only $\chi_i = O(h_i)$ on their contracted slots; the scalar weight uses $L_i \log a = -\kappa + O(h_i)$. Thus the constant part is favorable damping and the remaining terms are among the displayed costs. These placements account for every same-order interaction.

For use on a fixed finite slab, boundedness of h_i already suffices: substitute Equation (5.14) and multiply all currents by e^{-2Kt} , with $K > C_M$ for the bounded low coefficient set. Both directions gain damping K , giving finite flux bounds. Thus the estimates are a priori estimates on existing subslabs; their derivation does not require continuation or exhaustion.

These fixed-slab bounds ensure finiteness, but interpolation later needs more: high derivatives must cost only an arbitrarily small exponential rate in T . We now obtain that rate by rescaling the one-way limiting couplings and assigning stronger damping to the bounded optical direction.

For the sharper rate fix $0 < \delta \leq 1$. The upper-bounded direction is $d = 0$, and the long direction is $l = 1$. When $\sigma = h_0 = h_1 = 0$, the only nonzero arrows in Equations (5.17) and (5.18), after substitution of Equation (5.14), are

$$\mathbf{Y}_i, \mathbf{z}_i \longrightarrow \mathbf{P}_i, \quad \mathbf{D} \longrightarrow \mathbf{P}_0, \quad \mathbf{Y}_1, \mathbf{z}_1 \longrightarrow \mathbf{P}_\gamma.$$

There is also an asymmetric regime. Give the along-0 variables $\mathbf{Y}_1, \mathbf{Z}_0, \mathbf{z}_0, \mathbf{P}_0$ arbitrarily large damping and temporarily remove every product containing one of them. At $\sigma = h_1 = 0$, with h_0 bounded, the remaining arrows are exactly

$$\mathbf{z}_1 \longrightarrow \mathbf{Y}_0, \quad \mathbf{z}_1 \longrightarrow \mathbf{P}_1, \quad \mathbf{z}_1 \longrightarrow \mathbf{P}_\gamma.$$

For example, the $\mathbf{P}_1$ source contains $\mathbf{Y}_1$, which has been removed, rather than $\mathbf{Y}_0$; and $\mathbf{d}_1 = 0$. The retained trace and shift equations have no unsuppressed source returning to $\mathbf{z}_1$.

Give each paired current multiplier one, the trace currents multiplier Λ^2 , and the $\mathbf{P}$ currents multiplier ϵ_P^2 . Take Λ large and ϵ_P small, depending only on M, δ and the bounded low coefficients. Both displayed limiting quadratic forms then have norm at most $\delta/4$ in the rescaled energy. After those choices, continuity on the bounded low coefficient set gives a fixed threshold ϵ_* : the whole form has cost at most $\delta/2$ when $\sigma, h_0, h_1 \leq \epsilon_*$, and the retained long-direction form has that cost when $\sigma, h_1 \leq \epsilon_*$.

Use the common Lipschitz weight

$$\mathfrak{w}_T = \exp\{-2\delta(u + v) - 2K_0(u + B_{\text{cut}})_+ - 2K_0 \min(t, S_0)\}. \tag{5.19}$$

Uniform tail smallness from Equation (5.11) permits B_{cut}, S_0 independent of large T such that the all-small regime holds for $t > S_0, u < -B_{\text{cut}}$ and the asymmetric regime holds for $t > S_0, -B_{\text{cut}} \leq u \leq u_2$. Its directional damping is, almost everywhere,

$$-L_k \log \mathfrak{w}_T = 2\delta + 2K_0 \mathbf{1}_{\{k=0, u > -B_{\text{cut}}\}} + K_0 \mathbf{1}_{\{t < S_0\}}.$$

If C_{bd} bounds the complete rescaled form, the products containing an along-0 factor are absorbed by $K_0 > C(C_{\text{bd}} + C_{\text{bd}}^2/\delta)$. In $t < S_0$ both directions have this damping. The choice order is

$$(\Lambda, \epsilon_P), \quad \varepsilon_*, \quad (B_{\text{cut}}, S_0), \quad K_0.$$

In particular choosing K_0 does not change a previously selected tail threshold. All constants use only the dependencies stated in the Lemma.

Integration gives positive null and slice fluxes bounded by the initial fluxes. The tilted sides of Equation (5.1) have the same outgoing signs. Since $u \leq u_2$ and $v \leq C(T + 1)$,

$$1 \geq \mathfrak{w}_T \geq \exp\{-C\delta T - 2K_0(u_2 + B_{\text{cut}})_+ - 2K_0S_0 - C\delta\}.$$

Removing the weight loses an arbitrarily small exponential rate. Initial integration over intervals of length $O(T)$ costs only a polynomial. In particular the transverse fluxes satisfy

$$\sup_{y_j} \|\mathbf{Y}_i + \mathbf{d}_i\|_{L^2(dy_i)} + \sup_{y_i} \|\mathbf{Z}_i\|_{L^2(dy_j)} \leq e^{o(T)}. \quad (5.20)$$

The intervals here are the available intersections with the initial partial slab. Every past coordinate segment stays inside it and reaches $t = 0$, because the side defining functions are nondecreasing along both L_0, L_1 .

The weighted angular fluxes are now controlled. Unweighted fields and ordinary longitudinal jets remain to be recovered. Recover the former by transport, rather than dividing the $\mathbf{Z}$ estimate by σ . The damped trace equation and Equation (5.14) give

$$\begin{aligned} \mathbf{z}_i(y) &\leq e^{o(T)} e^{-\kappa(y_i + y_j)} \\ &\quad + C'_M \int_{-y_j}^{y_i} e^{-\kappa(y_i - r)} h_i(\mathbf{Y}_i + \mathbf{d}_i + h_i \mathbf{P})(r, y_j) dr. \end{aligned} \quad (5.21)$$

The homogeneous $C_M h_i$ factor has bounded directional integral. To see explicitly how the trace enters unweighted transport, fix y_j and integrate Equation (5.21) in its terminal variable. Reversing the integration triangle gives

$$\int_{-y_j}^{y_i} \mathbf{z}_i(s, y_j) ds \leq e^{o(T)} + \frac{C'_M}{\kappa} \int_{-y_j}^{y_i} h_i(\mathbf{Y}_i + \mathbf{d}_i + h_i \mathbf{P})(r, y_j) dr.$$

The first transverse flux in Equation (5.20) bounds the $\mathbf{Y}_i + \mathbf{d}_i$ integral by Cauchy–Schwarz, with only a polynomial interval factor. Since h_i is uniformly bounded, $h_i^2 \leq Ch_i$. The Hodge estimate for $\mathbf{R}_i$ consequently gives

$$\int_{-y_j}^{y_i} (\mathbf{z}_i + \mathbf{R}_i)(r, y_j) dr \leq e^{o(T)} + C_M \int_{-y_j}^{y_i} h_i(r, y_j) \mathbf{P}(r, y_j) dr. \quad (5.22)$$

The norm inequalities for metric and torsion transport are

$$\begin{aligned} \mathbf{P}_\gamma(y) &\leq e^{o(T)} + C_M \int_{-u}^v (\mathbf{R}_1 + h_1 \mathbf{P}_\gamma)(u, r) dr, \\ \mathbf{P}_i(y) &\leq e^{o(T)} + C_M \int_{-y_j}^{y_i} (\mathbf{Y}_i + \mathbf{R}_i + \mathbf{d}_i + h_i \mathbf{P})(r, y_j) dr. \end{aligned}$$

Substitution of Equation (5.22) and the transverse flux therefore gives

$$\mathbf{P}(y) \leq e^{o(T)} + C''_M \sum_i \int_{-y_j}^{y_i} h_i(r, y_j) \mathbf{P}(r, y_j) dr. \quad (5.23)$$

Multiply by $e^{-C\Phi_T(y)}$ and take a supremum. Along either path,

$$\int_{-y_j}^{y_i} h_i(r, y_j) e^{-C(\Phi_T(y) - \Phi_T(r, y_j))} dr \leq C^{-1}.$$

Choose $C > 4C_M''$ and absorb the two integrals. The uniform bound on Φ_T proves $\mathbf{P}, \mathbf{z}_i \leq e^{o(T)}$. Equation (5.22) now bounds the pure-direction integral of $\mathbf{R}_i$ by $e^{o(T)}$ as well.

We record the angular order obtained by the last transport step. The pointwise $\mathbf{P}$ bound controls γ in H^{M+1} and ξ_i in H^M . Tame differentiation of angular curvature and connection therefore gives

$$|a \operatorname{Sym} \nabla \xi_j|_{M-1} + |a K_\gamma \gamma|_{M-1} + |a \xi_j^{\otimes 2}|_{M-1} \leq C_M q \mathbf{P}.$$

Commuting the cross equations with D through order $M-1$ gives top homogeneous coefficient $C_M h_j$ on the χ_i, ℓ_i norms. The other shape and shift terms are bounded by $C_M h_i(\mathbf{R}_j + \mathbf{d}_j) + C_M h_i h_j \mathbf{P}$. Their integrals in direction j are $e^{o(T)}$ by Equations (5.20) and (5.22). Gronwall thus gives pointwise H^{M-1} bounds for χ_i, ℓ_i . Finally $[\partial_v, D^r]b = 0$ and the shift source has H^M norm at most $C_M q \mathbf{P}$; its norm derivative adds only $C_M h_1 |b|_M$. Integration gives a pointwise H^M bound for b . Thus the order- M current gives pointwise H^{M-1} bounds for all relative fields, and running it with $M \geq A+1$ supplies an angular target order A . The shift step uses the pointwise torsion bound: the transverse integration direction of the $\mathbf{Z}_1$ flux in Equation (5.20) is 0.

We next specify the longitudinal induction and its finite angular buffer. Fix a largest longitudinal order $n_\parallel$ and a desired final angular order $A_{\text{out}} \geq 5$, and put

$$A_k = A_{\text{out}} + 2(n_\parallel - k), \quad 0 \leq k \leq n_\parallel.$$

At level k the assertion is an $e^{o(T)}$ bound in reference H^{A_k} for every ordered transport jet of the relative fields listed in the Lemma with exactly k longitudinal derivatives: use L_i on scalars and $\mathfrak{L}_i$ on angular tensors. For b we also keep its ordinary longitudinal jets at that order. All earlier levels retain their larger H^{A_ℓ} bounds. Run the preceding angular argument with $M \geq A_0 + 1$ to establish level zero. The first pure jets in the bootstrap supply low coefficient bounds; the high-angular induction begins at $k = 1$ from this level-zero conclusion.

For $i \neq j$, the exact identities needed for the induction are

$$\begin{aligned} [\mathfrak{L}_j, \mathfrak{L}_i] &= \mathcal{L}_{X_{ij}}, & X_{ij} &= a(\xi_j - \xi_i)^\sharp, \\ \mathfrak{L}_i \gamma &= 2\chi_i, & L_i \log \mathcal{A} &= \ell_i + \kappa, & \mathfrak{L}_i(\xi_0 + \xi_1) &= d\ell_i. \end{aligned} \tag{5.24}$$

For $k \geq 1$, Raychaudhuri, torsion transport, and angular curvature variation give the reductions

$$\begin{aligned} L_i^k \tau_i &= L_i^{k-1}(\ell_i \tau_i - |\chi_i|^2), \\ \mathfrak{L}_i^k \xi_j &= \mathfrak{L}_i^{k-1}(\beta_i + \tau_i \xi_i), \\ \mathfrak{L}_i^k \xi_i &= dL_i^{k-1} \ell_i - \mathfrak{L}_i^k \xi_j, \\ L_i^k K_\gamma &= L_i^{k-1}(\operatorname{div} \beta_i - \tau_i K_\gamma), \\ [\mathfrak{L}_j, \mathfrak{L}_i^k] &= \sum_{r=0}^{k-1} \mathfrak{L}_i^r \mathcal{L}_{X_{ij}} \mathfrak{L}_i^{k-1-r}. \end{aligned} \tag{5.25}$$

The scalar commutator has the same formula with X_{ij} acting by directional differentiation. Opposite shape and lapse derivatives use their cross equations. The variation of the angular Levi-Civita connection under $\mathfrak{L}_i \gamma = 2\chi_i$ is

$$(\mathfrak{L}_i \nabla)^C_{AB} = \gamma^{CD}(\nabla_A \chi_{i,BD} + \nabla_B \chi_{i,AD} - \nabla_D \chi_{i,AB}),$$

and every derivative of a retains its a factor times relative fields. Thus a reduction to a preceding longitudinal level adds at most two angular derivatives: the largest costs are $\text{div } \beta_i$ and the outer ∇ on the torsion reduction. The buffer A_k covers these costs.

Set $U_{i,k} = \mathfrak{L}_i^k \chi_i$ and $V_{i,k} = L_i^k \ell_i$. Commuting the cross equations and using Equation (5.25) gives

$$\begin{aligned}\mathfrak{L}_j U_{i,k} &= -\frac{1}{2} \tau_j U_{i,k} + U_{i,k} \chi_j + \chi_j U_{i,k} + \mathcal{F}_{i,k}, \\ L_j V_{i,k} &= -U_{i,k} \cdot \chi_j + \mathcal{G}_{i,k}.\end{aligned}\tag{5.26}$$

Here $\mathcal{F}_{i,k}, \mathcal{G}_{i,k}$ contain only longitudinal levels below k and their buffered angular derivatives. In particular the first reduction in Equation (5.25) replaces the differentiated term $-\frac{1}{2}(L_i^k \tau_i) \chi_j$ by lower levels. This is why that term is absent from the displayed top system. The new lapse jet does not occur on its right side.

For completeness, commute the first equation of Equation (5.26) with D in increasing angular order. For $r \geq 4$ the tame product and shift commutator inequalities give

$$\begin{aligned}\frac{d}{dy_j} |U_{i,k}|_r &\leq C_r h_j |U_{i,k}|_r + C_r \mathcal{C}_{j,r} \sum_{\ell < r} |U_{i,k}|_\ell + |\mathcal{F}_{i,k}|_r, \\ \mathcal{C}_{j,r} &= 1 + |\gamma|_{r+1} + |\gamma^{-1}|_{r+1} + |\chi_j|_r + |B_j|_{r+1}.\end{aligned}$$

This norm inequality is understood in its regularized form at zero. At the first four angular orders the low H^4 shape and H^5 shift bounds give the same top coefficient directly. At higher orders $\mathcal{C}_{j,r}$ multiplies already controlled lower angular orders of $U_{i,k}$; it is a source in this subsidiary induction. It is $e^{o(T)}$ by level zero. The lower longitudinal induction bounds $\mathcal{F}_{i,k}$ using $A_\ell \geq A_k + 2(k - \ell)$. Consequently Gronwall exponentiates only the uniformly bounded directional integral of $C_r h_j$. Finite products of earlier $e^{o(T)}$ bounds and integration over length $O(T)$ remain $e^{o(T)}$. After $U_{i,k}$ is controlled, the second equation has the same transport commutators and the known source $U_{i,k} \cdot \chi_j$, so it controls $V_{i,k}$ in the same way.

The reductions also cover mixed words. In an ordered word of length k acting on a torsion, apply its innermost transport derivative first: the torsion equation applies to the opposite torsion, and the torsion-sum identity gives the derivative of the same-index torsion. The remaining $k - 1$ operators then act on β_i, ℓ_i and products of fields. Expanding their angular commutators uses the same two-derivative buffer. The metric and relative lapse words reduce in the same way by Equation (5.24). For a mixed shape or lapse word, commute a cross derivative inward; the commutator in Equation (5.25) has shorter words, and the cross equation reduces the resulting innermost derivative. This gives the already controlled pure or lower levels. For the shift, differentiate its exact direction-1 equation k times in u and integrate to obtain $\partial_u^k b$. Conversion of its source from transport to ordinary derivatives uses only longitudinal orders below k of b : expanding $\partial_u = \mathfrak{L}_0 - \mathcal{L}_b$ on tensors, or $\partial_u = L_0 - b \cdot d$ on scalars, differentiates b at most $k - 1$ times. Each replacement adds at most one angular derivative, which is covered by the same buffer. Mixed shift jets follow directly from its equation. Finally $q^{-1} \partial_v b = \mathcal{A} \gamma^{-1} (\xi_1 - \xi_0)$ is algebraic. This completes level k and the conversion to ordinary jets.

The required initial pure jets are finitely many entry jets. They can also be expressed using tangential x jets and the cross equations on the noncharacteristic cylinder, where $q = q_0 > 0$. For an output order n , take $n_\parallel = n$ and $A_{\text{out}} = \max(n, 5)$, then choose $M \geq A_0 + 1$ as above. Include the derivatives defining K, β and the initial cross identities and take their finite maximum as $r(n)$. All sources in the argument are polynomial in the finitely many earlier high norms; only low directional integrals enter an exponential. This gives the asserted uniform dependence and leaves $r(n)$ independent of later frequency-accuracy choices. The Lemma follows. $\square$

5.3 The unweighted low error system

Write $\Delta f = f' - \bar{f}$. The following sizes on a sphere use angular norms in uniformly equivalent low geometry; for the paired energies we use the primed metric exactly:

$$\begin{aligned} Y_i &= |\Delta\beta_i|_0, & Z_i &= \sigma|(\Delta Q_i, \Delta m_i)|_0, & z_i &= |\Delta w_i|_1, \\ N_i &= |\Delta\chi_i|_0 + |\Delta\ell_i|_0, & D_b &= |\Delta b|_1, & R_i &= |\Delta\chi_i|_1, \\ P &= |\Delta\gamma|_1 + |\Delta \log \mathcal{A}|_0 + |\Delta\xi_0|_0 + |\Delta\xi_1|_0. \end{aligned} \quad (5.27)$$

A squared sum below means the sum of the component squares, with each component assigned its indicated transport direction. This convention matters for the components of P . Set

$$V = P + D_b + \sum_i (N_i + Y_i + Z_i + z_i), \quad H_s = \sigma + h_0 + h_1.$$

Proposition 5.3 (Quadratic low error). *For every $\eta_2 > 0$, the input loss η and the support fraction ν may be chosen sufficiently small, with the finite base orders fixed, so that on every existing bootstrap subslab*

$$\sup_{0 \leq t \leq L_T} \|(\Delta\gamma, \Delta \log \mathcal{A}, \Delta b, \Delta\chi_i, \Delta\ell_i, \Delta\xi_i)\|_{L^2(I_T(t) \times \mathbb{S}^2)} \leq A_T^2 e^{\eta_2 T}. \quad (5.28)$$

The supremum is restricted to available slices. At the same times,

$$\sum_i \|Y_i + Z_i + z_i\|_{L^2(I_T(t))} + \|(\nabla\Delta\gamma, \nabla\Delta b)\|_{L^2(I_T(t) \times \mathbb{S}^2)} \leq A_T^2 e^{\eta_2 T}. \quad (5.29)$$

Proof. We give the operator differences first, to check derivative counts before estimating the currents. Put $h_{AB} = \Delta\gamma_{AB}$ in the next two displays only. The difference of angular connections is exactly

$$C^C{}_{AB} := (\Gamma')^C{}_{AB} - (\bar{\Gamma})^C{}_{AB} = \frac{1}{2}(\gamma'^{-1})^{CD}(\bar{\nabla}_A h_{BD} + \bar{\nabla}_B h_{AD} - \bar{\nabla}_D h_{AB}). \quad (5.30)$$

Thus $|C|_0 \leq C|\Delta\gamma|_1$. For a barred one-form f ,

$$\begin{aligned} (\operatorname{div}' - \overline{\operatorname{div}})f &= ((\gamma'^{-1})^{AB} - (\bar{\gamma}^{-1})^{AB})\bar{\nabla}_A f_B - (\gamma'^{-1})^{AB} C^D{}_{AB} f_D, \\ (\operatorname{curl}' - \overline{\operatorname{curl}})f &= \frac{1}{2}((\epsilon')^{AB} - \bar{\epsilon}^{AB})(df)_{AB}. \end{aligned} \quad (5.31)$$

The analogous divergence formula on two-tensors has one connection difference for each covariant slot. No second derivative of h occurs. Against $\bar{\beta}_i$ or $\bar{\chi}_i$, these differences cost $Ch_i P$, since their required low angular derivatives are $O(h_i)$; against bounded barred torsion they cost CP .

The exact torsion-sum identity gives

$$|\Delta \log \mathcal{A}|_1 \leq CP, \quad |\Delta \mathcal{A}|_1 \leq CP. \quad (5.32)$$

For the trace conversion, use the exact identity

$$\Delta\tau_i = \mathcal{A}'\Delta w_i + (\Delta\mathcal{A})\bar{w}_i, \quad |\Delta\tau_i|_1 \leq C(z_i + h_i P). \quad (5.33)$$

The trace of $\Delta\chi_i$ in the primed metric is $\Delta\tau_i - (\gamma'^{-1} - \bar{\gamma}^{-1})\bar{\chi}_i$. Writing $C_{\text{inv}} = \gamma'^{-1} - \bar{\gamma}^{-1}$, the full shape Hodge datum is exactly

$$\operatorname{div}' \Delta\chi_i - \operatorname{d} \operatorname{tr}' \Delta\chi_i = \Delta\beta_i - (\operatorname{div}' - \overline{\operatorname{div}})\bar{\chi}_i + \operatorname{d}(C_{\text{inv}} : \bar{\chi}_i).$$

Apply the primed trace-free divergence estimate, including the undifferentiated tensor norm to cover its kernel, and then the primed one-form div-curl estimate. Equation (5.31) and Equation (5.33) yield

$$R_i \leq C(Y_i + N_i + z_i + h_i P), \quad |\Delta \xi_j|_1 \leq C(\sigma^{-1} Z_i + P). \quad (5.34)$$

For the latter, the divergence data are $\Delta K_\gamma - \Delta m_i$ and the curl datum is Δc_i , plus the displayed coefficient differences on $\bar{\xi}_j$. Precisely,

$$\operatorname{div}' \Delta \xi_j = \Delta K_\gamma - \Delta m_i - (\operatorname{div}' - \overline{\operatorname{div}}) \bar{\xi}_j, \quad \operatorname{curl}' \Delta \xi_j = \Delta c_i - (\operatorname{curl}' - \overline{\operatorname{curl}}) \bar{\xi}_j.$$

These first-order elliptic constants need only the uniform low Lipschitz geometry. The estimate does not assert an unweighted bound for the torsion derivative by Z_i .

When subtracting a transport equation, keep the actual operator:

$$\begin{aligned} \mathfrak{L}'_i f' - \bar{\mathfrak{L}}_i \bar{f} &= \mathfrak{L}'_i \Delta f + \mathcal{L}_{\Delta B_i} \bar{f}, \\ L'_i f' - \bar{L}_i \bar{f} &= L'_i \Delta f + \Delta B_i \cdot d\bar{f}. \end{aligned} \quad (5.35)$$

For a one-form, the last Lie term consists of $\Delta B_i \cdot \partial \bar{f} + (\partial \Delta B_i) * \bar{f}$. It costs $Ch_i D_b$ on $\bar{\beta}_i, \bar{\chi}_i, CD_b$ on bounded barred torsion, and CD_b on $\bar{Q}_i, \bar{m}_i$ before their σ weight. On $\bar{\ell}_i = -\kappa + O(h_i)$ it costs $Ch_i D_b$, since the constant $-\kappa$ has zero angular derivative.

The paired currents. Writing $k = \Delta K_\gamma$ and $c = \Delta c_i$, the primed principal part of the difference system is

$$\mathfrak{L}'_j \Delta \beta_i = \frac{a'}{2} (dk + J' dc) + F_i, \quad L'_i \Delta Q_i = (\operatorname{div}', \operatorname{curl}') \Delta \beta_i + G_i, \quad L'_i \Delta m_i = G_i^m.$$

There are no angular derivatives in the last principal equation. Use the currents with densities

$$e_{Y_i} = \frac{1}{2} |\Delta \beta_i|_{\gamma'}^2 \quad \text{in direction } j, \quad e_{Z_i} = \frac{a'}{4} (|\Delta Q_i|^2 + |\Delta m_i|^2) \quad \text{in direction } i. \quad (5.36)$$

The ratio of scalar to one-form weight is exactly $a'/2$. Since d and $J'd$ have adjoints $-\operatorname{div}'$ and $-\operatorname{curl}'$, their principal contribution, integrated on the full sphere, is

$$-\frac{1}{2} \int_{\mathbb{S}^2} \langle \Delta \beta_i, (da') k + J'(da') c \rangle_{\gamma'} d\mu_{\gamma'}. \quad (5.37)$$

This has size $C\sigma Y_i Z_i$, because $da' = a'(\xi'_0 + \xi'_1)$ and $a' \asymp q$. The derivative of the scalar weight contains $L'_i a' = a' \ell'_i = a'(-\kappa + O(h_i))$, whose constant part is favorable damping. The area and tensor-norm derivatives contain only low shapes and shifts. They cost $C(h_j Y_i^2 + h_i Z_i^2)$. Indeed, after the angular transport divergence is integrated out, their exact contributions to the two densities are

$$\frac{\tau'_j}{2} |\Delta \beta_i|^2 - \chi'_j ((\Delta \beta_i)^\sharp, (\Delta \beta_i)^\sharp), \quad \frac{a'}{4} (\ell'_i + \tau'_i) (|\Delta Q_i|^2 + |\Delta m_i|^2).$$

Here are all nonprincipal placements in F_i, G_i, G_i^m . The change of the angular principal coefficient on the barred field is

$$\frac{1}{2} \{ (a' - \bar{a})(d\bar{K} + \bar{J}d\bar{c}_i) + a'(J' - \bar{J})d\bar{c}_i \}. \quad (5.38)$$

It costs CqP , since $\Delta a = q\Delta \mathcal{A}$ and $J' - \bar{J}$ is an algebraic metric difference. The divergence and curl coefficient differences against $\bar{\beta}_i$ cost $Ch_i P$ and therefore $C\sigma h_i P$ after scalar weighting. Shift differences cost $Ch_i D_b$ in the first equation and $C\sigma D_b$ in the other two. The remaining a terms cost at most $C\sigma(Z_0 + Z_1 + P)$ in the first equation. Indeed they consist of $a'\Delta K$, $a'\nabla' \Delta \xi$, bounded

low factors times $a'\Delta\xi$, and coefficient differences on bounded barred factors. The shape products have bound

$$C(h_i R_j + h_j R_i + h_i h_j P). \quad (5.39)$$

In particular the angular connection difference in $\chi_i * \nabla \chi_j$ has the explicit factor $\chi'_i * C * \bar{\chi}_j$ and costs $Ch_i h_j P$. In the scalar equations, the weighted remainder has bound

$$C\{\sigma(R_i + D_b + P) + h_i(Z_0 + Z_1)\}.$$

These lists exhaust the factors of Equation (5.7). Put $Z = Z_0 + Z_1$. In particular all uses of the apparently dangerous second estimate in Equation (5.34) occur with one of the factors

$$a'|\Delta\xi|_1 \leq C(\sigma Z + qP), \quad \sigma h_i |\Delta\xi|_1 \leq C(h_i Z + \sigma h_i P). \quad (5.40)$$

There is no naked $\sigma^{-1}Z$ in a current source. The combined remaining paired cost is consequently bounded by

$$CH_s(Y_i + Z_i)(V + R_0 + R_1), \quad (5.41)$$

apart from the defects paired with the current variables.

Trace, shape, lapse, and shift currents. Subtract Equation (5.6) with the actual L'_i :

$$(L'_i + \kappa)\Delta w_i = -\Delta(\mathcal{A}^{-1}|\chi_i|^2) - \Delta B_i \cdot d\bar{w}_i + \mathcal{F}_i^w. \quad (5.42)$$

The defect $\mathcal{F}_i^w$ has the input bound. Its one angular commutation has cost at most $Ch_i(z_i + R_i + D_b + h_i P)$, in addition to the homogeneous damping: the differentiated quadratic shape has a low h_i factor, $\Delta\mathcal{A}$ is controlled in H^1 by Equation (5.32), and the shift term uses $\bar{w}_i$ through two low angular derivatives, all $O(h_i)$. The commutator with L'_i uses only the low first angular derivative of B'_i . Thus the remaining z_i^2 current cost is $CH_s z_i(V + R_i)$; the constant $2\kappa z_i^2$ has favorable sign. In particular no $\Delta\ell_i$ term is generated by this normalized trace equation.

For N_i use the shape and lapse cross equations in direction j . Their source factors are precisely

$$a\nabla\xi, \quad aK_\gamma, \quad a\xi^2, \quad \chi_i * \chi_j.$$

Their differences are bounded by $C\{\sigma(Z_0 + Z_1) + qP + h_i N_j + h_j N_i + h_i h_j P + h_i D_b\}$. For example, using the connection difference in Equation (5.30),

$$\begin{aligned} \Delta(a \text{Sym} \nabla \xi_j) &= a' \text{Sym} \nabla' \Delta \xi_j + (\Delta a) \text{Sym} \bar{\nabla} \bar{\xi}_j - a' \text{Sym}(C * \bar{\xi}_j), \\ \Delta(a K_\gamma \gamma) &= a'(\Delta K_\gamma) \gamma' + a' \bar{K}_\gamma \Delta \gamma + (\Delta a) \bar{K}_\gamma \bar{\gamma}. \end{aligned}$$

This uses Equation (5.40) and Equation (5.35); the constant part of ℓ_i does not contribute. The squared N_i currents therefore cost at most $CH_s N_i(V + R_0 + R_1)$. In particular there is no unsuppressed return from a lapse or metric error to N_i .

For the H^1 shift current in direction 1, subtract its exact equation:

$$\partial_v \Delta b = q \Delta \{\mathcal{A} \gamma^{-1}(\xi_1 - \xi_0)\}.$$

One angular derivative uses Equation (5.32) and Equation (5.40), giving a source $C(\sigma(Z_0 + Z_1) + qP)$ and a squared-current cost $CH_s D_b V$. No differentiated shift is needed on its right side.

The unweighted recovery currents. Use direction 1 for $\Delta\gamma$ in H^1 and ΔH in L^2 , where $H = \log \mathcal{A}$:

$$\partial_v \Delta\gamma = 2\Delta\chi_1, \quad \partial_v \Delta H = \Delta\ell_1. \quad (5.43)$$

For comparison, an arbitrary-direction equation would be $L'_i \Delta H = \Delta\ell_i - \Delta B_i \cdot d\bar{H}$; the chosen direction has $\Delta B_1 = 0$. For $\Delta\xi_j$ use direction i and

$$\mathfrak{L}'_i \Delta\xi_j = \Delta\beta_i + \tau'_i \Delta\xi_i + (\Delta\tau_i)\bar{\xi}_i - \mathcal{L}_{\Delta B_i} \bar{\xi}_j + \mathcal{F}_i^\xi. \quad (5.44)$$

In its undifferentiated norm the trace difference costs $C(N_i + h_i P)$ directly from the definition of τ_i . The last Lie derivative involves bounded barred torsion and its first angular derivative, giving CD_b . Use fixed-background angular norms for these recovery currents to avoid any unnecessary time differentiation of their norm coefficients. Their total cost, including harmless low transport commutators, is

$$CP\left\{\sum_i (Y_i + N_i + R_i) + D_b + H_s P\right\}. \quad (5.45)$$

All occurrences of unweighted first angular torsion derivatives have already been confined to Equation (5.40).

The growth rate and the late support. Substitute Equation (5.34) in the preceding costs. At $H_s = 0$, all currents except the P currents have zero unfavorable cost. The surviving directed interactions are

$$(Y_i, N_i, z_i, D_b) \longrightarrow P. \quad (5.46)$$

There is no reverse arrow. Give the component P currents a small fixed multiplier δ_P^2 and every other current fixed positive multipliers, equal on each Bianchi pair. Let $\mathcal{E}_T(t)$ be the resulting spatial slice energy. Its density is uniformly equivalent, with constants depending on δ_P but not T , to

$$\sum_i (Y_i^2 + Z_i^2 + z_i^2 + N_i^2) + D_b^2 + \delta_P^2 P^2. \quad (5.47)$$

In particular it controls all the unweighted quantities claimed in Equation (5.28). The factors $1/2$ from the time components of L'_i merely change this equivalence by constants.

Given a small rate $\rho_* > 0$, first take δ_P small so that the form in Equation (5.46), expressed in scaled variables, costs at most $\rho_* \mathcal{E}_T/2$. Next choose a fixed $h_* > 0$ so that every term with factor H_s costs at most $\rho_* \mathcal{E}_T/2$ when $H_s \leq h_*$. On the remaining bounded low coefficient set the rate is bounded by a constant C_* independent of T . This constant is fixed before ν is chosen.

All difference fields vanish in $v < (1 - \nu)T$ by protected agreement, causal monotonicity and uniqueness of the optical transports. Fix a large R so that the background envelope suprema are small outside $[-R, R]$, and a fixed t_0 so that both σ and $q = \sigma^2$ are below their required thresholds for $t > t_0$. For large T , h_1 is small on the support for $t > t_0$. At those times, the only possible failure of $H_s \leq h_*$ is where $u \geq -R$. At any such supported point,

$$2t = u + v \geq (1 - \nu)T - R, \quad t \leq L_T = T/2 + O(1).$$

Consequently the slice times containing a bad supported point lie in

$$[0, t_0] \cup \left[\frac{(1 - \nu)T - R}{2}, L_T \right]$$

after omitting any empty interval. Their total length is at most

$$t_0 + \nu T/2 + O(1). \quad (5.48)$$

The added e_T in the majorants is uniformly below the threshold for large T . Thus the integrated homogeneous rate is bounded by $C\rho_*T + C_*\nu T/2 + O(1)$, and is an arbitrarily small multiple of T after choosing ρ_* and then ν .

Both optical directions have nonnegative outgoing flux on the tilted sides: the derivatives of $u + c_0t/T$ and $v + c_0t/T$ along L'_0, L'_1 are nonnegative, and are strictly positive where needed. There are no angular boundary terms because the sphere is complete. The defects have magnitude F_T on a spatial interval of length $O(T)$ and therefore add at most $C \text{poly}(T)F_T\sqrt{\mathcal{E}_T}$ to the squared energy inequality. The initial discrepancy has the same bound after the finite spatial integration. The square-root inequality (or its regularization at zero) and Gronwall yield

$$\sqrt{\mathcal{E}_T(t)} \leq C \text{poly}(T) A_T^2 e^{\eta T} \exp\{C\rho_*T + C_*\nu T/2 + O(1)\} \leq A_T^2 e^{\eta_2 T}. \quad (5.49)$$

Choose the several positive losses strictly below η_2 to absorb the fixed and polynomial factors. This proves the Proposition. $\square$

5.4 Interpolation, improvement, and continuation

Proof of Theorem 5.1. We first make explicit the finite order choices. Choose a spatial target order $k \geq 2s$, enlarged if necessary to improve every bootstrap and continuation norm. The time recovery below uses at most two spatial derivatives for each time derivative. Choose an integer M much larger than $k + 2$ so that, with $\vartheta = (k + 2)/M$,

$$2\alpha(1 - \vartheta) > (1 + c_1)\alpha \quad \text{for some } 0 < c < c_1 < 1/2. \quad (5.50)$$

The constant 2 in $k + 2$ supplies Sobolev embedding on the three-dimensional slices. Here M is the requested slice Sobolev order; the proof of Lemma 5.2 runs its angular currents at the larger order specified there. That Lemma determines a finite initial buffer $r(M)$ and its finite template/exterior jet requirements. Include the finite order m in that base budget. Choose the rate tolerance η_2 small enough that Equation (5.50) retains a strict gap after adding all interpolation and later time-recovery losses. In Proposition 5.3 choose the fixed current multipliers, the small-region threshold, the growth tolerance, then ν , and finally sufficiently small input/exterior losses.

Let $P_{\text{base}} \geq 1$ be the maximum of the finitely many positive frequency exponents required at these fixed base orders. Choose $\zeta > 0$ so small that each has its allocated loss, in particular $\zeta P_{\text{base}} < \alpha/2$. Only now increase the finite hierarchy and beam accuracy orders to make all linear defects as small as required, including the finite inverse-lapse losses in deriving linearized double-null identities. The base-loss statement in Proposition 4.1 ensures that this last operation changes constants and required fixed background jets, but does not increase any previously budgeted positive power bounded by P_{base} of λ . Finally take T large. Every high input is a finite entry norm already supplied by the preparation. This order of choices has no unknown-high-norm smallness requirement.

Local smooth vacuum theory and noncharacteristic eikonal and angle transport start the primed solution at the actual cylinder data. On each available slice, Equation (5.28) and Lemma 5.2 give, for each component error f ,

$$\|f\|_{L^2} \leq e^{-(2\alpha - \eta_2)T}, \quad \|f\|_{H^M} \leq e^{o(T)}.$$

For the second inequality the barred fields have their individual high bounds directly from the linear construction. Sphere charts and extension in the interval variable give

$$\|f\|_{C^k(I_T(t) \times \mathbb{S}^2)} \leq C \text{poly}(T) \|f\|_{L^2}^{1-\vartheta} \|f\|_{H^M}^{\vartheta} \leq e^{-(1+c_1)\alpha T} \quad (5.51)$$

after reducing c_1 within its fixed margin. The lower bound for the interval length ensures uniform local extension constants; covering a longer interval has at worst a polynomial cost. This argument is performed on each slice and needs no positive lower bound for the available time thickness.

To obtain time derivatives, use $\partial_v = (\partial_t + \partial_x)/2$ in Equation (5.5):

$$\begin{aligned}\partial_t \gamma &= 4\chi_1 - \partial_x \gamma, \\ \partial_t \log \mathcal{A} &= 2(\ell_1 + \kappa) - \partial_x \log \mathcal{A}, \\ \partial_t b &= 2q\mathcal{A}\gamma^{-1}(\xi_1 - \xi_0) - \partial_x b.\end{aligned}$$

Use the shape/lapse cross equations and the torsion equations for the other fields. For a tensor transported in direction 0 use $\mathfrak{L}_0 = (\partial_t - \partial_x)/2 + \mathcal{L}_b$; for a scalar replace the Lie derivative by $b \cdot d$. Thus each time derivative is twice the corresponding right side, plus a spatial derivative and, when appropriate, an angular Lie term. A shape or lapse cross equation contains at most two angular derivatives of the metric and one of the torsion; the torsion equation contains one angular shape derivative. Thus r time derivatives and p spatial derivatives require at most spatial order $p + 2r$. The choice $k \geq 2s$ covers every requested ordinary derivative of total order at most s . Subtract the equations with the actual principal operators, as above. All products use already controlled lower error jets and individual jets from Lemma 5.2; barred defects have the prescribed finite differentiated bounds. There is no division by q . Finite product losses can be assigned arbitrarily small rates in that Lemma, leaving Equation (5.2) with $c < c_1$. The relative shift conclusion follows by differentiating the algebraic identity

$$q^{-1}\partial_v b = \mathcal{A}\gamma^{-1}(\xi_1 - \xi_0).$$

The template change at the fixed bootstrap orders is at most $e^{-(\alpha-\eta_0)T}$ with $\eta_0 < \alpha/2$. Together with Equation (5.2), this is strictly smaller than $\frac{1}{2}e^{-\alpha T/2}$ for large T . It improves Equation (5.8), preserves the nondegeneracy margins, and provides the finite Sobolev buffers required for restart.

We finish the existence argument, since an a priori comparison on a putative full domain would not suffice. On each fixed finite experiment, $q \geq q_0 e^{-2\kappa L T} > 0$. All requisite finite norms and inverse metrics are bounded up to a possible terminal slice. Use the finite-domain restart in [23, Proposition 7.7, proof] as follows. On an earlier slice take its induced metric and second fundamental form, initialize wave gauge, and extend the smooth slice jets across the interval endpoints on an enlarged margin. Their required finite Sobolev norms and positivity have controlled extensions. Constraints are needed only on the original margin. The reduced wave equation has a positive local existence time depending on those finite norms and the finite positive lapse. Solve the eikonal pair and angular transport from the slice with their inherited values. Transversality and invertibility persist on a smaller margin for a time bounded below at this fixed T .

Restrict the new solution to the region with exiting sides. There $du, dv \geq 0$ on future causal vectors, so artificial extension data cannot enter the retained domain. Constraint and gauge propagation give the vacuum equations there. Local geometric uniqueness, followed by uniqueness for the eikonal and angle transports, identifies this solution with the existing one on their overlap. Restart before any proposed terminal time; the uniform bounds at that fixed T contradict termination. Slightly enlarged intervals and sphere patches give smooth endpoint jets throughout. The restart time need not be uniform as $T \rightarrow \infty$.

Finally, smoothness at every order is preserved. At each fixed T the fixed-region version of the high-current proof, with the weight e^{-2Kt} , propagates any larger finite order from the smooth entry data. Its constants may depend on that order and on T ; no simultaneous subexponential estimate at all orders is asserted or needed. The constructed solution therefore reaches all of Equation (5.1)

smoothly. On the initial collar it is the same vacuum solution as the prepared exterior solution in its newly initialized optical coordinates, by the same local uniqueness. The whole-bridge attachment is carried out in Proposition 6.1. This proves the Theorem. $\square$

6 Attachment, curvature signal, and matched observing states

Apply Theorem 5.1 to the actual entry data and the template from Proposition 4.1. Write g_T for the resulting exact finite double-null metric. Its fields differ from the template by $e^{-(1+c)\alpha T}$ in every preassigned finite observation order, for a fixed $c > 0$. All orders used below, including thirty ordinary observation derivatives, are included before choosing ζ . We first place this metric in the full development of d_T .

6.1 Attaching the actual exterior and interior

Proposition 6.1 (Attachment to the full development). *For sufficiently large finite T , the exact exterior and deep experiments, restricted inside their construction margins, lie in a smooth globally hyperbolic vacuum development of the entire datum d_T . This development embeds isometrically in its full MGHD M_{d_T} . The full initial bridge and the exterior correction tail are retained. The common earlier branch tubes and the compact observation region have interior margins in the attached region.*

Proof. The lower and exterior joins are the qualitative construction in [22, Section 7.1, Remark 7.2], based on [23, Lemma 8.1, Proposition 8.2, Proposition 9.10]. Its hypotheses are full-bridge p_{10} closeness, exact protected agreement on the compact causal shadows of the early joins, actual corrected partial-slice data, low exterior closeness on enlarged matching bands, and strict outflow and overlap margins. We verify these hypotheses and the new double-null join separately.

The finite lower-block construction in [23, Lemma 8.1] applies to d_T since $p_{10}(d_T - d) \rightarrow 0$. It contains every spatial radius of the new bridge, a past collar, the two bent exterior initial slices, and fixed forward bands on them. On the ends the bent slices equal the original bridge outside a fixed compact set. The protected ball (4.7) contains a fixed-origin ball of radius

$$(1 - 2\beta - s_*)t_T - O(1) \longrightarrow \infty.$$

Consequently every fixed compact causal shadow needed for the bent pieces and bounded connecting pieces is exactly unchanged. Domain of dependence in a common early Cauchy gauge identifies these pieces with the background. At large radii the horizontal selected-end slice carries the actual new datum, including its tail; no replacement by the old datum is made there.

Choose a fixed exterior joining time t_a inside the enlarged lower block. On the selected end attach the exact exterior from Proposition 4.1, computed from these same partial-slice data, using the shrinking outer cutoff (4.10). The identifying wave map on an enlarged fixed band about t_a has the prescribed initial slice map and matching unit normals. Fixed-time wave-map stability gives a diffeomorphism on the smaller band, uniformly through the large outer cutoff: the exterior charts have uniform low regular geometry, the new metrics are low-order close, and artificial sides are strictly outgoing. The induced metric and second fundamental form agree; the wave gauge then determines the remaining normal metric jets. Reduced uniqueness therefore makes this map an isometry. These are exactly the finite-band hypotheses checked in [22, Section 7.1, Remark 7.2]. The unselected end uses the unchanged exterior. Straighten and identify inside the matching margins, without keeping a second copy of overlapping points.

It matters that the lower block has not been spatially truncated. In a selected-end overlap the retained region has, in addition to its finite future cuts, the condition

$$t_e < t_a \quad \text{or} \quad r_{\text{co}} < R_{\text{out}}(t_e). \quad (6.1)$$

Both t_e and $r_{\text{co}} - R_{\text{out}}(t_e)$ increase strictly along future causal curves at their active boundaries. The derivative of the second quantity with respect to t_e is at least $V_{\text{out}} - \sup |dr_{\text{co}}/dt_e| > 0$. Their strict sublevels and their union are preserved to the past. A past curve in the added piece stays inside its outer cutoff until reaching t_a , after which all radii are present. A past curve already below t_a stays there. The omitted corner at $t_e = t_a$, $r_{\text{co}} = R_{\text{out}}(t_a)$ is consequently not a missing past endpoint.

The required interval of $\mathcal{S}$ and a margin are present in this joined lower/exterior region. At its selected late end, $t_e \leq t_T + O(1)$ and the co-radius is bounded; its middle and unselected portions are unchanged. Low closeness preserves its spacelike character. Use the *actual exterior induced data* there. The nearby eikonal roots with the prescribed entry values, and the shift-free angular transport, define the new coordinates on a collar. They are noncharacteristic at this fixed depth. Local geometric vacuum uniqueness identifies this collar with the initial collar of the exact solution from Theorem 5.1; equivalently that solution may be started on the collar itself. The quadratic template comparison does not substitute for matching actual data. For every fixed finite T the collar widths can be chosen inside the available margins; no lower width uniform in T is required.

Above the join all cuts are made in the *new* double-null coordinates. For a causal vector V in their future cone,

$$-2a_T du(V)dv(V) + \gamma_T(d\theta(V) - b_T du(V), d\theta(V) - b_T du(V)) \leq 0.$$

Thus $du(V)$ and $dv(V)$ have the same weak sign, and the future choice is the nonnegative sign. Also $g_T^{-1}(dt, dt) = -1/(2a_T) < 0$; hence t is temporal. For a nonzero future causal vector, $dt(V) > 0$, and

$$\frac{d}{dt} \left(u + \frac{c_0 t}{T} \right) \geq \frac{c_0}{T}, \quad \frac{d}{dt} \left(v + \frac{c_0 t}{T} \right) \geq \frac{c_0}{T}. \quad (6.2)$$

The strict side and terminal sublevels of (4.1) are therefore preserved into the past. In particular a past-inextendible causal curve from its strict interior reaches $t = 0$ strictly inside the attached interval. Indeed its side functions never exceed their strictly smaller starting values. On each fixed experiment, the retained closure lies inside a compact enlargement on which smooth nondegeneracy bounds $a_T, a_T^{-1}, b_T, \gamma_T^{\pm 1}$. In t parameter, $0 \leq du/dt, dv/dt \leq 2$, and the displayed cone inequality then bounds the angular speed using the upper bound for a_T and the bounds for b_T, γ_T^{-1} . A finite interior endpoint is therefore handled by ordinary continuation. This argument uses the exact new coordinates and does not require small relative cone error in a deep background wave chart.

Omit every unattached future face of a lower piece, including its edges. It is a future exit by the spacelike sign, so a past curve below it cannot end there. At intersections of cuts, the retained sets are finite unions and intersections of the same past-preserved strict sublevels; (6.1) handles the only change from unrestricted to truncated radii. Matching collars give product neighborhoods across the included faces. One first joins the lower bands and then the deep collar at their intersections, as in [23, Proposition 8.2]; this gives a Hausdorff smooth manifold with only one copy of each identified point.

Every past-inextendible causal curve from an added piece therefore reaches the complete initial bridge. In the lower pieces bounded speeds exclude escape to spatial infinity in bounded time, and their Cauchy construction gives the same conclusion. The past collar gives the future crossing for curves starting below the bridge. There is at most one crossing, by the lower-block Cauchy

property and the one-way signs of all joins. Thus the complete Σ is Cauchy in the assembled vacuum region. Maximal-development uniqueness embeds this region in M_{d_T} , proving the claim. All restrictions were made inside preassigned enlarged margins, so the earlier flow tubes and observation neighborhood remain interior. $\square$

6.2 A curvature difference on the whole observation patch

Use the product observation box $\mathcal{Q}_T$ fixed before the packet construction in Section 4. It contains all hits of the chosen seed-neighborhood closure at $v \in T + J$, and the feet of its enlarged closure lie in the launch-bump plateau where the bump is at least one. The local polarization there has unit screen norm. These properties remain available when the seed ball and interval are trimmed below. Adopt $R(X, Y) = [\nabla_X, \nabla_Y] - \nabla_{[X, Y]}$ and set

$$D(g)_{AB} = g(R(\partial_v, \partial_A)\partial_v, \partial_B). \quad (6.3)$$

Changing the overall curvature convention changes all ensuing signs together and leaves the norm estimates unchanged.

Proposition 6.2 (Whole-patch curvature signal). *At every point of $\mathcal{Q}_T$ with $|\cos(\lambda v)| \geq 1/2$, one has, for all large T ,*

$$|D(g_T) - D(g)| > 2A_T. \quad (6.4)$$

The norm is the ordinary angular component norm. No bound on $D(g)$, or on background parallel-frame curvature, is assumed.

Proof. For a shift-free direction $L_1 = \partial_v$, the null connection identities are

$$\nabla_{L_1} L_1 = \ell_1 L_1, \quad \nabla_{\partial_A} L_1 = \chi_{1,A}{}^B \partial_B + \xi_{1,A} L_1.$$

Since $[\partial_v, \partial_A] = 0$, applying the curvature commutator and pairing with ∂_B gives

$$D(g)_{AB} = \gamma_{BC} \partial_v (\chi_{1,A}{}^C) + (\chi_1 \gamma^{-1} \chi_1)_{AB} - \ell_1 \chi_{1,AB}.$$

Use $\partial_v \gamma = 2\chi_1$ to lower the differentiated index. This gives the exact formula

$$D(g) = \partial_v \chi_1 - \chi_1 \gamma^{-1} \chi_1 - \ell_1 \chi_1 = \frac{1}{2} \partial_v^2 \gamma - \frac{1}{4} (\partial_v \gamma) \gamma^{-1} (\partial_v \gamma) - \frac{1}{2} (\partial_v \log a) (\partial_v \gamma). \quad (6.5)$$

There is no inverse radial lapse in this formula.

For the template, $\bar{\chi}_1 = \chi_1 + \frac{1}{2} \partial_v k_\gamma$ and $\bar{\ell}_1 = \ell_1 + \partial_v k_H$ exactly. The linear variation of (6.5) is therefore

$$\begin{aligned} \delta D = & \frac{1}{2} \partial_v^2 k_\gamma - \frac{1}{4} ((\partial_v k_\gamma) \gamma^{-1} \partial_v \gamma + (\partial_v \gamma) \gamma^{-1} \partial_v k_\gamma) \\ & + \frac{1}{4} (\partial_v \gamma) \gamma^{-1} k_\gamma \gamma^{-1} (\partial_v \gamma) - \frac{1}{2} \ell_1 \partial_v k_\gamma - \frac{1}{2} (\partial_v k_H) \partial_v \gamma. \end{aligned} \quad (6.6)$$

The angular Lie term in (4.18) contains only Z and angular derivatives of Z , without a longitudinal derivative of Z . Hence (4.15) gives

$$\partial_v^2 (\mathcal{L}_Z g)_{AB} = O(A_T \lambda e^{o(T)}).$$

The formula for k_H has at most one longitudinal derivative of Z , so $\partial_v k_H = O(A_T \lambda e^{o(T)})$ as well. All other linear terms in (6.6) have this bound. Twice differentiating the real leading oscillation in (4.3), whose trace-free leading tensor is unchanged by trace reversal, yields

$$D(\bar{g}) - D(g) = -\frac{1}{2} A_T \lambda^2 \cos(\lambda v) (p_0)_{AB} + O(A_T \lambda e^{o(T)}) + O(A_T^2 \lambda^P e^{o(T)}). \quad (6.7)$$

The last term comprises quadratic template products in the smooth relative-field expression (6.5). Choose ζ small enough that this term is $o(A_T)$. The exact-minus-template error from Theorem 5.1, at the finite derivative orders in (6.5), is also $o(A_T)$; multiplication by the individual $e^{o(T)}$ coefficient bounds preserves this conclusion.

By (4.4), on the cosine-good set the leading norm is at least $\frac{1}{4}A_T\lambda^2e^{-o(T)}$ uniformly throughout $\mathcal{Q}_T$. Its ratio to the linear lower-order error tends to infinity, since $\lambda e^{-o(T)} \rightarrow \infty$, and its ratio to A_T also tends to infinity. This proves (6.4). The argument compares two curvature tensors at common numerical coordinate positions; the background curvature is never presumed small or bounded independently of T . $\square$

6.3 Canonical terminal states and finite persistence

Let $O \in \mathcal{B}$ be the fixed test ball from Definition 3.5, with $\bar{O} \subset \mathcal{N}$ for the chosen observation neighborhood. Choose a closed concentric subball $O_0 \Subset O$ such that

$$\mu_d(O_0) > .95\mu_d(O). \quad (6.8)$$

Such a choice follows by monotone convergence of the smooth positive density. Choose a smaller closed interval $J_0 \Subset J$ of positive length. For every late level $v \geq v_0$ reached by the tube, let $\mathcal{H}_v$ be the background hitting map of O_0 , where v_0 is the fixed late starting level. It records the canonical state

$$z = (u, \theta, \pi_u, P_1, P_2). \quad (6.9)$$

The original seed-to-covector identification uses $w \mapsto h(w, \cdot)$ on Σ . Thus μ_d is exactly the canonical section measure, not a position-only measure.

Proposition 6.3 (Matched seeds and finite observations). *For $v \in T + J_0$, use the same numerical states (6.9) in g_T on its future unit shell. Their backward geodesics give maps $\Psi_{T,v} : O_0 \rightarrow O$ with*

$$\sup_{v \in T + J_0} \|\Psi_{T,v} - \text{incl}\|_{C^1} \longrightarrow 0. \quad (6.10)$$

Write δ_T for the left side of (6.10). Each map is injective and preserves the corresponding section measure: $\Psi_{T,v}^* \mu_{d_T} = \mu_d$ on O_0 . In particular $\mathcal{H}_v(O_0)$ has canonical volume $\mu_d(O_0)$ for every such v .

Put

$$r_T = e^{-\kappa T}, \quad F = (\partial_u, r_T^{-1}\partial_v, \partial_{\theta^1}, \partial_{\theta^2}). \quad (6.11)$$

For g and g_T on the larger observation box, the Gram matrices in F and their inverses are uniformly bounded and nondegenerate. Their ordinary coordinate jets through order thirty are $e^{o(T)}$. The selected unit tangent is bounded in F , all state momenta lie in one bounded box, and

$$cr_T du dv d^2\theta \leq d \text{vol}_g, \quad d \text{vol}_{g_T} \leq Cr_T du dv d^2\theta. \quad (6.12)$$

For each fixed finite T and each $\eta_{\text{map}} > 0$, these observations persist for all data in a sufficiently small relative weighted smooth neighborhood of d_T , with signal lower bound A_T in place of $2A_T$. The persisted common coordinates need not be optical. The canonical section volume, injective maps into O , and frame and density bounds persist. Their backward maps satisfy

$$\sup_{v \in T + J_0} \|\hat{\Psi}_{T,v} - \Psi_{T,v}\|_{C^1} < \eta_{\text{map}}, \quad \sup_{v \in T + J_0} \|\hat{\Psi}_{T,v} - \text{incl}\|_{C^1} \leq \delta_T + \eta_{\text{map}}.$$

The neighborhood may depend on T and η_{map} .

Proof. The unit mass shell in double-null variables reads

$$p = -(\pi_u + b^A P_A) > 0, \quad 2p\pi_v + a(1 + \gamma^{AB} P_A P_B) = 0.$$

Thus the reduced v -Hamiltonian is

$$\mathcal{H} = -\pi_v = \frac{a(1 + \gamma^{AB} P_A P_B)}{2p}. \quad (6.13)$$

The same expression with primed fields defines the new future unit root at the identical numerical terminal state. The background tube keeps p in a compact positive interval and the other momenta bounded. Since $b_T - b \rightarrow 0$ uniformly, the new $p_T = -(\pi_u + b_T^A P_A)$ remains bounded away from zero. Choosing the root with $\dot{v} = p_T/a_T > 0$ fixes its future orientation.

Write V for the background Hamiltonian vector field. On the common late state tube this field and its first two state derivatives are bounded by $ae^{o(v)}$, by Lemma 2.4. This bound is integrable over the late v interval. Differentiating (6.13) shows that the new field and its first two state derivatives differ by an absolutely exponentially small quantity, uniformly on that tube. Indeed $a_T = q\mathcal{A}_T$ retains its factor q under ordinary differentiation; the denominators are powers of p_T separated from zero; and all relative coefficient differences and their needed jets are exponentially small by Proposition 4.1 and Theorem 5.1. Fixed higher background factors cost $e^{o(T)}$, which is absorbed by that exponential.

The terminal state difference is exactly zero. Backward Gronwall on a late interval of length $O(T)$ therefore gives state differences tending to zero. For first variations the equation is $J' = D_z V J$. Its coefficient difference is bounded by the small difference of vector-field derivatives plus $|D_z^2 V|$ times the already small state difference. The latter background bound is integrable. A second Gronwall estimate therefore gives uniform C^1 convergence of the backward maps. The background forward hitting map also has a uniform derivative bound: if v_0 is the fixed late starting level, then

$$\|D_w \mathcal{H}_v\| \leq \|D_w \mathcal{H}_{v_0}\| \exp\left(\int_{v_0}^v \|D_z V(s)\| ds\right) \leq C.$$

Composing the terminal-state estimates with $\mathcal{H}_v$ therefore preserves their C^1 convergence in the original seed variables. The strict tube and construction margins prevent exit during this comparison. Below any fixed late v the remaining flow tubes to $\mathcal{S}$ and then Σ are compact and transverse. Smooth finite-flow dependence handles their hitting maps; protected agreement makes their metrics exactly the background metrics for sufficiently large T . This proves (6.10) and, by the margin $O_0 \in O$, places the new seeds in O .

Here is the exact volume statement. On the cotangent bundle, let

$$\omega = d\pi_u \wedge du + d\pi_v \wedge dv + dP_A \wedge d\theta^A.$$

On the unit Hamiltonian shell its restriction has the geodesic flow direction as kernel. A single-valued transverse hitting map therefore preserves the restrictions of ω to two sections: flowing by a variable time only adds multiples of its kernel to tangent vectors. On $v = \text{constant}$ this restriction is exactly

$$\omega_v = d\pi_u \wedge du + dP_A \wedge d\theta^A,$$

independently of the shell root π_v . Its volume is $du d^2\theta d\pi_u d^2P$. Apply this to the background hit and to the new backward hit. It gives $\Psi_{T,v}^* \mu_{d_T} = \mu_d$ exactly. The hits are single-valued on these compact transverse flow tubes; uniqueness also gives injectivity. Alternatively, in the ball coordinates (6.10) makes the map $\text{Id} + f$ with $\|Df\| < 1/2$, which is injective on the convex closed

ball. Nothing in this argument assumes injectivity of the projection from seed states to observation positions.

At an observation state the unit tangent satisfies

$$\dot{u} = \frac{1 + \gamma^{AB} P_A P_B}{2p}, \quad r_T \dot{v} = \frac{r_T p}{a}, \quad \dot{\theta} = b\dot{u} + \gamma^{-1} P, \quad (6.14)$$

and likewise for g_T . On the whole observation box, u and $v - T$ are bounded, so $a/r_T = q_0 \mathcal{A} e^{-\kappa(u+v-T)}$ is bounded above and below. These formulas prove the tangent bounds in F . The Gram matrix has the block form

$$G_F = \begin{pmatrix} \gamma(b, b) & -a/r_T & -b^C \gamma_{CB} \\ -a/r_T & 0 & 0 \\ -\gamma_{AC} b^C & 0 & \gamma_{AB} \end{pmatrix}, \quad \det G_F = -(a/r_T)^2 \det \gamma. \quad (6.15)$$

The low field bounds give bounded entries and a determinant bounded away from zero, hence a bounded inverse. The same holds for g_T . All ordinary derivatives of a/r_T retain this bounded relative factor, and the other required field jets are $e^{o(T)}$ by the individual estimates. This proves the thirty-jet assertion on the larger box, not just on the hitting positions. Finally $\sqrt{|\det g|} = a\sqrt{\det \gamma}$ proves (6.12).

We finish with the finite persistence assertion. For a fixed T the closures of the observation box, a small enlargement, and all backward flow tubes just used are compact subsets of the smooth attached development with an interior margin. The compact-subset Cauchy stability and geometric uniqueness of [23, Lemma 2.2], also used in the closed-test proof of [22, Lemma 3.3, proof], supply corresponding compact coordinate witnesses for all sufficiently close smooth data. Require closeness in the finite C^k norms needed here. Since T is fixed, it can be arbitrarily accurate even in the rescaled frame F . Preserve the Gram and inverse bounds, the thirty derivative bounds (for example allowing additive error at most one), the density comparison, and the curvature margin, so that

$$|D(\hat{g}_T) - D(g)| > A_T \quad \text{when} \quad |\cos(\lambda v)| \geq 1/2. \quad (6.16)$$

The common coordinates for $\hat{g}_T$ need not retain double-null form. To define its terminal state solve $\frac{1}{2}\hat{g}_T^{-1}(\pi, \pi) = -\frac{1}{2}$ for π_v with the other six numbers fixed as in (6.9). At the original terminal state the derivative with respect to π_v is $U^v > 0$. On the fixed compact state set it is separated from zero, so the implicit function theorem supplies the same smooth simple future root for close metrics. The level- v section remains transverse to this flow. Smooth backward-flow dependence lets us impose $\sup_{v \in T+J_0} \|\hat{\Psi}_{T,v} - \Psi_{T,v}\|_{C^1} < \min\{\eta_{\text{map}}, \eta_0\}$, where a fixed $\eta_0 > 0$ preserves injectivity and the image margin in O . The restriction of the canonical symplectic form to that section is still ω_v , regardless of optical gauge. Its transported volume is therefore exactly the new initial seed measure. All statements hold for this fixed experiment, with no assertion of a data-neighborhood radius uniform in T . $\square$

7 Averaged holonomy and the category argument

We turn the finite curvature difference of Proposition 6.2 into an obstruction to a square-integrable connection. The estimate is averaged over four position variables and three momentum variables. The loops themselves remain in the smooth development. In particular, no restriction of the weak ambient connection to a prescribed curve is required. Sbierski developed holonomy obstructions at Lipschitz regularity [30] and a related nonsymmetric obstruction based on integrated curvature

[31]. Here the ambient hypothesis controls an integral over positions; averaging must therefore precede any curvewise transport estimate. The stopping argument below establishes the needed chart control for the sampled loops.

7.1 Canonical parameters and the observation frame

Write

$$r = r_T = e^{-\kappa T}, \quad l_T = e^{-\kappa T/16}, \quad A_T = e^{-\kappa T/4}. \quad (7.1)$$

The observation coordinates are $(u, v, \theta^1, \theta^2)$, and the frame used throughout this section is

$$F_0 = \partial_u, \quad F_1 = r^{-1} \partial_v, \quad F_2 = \partial_{\theta^1}, \quad F_3 = \partial_{\theta^2}. \quad (7.2)$$

For each fixed experiment, r is a constant on the entire observation box. Thus the four fields in (7.2) commute.

Let $O_0 \Subset O$ be the closed trimmed ball from Proposition 6.3, chosen so that

$$\mu_d(O_0) > .95\mu_d(O). \quad (7.3)$$

For $v \in T + J_0$, where J_0 is a fixed compact interval of positive length, let $\mathcal{P}_{T,v}$ be the set of terminal states (u, θ, π_u, P) obtained by following the background seeds in O_0 to that level. The full parameter set and its measure are

$$\mathcal{P}_T = \{(v, u, \theta, \pi_u, P) : v \in T + J_0, (u, \theta, \pi_u, P) \in \mathcal{P}_{T,v}\}, \quad d\nu_T = dv du d^2\theta d\pi_u d^2P. \quad (7.4)$$

The three momentum labels range in a fixed bounded box $Q \subset \mathbb{R}^3$. Here and below, measures may be completed without changing any estimate.

We recall why these are canonical measures, also for the nearby metrics used later. On the future unit shell of $\mathcal{H}(x, \pi) = \frac{1}{2}g^{\alpha\beta}(x)\pi_\alpha\pi_\beta$, a transverse v section has a simple root for π_v because $\partial_{\pi_v}\mathcal{H} = \dot{v} > 0$. On that section the pullback of the symplectic form is

$$d\pi_u \wedge du + \sum_{A=1}^2 dP_A \wedge d\theta^A.$$

The Hamiltonian vector field is the kernel of the symplectic form restricted to the shell. Its transverse hitting maps consequently preserve this two-form and its canonical volume, the absolute value of its third exterior power divided by $3!$. The corresponding section on Σ , pulled back by $w \mapsto h(w, \cdot)$, is precisely μ_d . Hence

$$\int_{\mathcal{P}_{T,v}} du d^2\theta d\pi_u d^2P = \mu_d(O_0), \quad \nu_T(\mathcal{P}_T) = |J_0|\mu_d(O_0). \quad (7.5)$$

This uses injectivity of transverse phase-space hitting maps; it makes no injectivity assertion about their projection onto positions.

The following argument applies separately to the background metric and to any persisted finite experiment satisfying Proposition 6.3. Denote either metric temporarily by g and put $G_{ij} = g(F_i, F_j)$. On an observation box containing the positions in (7.4) with a fixed margin, we have

$$\|G\| + \|G^{-1}\| \leq C, \quad \sum_{|\beta| \leq 30} (\|\partial^\beta G\| + \|\partial^\beta G^{-1}\|) \leq Q_T, \quad Q_T = e^{o(T)}. \quad (7.6)$$

All derivatives here are ordinary coordinate derivatives. The selected future unit tangent U at every parameter in $\mathcal{P}_T$ has bounded F components. Constants in this section may depend on the fixed

observation tube, O , and the chart bound B , but not on T . Finite products of subexponential bounds are absorbed into a new Q_T as in Convention 2.1.

If g_{coord} is the ordinary coordinate matrix, the constant frame change gives $\det G = r^{-2} \det g_{\text{coord}}$. Thus

$$d \text{vol}_g = r \sqrt{|\det G|} du dv d^2\theta, \quad cr \leq \sqrt{|\det g_{\text{coord}}|} \leq Cr. \quad (7.7)$$

For an optical metric these bounds follow directly from $a/r \asymp 1$, the bounded angular metric and shift. Formula (7.7) also holds for the nearby metrics without requiring the observation coordinates to remain optical.

Lemma 7.1 (Comparison at a good base point). *Suppose an extension chart has bounded metric and inverse metric, and the components of U are bounded both in that chart and in F . Then the matrix M with columns F_i in chart coordinates and its inverse are bounded by a constant depending only on these bounds and (7.6).*

Proof. For any Lorentzian Gram matrix G and unit timelike column U , set

$$K(G, U) = G + 2(GU)(GU)^\dagger.$$

The associated form is positive definite: writing a vector as $aU + Z$ with $g(U, Z) = 0$ gives $a^2 + g(Z, Z) > 0$ unless the vector vanishes. The determinant lemma and $U^\dagger GU = -1$ give the exact identity

$$\det K(G, U) = \det G (1 + 2U^\dagger GU) = -\det G. \quad (7.8)$$

Bounded G, G^{-1}, U give an upper eigenvalue bound for K and a positive lower bound for its determinant. Its smallest eigenvalue is therefore bounded below by the determinant divided by the cube of the upper eigenvalue bound. Apply this in both bases. If g_c is the chart matrix and $U_c = MU_F$, then

$$K(G, U_F) = M^\dagger K(g_c, U_c) M.$$

The two uniform positive definite comparisons bound both M and M^{-1} . In particular, bounded Lorentzian Gram matrices alone are not being used to bound a Lorentz boost. $\square$

7.2 Finite jets of coordinate rectangles

The scale choices can be read from the estimates proved below. With a stopped edge-integral cutoff R , position averaging will leave exceptional measure $O((rR^2)^{-1})$, while the extracted quadratic coefficient will be bounded by $O(rR/l_T) + Q_T l_T^{16}$. Taking

$$R = r^{-17/32}, \quad l_T = r^{1/16}, \quad A_T = r^{1/4}$$

gives exceptional measure $O(r^{1/16})$ and curvature errors $O(r^{13/32}) + Q_T r = o(A_T)$. The power l_T^{16} comes from an eighteenth-order remainder after extracting a quadratic coefficient.

Let

$$\mathcal{X} = \{\partial_{\theta^1}, \partial_{\theta^2}, \partial_{\theta^1} + \partial_{\theta^2}\}. \quad (7.9)$$

For a position x in the parameter set, $X \in \mathcal{X}$, and $0 \leq l \leq 17l_T$, take the coordinate rectangle with consecutive displacements $l\partial_v, lX, -l\partial_v, -lX$. Denote its holonomy in the based frame $F(x)$ by $H_{x,X}(l)$. Every rectangle lies in the enlarged observation box for large T , because its ordinary coordinate displacement is $O(l)$. Its velocity in F , on each unit-parameter edge, is lX times a fixed bounded column. The two ordinary side lengths have different scales, although both directions have this same F -component scale. Figure 2 illustrates these two scales. The quadratic coefficient of the loop holonomy recovers the curvature block D . We will extract that coefficient from the eighteen values at $l = 0, l_T, \dots, 17l_T$; a bound for the eighteenth derivative controls the interpolation remainder at the required $o(A_T)$ scale.

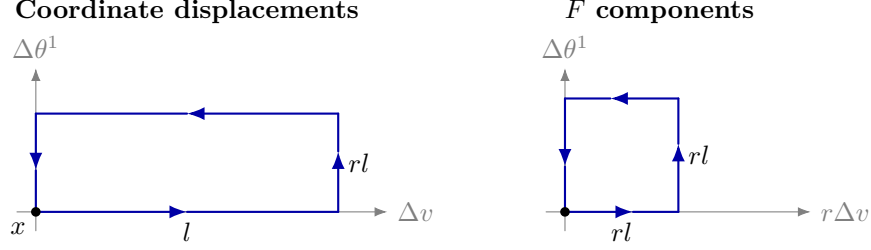


Figure 2: A coordinate rectangle for $X = \partial_{\theta^1}$, with u and θ^2 fixed (schematic). Its sides have equal component scale in F , even though their coordinate scales differ. The proof averages translated rectangles over all four base-position variables and the three bounded momentum labels.

Lemma 7.2 (Loop jets and the quadratic coefficient). *Uniformly in the observation positions and $X \in \mathcal{X}$,*

$$\sup_{0 \leq l \leq 17l_T} \|\partial_l^{18} H_{x,X}(l)\| \leq Q_T. \quad (7.10)$$

The quadratic Taylor coefficient $C_2(x, X)$ of $H_{x,X}(l) - \text{Id}$ is the matrix, in $F(x)$, of $\pm R(\partial_v, rX)$. Consequently

$$g(C_2(x, X)F_1, X) = \pm R(\partial_v, X, \partial_v, X) = \pm D(X, X). \quad (7.11)$$

The same curvature and loop-orientation convention fixes the signs.

Proof. Since F commutes, its connection coefficients are given by Koszul's formula

$$C^k_{ij} = \frac{1}{2} G^{kh} (F_i G_{jh} + F_j G_{ih} - F_h G_{ij}).$$

For a constant column V and $W = rV^i F_i$, the connection matrix on W is therefore

$$A(W)^k_j = \frac{1}{2} G^{kh} V^i ((rF_i) G_{jh} + (rF_j) G_{ih} - (rF_h) G_{ij}). \quad (7.12)$$

Each rF_i is one of $r\partial_u, \partial_v, r\partial_{\theta^A}$. Their coefficients are constant and bounded. Thus the connection contracted with each of ∂_v and rX , and its ordinary derivatives through order eighteen, are bounded by Q_T . This proves the required cancellation directly, including all shift and lapse terms; uncontracted F connection coefficients need not have such a bound.

For an edge parametrized by $s \in [0, 1]$, its position has the form $x + c_e(s, l)$, where c_e is linear in l and independent of momentum. Its transport equation in F has coefficient $lA(\pm W)(x + c_e(s, l))$ for $W = \partial_v$ or rX . Its integral norm is at most $lQ_T = o(1)$, uniformly in the indicated range. Differentiating this coefficient at most eighteen times in l uses only ordinary derivatives in bounded coordinate directions, so each derivative is bounded by Q_T . To see the corresponding assertion for transport, differentiate $P' = -A_e P$, $P(0) = \text{Id}$, k times in l :

$$(\partial_l^k P)' = -A_e \partial_l^k P - \sum_{j=1}^k \binom{k}{j} (\partial_l^j A_e) (\partial_l^{k-j} P).$$

Gronwall uses only the small integral of the undifferentiated A_e . Induction in $k \leq 18$ bounds every differentiated solution by finite products of Q_T , hence by another subexponential bound. Multiplying the four edge transports proves (7.10).

For completeness, pull the smooth connection back by $(s, t) \mapsto x + s\partial_v + tX$. Choose a smooth frame agreeing with F at $(0, 0)$ whose connection one-form vanishes there. This is possible by

prescribing the first derivatives of a frame change at that point. On the boundary of $[0, l]^2$ its connection one-form has integral norm $O(l^2)$. The ordered transport equals Id minus its boundary integral up to $O(l^4)$. Stokes' theorem identifies the quadratic coefficient with minus its exterior derivative at the origin; since the connection form vanishes there, this is the curvature on (∂_v, rX) , with the orientation sign. This auxiliary frame is used only to identify a Taylor coefficient, not to obtain a uniform estimate. Finally, multilinearity gives

$$g(R(\partial_v, rX)(r^{-1}\partial_v), X) = rr^{-1}R(\partial_v, X, \partial_v, X).$$

This proves (7.11) with no remaining factor of r . $\square$

7.3 Averaging before closing the chart estimates

Proposition 7.3 (Averaged holonomy obstruction). *Assume (7.6) on the observation box, bounded F components of the selected unit tangents, and a fixed bounded momentum box in (7.4). Fix one extension and one collection of at most B charts with the metric, inverse-metric, and connection bounds in Definition 3.5. Let $\mathcal{G}_T \subset \mathcal{P}_T$ be the measurable parameters whose base point and unit tangent satisfy the margin and velocity conditions in at least one chart of this same collection. There is a measurable exceptional set $\mathcal{N}_T \subset \mathcal{G}_T$ such that*

$$\nu_T(\mathcal{N}_T) \leq Ce^{-\kappa T/16}, \quad (7.13)$$

and, for all parameters in $\mathcal{G}_T \setminus \mathcal{N}_T$,

$$|D| \leq Ce^{-13\kappa T/32} + Q_T e^{-\kappa T} = o(A_T). \quad (7.14)$$

The constants use only B and the uniform observation bounds.

Proof. At each good parameter select the least chart index for which the coordinate margin is at least $1/B$ and $|U_c| \leq B$. The finite selector is measurable and may depend on all three momentum labels. Fix Borel representatives of the finitely many weak Christoffel arrays. By Lemma 7.1, the base frame matrices obey $\|M(0)\| + \|M(0)^{-1}\| \leq C_0$. Choose a fixed $K > 2C_0 + 1$.

Fix $X \in \mathcal{X}$ and a sampled positive size $l = jl_T$, $1 \leq j \leq 17$. Use the selected base chart throughout the four concatenated edges. Define one first stop on this concatenation: stop when either $\|M\| = K$ or the chart position has moved Euclidean distance $1/(2B)$ from its base position. The latter displacement keeps the path in the chart because the base margin is $1/B$. An edge after an earlier stop contributes nothing to any integral below. Before this global stop the smooth interior path, frame matrix, and chart coordinates are all defined. More precisely, a parameter is before the stop at a given time when both strict thresholds hold at every earlier time in the concatenation. In a fixed chart, continuity expresses this condition using suprema over rational earlier times, so it is Borel on the smooth state domain. Intersecting with the measurable selected parameter set gives the stopped subsets used below.

Write $x = (v, u, \theta)$ and $m = (\pi_u, P)$. At a fixed edge parameter s the position is the translation $x \mapsto x + c_e(s, l)$, independent of m . For the four edges the translations are, successively,

$$sl\partial_v, \quad l\partial_v + slrX, \quad (1-s)l\partial_v + lrX, \quad (1-s)lrX.$$

For a selected chart c , restrict first to good parameters before the global stop. Enlarge this nonnegative integral to all $m \in Q$ and all translated observation positions lying in that chart. On positions the translation, the smooth open isometric embedding, and the chart map compose to an

injective smooth coordinate map. Its volume change is controlled without estimating its derivative matrix: isometry and (7.7) give

$$dx \leq Cr^{-1} d \text{vol}_g = Cr^{-1} \sqrt{|\det \tilde{g}_c|} dy_c \leq C_B r^{-1} dy_c. \quad (7.15)$$

Consequently, for each fixed s ,

$$\int_{\mathcal{G}_T: \text{chart } c \text{ selected before global stop}} |\tilde{\Gamma}_c(y_c(x + c_e(s, l)))|^2 d\nu_T \leq \frac{C_B |Q|}{r} \int_c |\tilde{\Gamma}_c(y)|^2 dy. \quad (7.16)$$

Here $d\nu_T = dx dm$; no seed Jacobian remains. The bounded momentum fiber contributes only $|Q|$. Momentum-dependent chart selection and stop conditions have only restricted the integral before it was enlarged. In particular, (7.15) has not used the unproved along-edge bound for M .

The weak chart Christoffel array agrees almost everywhere on the open image with its smooth interior array. For each s , the inverse image of the exceptional position null set under the preceding smooth coordinate map is null. Fubini in s, x, m shows that, outside a null set of parameters, the stopped trajectory integrals are unchanged by this choice of representative. We discard that null set for each of the finitely many sampled loops. This justifies the use of the smooth connection equation below, without asserting a Sobolev trace on an individually prescribed loop.

Integrate (7.16) in $s \in [0, 1]$ and sum the at most B charts. If

$$I_e(x, m) = \int_0^1 \mathbf{1}_{\{\text{before global stop}\}} |\tilde{\Gamma}_c(y_e(s))| ds,$$

Cauchy–Schwarz and the chart L^2 bounds imply

$$\int_{\mathcal{G}_T} I_e^2 d\nu_T \leq C/r. \quad (7.17)$$

Set $R_T = r^{-1/2} e^{\kappa T/32}$. Markov's inequality applied to I_e^2 gives

$$\nu_T\{I_e > R_T\} \leq (C/r) R_T^{-2} = C e^{-\kappa T/16}. \quad (7.18)$$

There are four edges, seventeen positive sizes, and three choices of X . The union of their exceptional sets, together with the null sets already removed, satisfies (7.13). Only this finite collection is used: no common exceptional set for a continuum of positive loop sizes is asserted.

We now prove that no stop occurs for a remaining parameter. On one edge write its F -velocity as $f = lrV$, with V a fixed bounded column, and let A be its internal connection matrix. By (7.12), $|A| \leq lQ_T$. The connection transformation law along the smooth interior curve is the exact matrix equation

$$M' = MA - \tilde{\Gamma}_c(Mf)M. \quad (7.19)$$

Here $\tilde{\Gamma}_c(Mf)$ means the chart connection matrix contracted with the chart velocity Mf . Before the stop,

$$|M'| \leq KlQ_T + CK^2 lr |\tilde{\Gamma}_c|, \quad |y'| \leq CKlr.$$

Summing over all four stopped edges and using $I_e \leq R_T$ yields

$$\begin{aligned} \sup \|M - M(0)\| &\leq 4KlQ_T + CK^2 lr R_T = o(1), \\ \sup |y - y(0)| &\leq CKlr = o(1). \end{aligned} \quad (7.20)$$

Indeed $l \leq 17e^{-\kappa T/16}$ and $lrR_T \leq 17e^{-17\kappa T/32}$. For large T these inequalities are strict improvements of both stop thresholds. Continuity rules out a first stop, including on any of the later three edges. Thus the entire loop stays in its selected base chart, with $\|M\| < K$. Only an upper matrix threshold was needed; the inverse at the base is already bounded. If desired, inverse bounds along the loop also follow from $\det(M)^2 = \det G / \det \tilde{g}_c$ and the cofactor formula.

Ordinary chart parallel transport now has total integrated coefficient at most $Clr \sum_e I_e \leq Clr^{1/2}e^{\kappa T/32}$. Its matrix differs from Id by at most the exponential of this quantity minus one. Since the rectangle closes exactly, its final frame matrix equals its base matrix. Thus if P_c is the full chart transport, $H = M(0)^{-1}P_cM(0)$. Conjugating by the bounded base matrix and its inverse gives

$$\|H_{x,X}(jl_T) - \text{Id}\| \leq Cr^{1/2}l_T e^{\kappa T/32}, \quad 1 \leq j \leq 17. \quad (7.21)$$

At $j = 0$ the left side is zero. All transports in this argument have taken place in the smooth original spacetime.

It remains to extract the quadratic coefficient from these eighteen values. We include the elementary interpolation argument, as in [22, Lemma 8.5]. Write

$$H_{x,X}(l) - \text{Id} = \sum_{k=0}^{17} C_k l^k + \mathcal{R}_{18}(l).$$

By Lemma 7.2, the Taylor integral remainder at $l = jl_T$ is bounded by $CQ_T l_T^{18}$. The Vandermonde matrix $(j^k)_{0 \leq j, k \leq 17}$ is invertible and independent of T . Therefore its inverse expresses $l_T^2 C_2$ as a fixed linear combination of the values of the polynomial at these nodes, giving

$$\begin{aligned} \|C_2\| &\leq Cl_T^{-2} \max_{0 \leq j \leq 17} \|H_{x,X}(jl_T) - \text{Id}\| + Q_T l_T^{16} \\ &\leq Cr^{1/2} l_T^{-1} e^{\kappa T/32} + Q_T l_T^{16} \\ &= Ce^{-13\kappa T/32} + Q_T e^{-\kappa T}. \end{aligned} \quad (7.22)$$

Both exponents have a strict gain over $-\kappa/4$. By (7.11) and boundedness of G , this controls $D(X, X)$ for each $X \in \mathcal{X}$. Curvature pair symmetry gives $D_{12} = D_{21}$, and

$$2D_{12} = D(\partial_{\theta^1} + \partial_{\theta^2}, \partial_{\theta^1} + \partial_{\theta^2}) - D_{11} - D_{22}.$$

These three estimates prove (7.14) for the entire symmetric 2×2 block. $\square$

7.4 Finite persistence and nowhere denseness

The preceding estimate makes the observed curvature block small on almost every parameter with an extension-chart witness. To contradict the packet signal, we will apply it to the original datum and a nearby datum at identical numerical terminal states. Canonical section volume ensures that their two good families still overlap.

Proposition 6.3 supplies the needed persistence for each fixed finite experiment: a relative open neighborhood preserves the signal with lower bound A_T , the frame and finite-jet bounds, and the injective measure-preserving backward maps into O . Its map tolerance can be chosen after T , and its data-neighborhood radius may depend on that experiment. The next argument uses precisely this finite persistence.

Theorem 7.4 (Nowhere denseness of the extension tests). *For the single neighborhood $\mathcal{U}_\varepsilon$ fixed in Proposition 2.2, every set $E(O, B)$ of Definition 3.5 is relatively nowhere dense in $\mathcal{U}_\varepsilon$.*

Proof. Suppose instead that a nonempty relative open set $V \subset \mathcal{U}_\varepsilon$ is contained in $\overline{E(O, B)}$. Since $E(O, B)$ is dense in V , choose $d \in E(O, B) \cap V$. The weighted seminorms are increasing, so there are a finite $m \geq 10$ and $\eta > 0$ such that

$$\{d' \in \mathcal{U}_\varepsilon : p_m(d' - d) < \eta\} \subset V. \quad (7.23)$$

Fix the branch tube and the extension witness belonging to this datum. Apply the finite packet construction at order m , along its unbounded sequence of admissible observation times. By Propositions 4.1 and 6.1 and Theorem 5.1, the exact full-bridge data d_T satisfy $p_m(d_T - d) \rightarrow 0$; hence they lie in V for all sufficiently large T .

The set $E(O, B)$ is dense in V , so Proposition 6.3 allows us, for every such fixed T , to choose $\hat{d}_T \in E(O, B) \cap V$ as close to d_T as required by all the finite observation conditions. We also require $p_0(\hat{d}_T - d_T) \rightarrow 0$. Write $\hat{g}_T$ for its observed metric. We then have, on the common position tube,

$$|D(\hat{g}_T) - D(g_d)| > A_T \quad \text{if } |\cos(\lambda v)| \geq 1/2, \quad (7.24)$$

and both metrics have the uniform hypotheses of Proposition 7.3. In particular the constants there are independent of T ; the required finite closeness was chosen to retain them. Smooth dependence of the initial section density on h gives

$$\frac{\mu_{\hat{d}_T}(O)}{\mu_d(O)} \rightarrow 1. \quad (7.25)$$

Let $S \subset O$ and $\hat{S}_T \subset O$ be the good seed families in the two existential extension witnesses. Each has more than 99 percent of its respective measure of O . Use (7.4) for the background hits of O_0 . At a fixed v , the loss from parameters whose background seed is outside S is at most $.01\mu_d(O)$. The matched backward map for $\hat{g}_T$ is injective into O and preserves its canonical section measure. Its loss from $O \setminus \hat{S}_T$ is therefore at most $.01\mu_{\hat{d}_T}(O)$. By (7.25), their combined loss is less than $.04\mu_d(O)$ for large T , uniformly on every v slice. Thus, if $\mathcal{G}_T^\cap$ denotes parameters good for both metrics,

$$\text{vol}_6(\mathcal{G}_T^\cap \cap \{v = \text{constant}\}) > (.95 - .04)\mu_d(O) = .91\mu_d(O). \quad (7.26)$$

The dependence on v is measurable by the smooth finite flow maps. The chart bounds from either witness apply at these observations because Definition 3.5 imposes them throughout the full future geodesic lifetime, and the matched observations are interior points of that geodesic.

Let

$$I_T = \{v \in T + J_0 : |\cos(\lambda v)| \geq 1/2\}.$$

The condition occupies $2/3$ of each full period. Only the two end periods of an interval can be incomplete, so $|I_T| = \frac{2}{3}|J_0| + O(\lambda^{-1}) > \frac{1}{2}|J_0|$ for large T . Integration of (7.26) over I_T gives

$$\nu_T(\mathcal{G}_T^\cap \cap \{v \in I_T\}) > .455\mu_d(O)|J_0|. \quad (7.27)$$

Apply Proposition 7.3 separately to g_d and $\hat{g}_T$. The union of the two additional exceptional sets has measure tending to zero. It cannot cover (7.27). At a remaining parameter both $|D(g_d)|$ and $|D(\hat{g}_T)|$ are $o(A_T)$, uniformly, whereas (7.24) says their difference exceeds A_T . This is a contradiction.

We have excluded nonempty interior of $\overline{E(O, B)}$. No closedness of $E(O, B)$ was assumed. $\square$

Proof of Theorem 1.2. By Proposition 3.6, every datum in $\mathcal{U}_\varepsilon$ admitting the future extension in the statement belongs to some $E(O, B)$. The seed basis and the positive integers B are countable. By Theorem 7.4, their union is meagre in the full relative weighted smooth topology on $\mathcal{U}_\varepsilon$. Explicitly,

$$\mathcal{R} = \bigcap_{O, B} (\mathcal{U}_\varepsilon \setminus \overline{E(O, B)})$$

is a countable intersection of relatively open dense sets and consists of future-inextendible data. The exit cover concerns all future exits of the full developments. Every background observation precedes the corresponding geodesic's first comparison runout. The exact and persisted observations lie in finite interior regions of their full developments, where the full-lifetime chart witnesses apply. Thus the conclusion excludes every extension specified in Theorem 1.2, for this single p_{10} neighborhood. $\square$

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